

Telflow HLD - FTTH Solution - Updated

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 - Bespoke Integration (Synchronous)
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 - Bespoke Integration (Synchronous)

Introduction

Mobily had launched a program - Mobily DBSS Transformation Program, which encompasses the implementation of both B2C and B2B functionality as part of this solution. Mobily is planning to execute this program in launches, wherein each launch is committed to deliver certain identified products and services which are part of their business.

CSG is already deploying the B2C stack and Mobily had decided to implement the solution for B2C FTTH products along with the existing B2C stack through the introduction of Telflow in this solution.

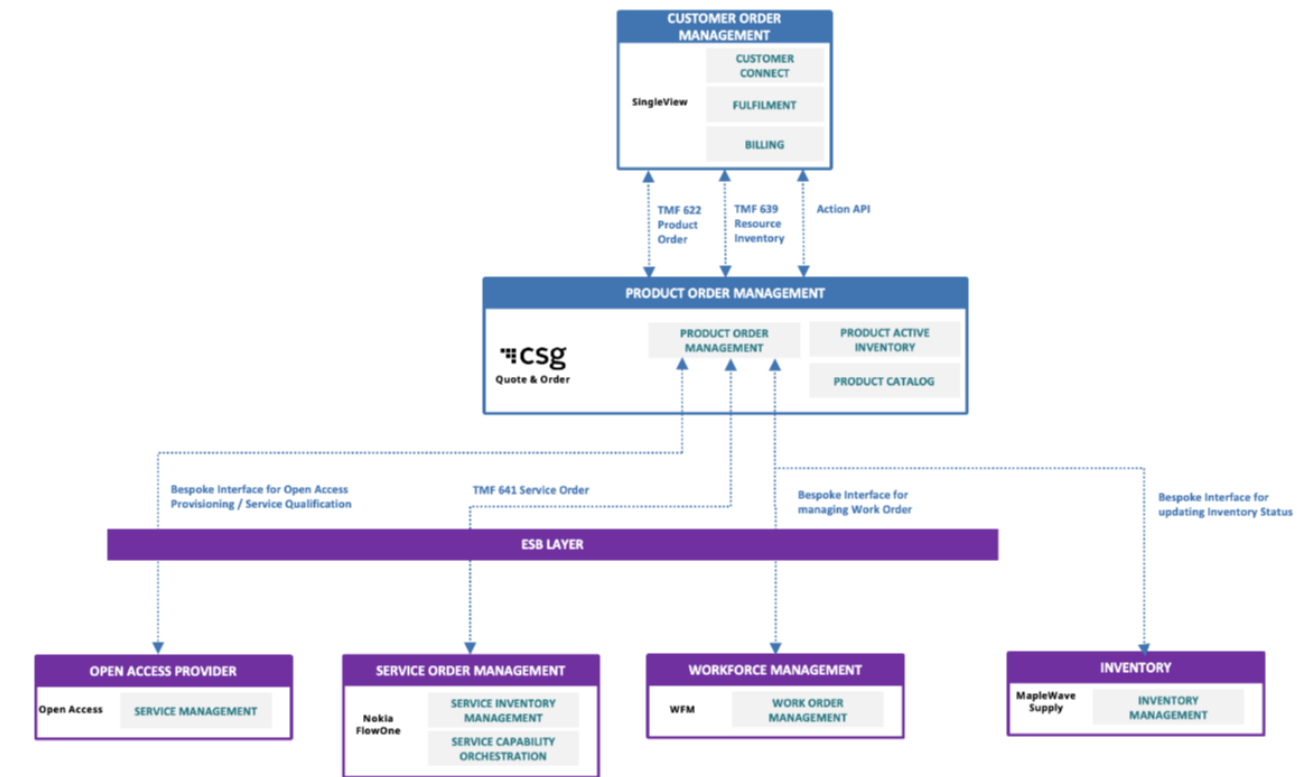
This document represents the solution proposed for L5 FTTH related use cases.

Glossary

Term	Definition
B2C	Business to Consumer
B2B	Business to Business
SV	SingleView
NFO	Nokia Flow One
WFM	Work Force Management
VAS	Value Added Services
ESB	Enterprise Service Bus
API	Application Programming Interface
POM	Product Order Management
SOM	Service Order Management
TMF	TM Forum
FTTH	Fiber to the Home

Solution Context

Context Diagram



Use Case Journeys

Use Case Overview

The following use cases are addressed in this solution design document,

S. No	Join & Buy	
	Use Case	Variant
1	L5-FTTH-Join&Buy-01 - Consolidated Join and Buy-Customer buys new FTTH Product via POS/Tafeely	1. Customer Buys New FTTH product in POS/Tafeely - Mobily Infrastructure 2. Customer Buys New FTTH product in POS/Tafeely - Open Access 3. Bundling FTTH product with Mesh extenders
2	L5-FTTH-Join&Buy-05 - Customer buys new FTTH product via Digital	1. Customer Buys New FTTH product via Digital - Mobily Infrastructure 2. Customer Buys New FTTH product via Digital - Open Access
3	L5-FTTH-Join&Buy-08 - Staff buys new FTTH product	
4	L5-FTTH-Join&Buy-09 - RCY FTTH Activation	
5	L5-FTTH-Join&Buy-03 - Auto Cancellation of FTTH Order	
6	L5-FTTH-Join&Buy-11 - Customer rejects the UAT-Mobily Infra	
7	L5-FTTH-Join&Buy-12 - Customer buys FTTH Add-ons in Store, Digital and CC	
8	L5-FTTH-Join&Buy-19 - Customer buys Mesh Extender after Activation using POS	
9	L5-FTTH-Join&Buy-21 Customer buys new prepaid FTTH Package using GCC ID or Border Control Number using Digital and POS	

	Change	
10	L5-FTTH-Change-02 - Customer requests FTTH Device Swap from POS	1. Mobily Infra 2. Open Access
11	L5-FTTH-Change-03 - Customer requests FTTH bandwidth upgrade/downgrade from POS, Digital and CC	
12	L5-FTTH-Change-06 - Customer requests FTTH Rewiring from POS	1. Mobily Infra 2. Open Access
13	L5-FTTH-Change-07 - Installation failure scenarios for Open Access partner	
14	L5-FTTH-Change-08 - Installation failure scenarios for Mobily Infra	
15	L5-FTTH-Change-15 - FTTH Device Swap from WFM	
16	L5-FTTH-Change-16 - FTTH Rewiring from WFM	
17	L5-FTTH-Change-11 - Customer voluntary suspends FTTH services in POS, CC and Digital	1. Customer has fixed voice service
18	L5-FTTH-Change-12 - Customer resumes FTTH service via CC/POS/Digital	1. Customer has fixed voice service
19	L5-FTTH-Change-04 - Customer requests to cancel FTTH order from POS and CC	1. Customer requests to cancel FTTH activation order from POS and CC 2. Customer requests to cancel FTTH relocation order from POS and CC 3. Customer requests to cancel FTTH rewiring order from POS and CC 4. Customer requests to cancel FTTH CPE/HAG swap order from POS and CC 5. Customer requests to cancel FTTH ONT swap order from POS and CC
20	L5-FTTH-Change-05 - Customer request update on open order from POS and CC	1. Customer requests update on FTTH activation order from POS and CC 2. Customer requests update on FTTH relocation order from POS and CC 3. Customer requests update on FTTH rewiring order from POS and CC 4. Customer requests update on FTTH CPE/HAG swap order from POS and CC 5. Customer requests update on FTTH ONT swap order from POS and CC
21	L5-FTTH-Change-01 - Customer requests FTTH Relocation from POS, CC	1. Relocation within Mobily Infrastructure 2. Relocation from Open Access to Mobily Infrastructure 3. Relocation within Open Access Provider 4. Relocation from Mobily to Open Access Infrastructure 5. Relocation from Open Access to another Open Access Infrastructure
22	L5-FTTH-Change-10 - Customer changes from Postpaid to Prepaid FTTH package in POS	1. Customer has fixed voice service
23	L5-FTTH-Change-09 Consolidated Change - Customer changes from Prepaid to Postpaid FTTH package in POS	
24	L5-FTTH-Change-13 Consolidated Change - Customer changes Fixed Number in POS	
25	L5-FTTH-Change-18 Consolidated Change - Customer changes account ownership in POS	

26	L5-FTTH-Change-22 Customer requests FTTH Rewiring using CC.	
Get Help		
27	L5-FTTH-Get Help-63 - Customer raises issue in POS, Digital and CC	
28	L5-FTTH-Get Help-68 FTTH Complaints - Negative Scenarios	1. Mobily Infra 2. Open Access
Leave		
29	L5-FTTH-Leave-01 - Customer terminates FTTH service in POS, Digital and CC	1. Customer does not have fixed voice service along with FTTH - Mobily Infrastructure 2. Customer does not have fixed voice service along with FTTH - Open Access
30	L5-FTTH-Leave-04 - Customer request FTTH service and Fixed Voice disconnection	1. Customer has fixed voice service along with FTTH - Mobily Infrastructure 2. Customer has fixed voice service along with FTTH - Open Access
31	L5-FTTH-Leave-02 - Customer requests Fixed Voice disconnection in POS	
32	L5-FTTH-Leave-06 - Fixed voice disconnection due to FTTH Postpaid to Prepaid package change	
33	L5-FTTH-Leave-05 Consolidated Leave - Fixed Voice Disconnection due to FTTH relocation	1. Relocation within the same fixed voice geographical zone 2. Relocation outside the fixed voice geographical zone
Pay or Not		
34	L5-FTTH-PayOrNot-06 - Customer fails to pay FTTH invoice and gets treated in Collections Treatment Flow	
35	L5-FTTH-PayOrNot-07 - Customer agrees Promise-to-Pay to maintain FTTH service while in Treatment	
36	L5-FTTH-PayOrNot-09 - Impact of dunning life cycle in case customer has Fixed voice	
37	L5-FTTH-PayOrNot-02 - Customer Prepaid FTTH service recharge with auto renewal using Best Fit Method	
38	L5-FTTH-PayOrNot-03 - Customer with Prepaid FTTH service receives Grace Period	
Get Ready		
39	L5-FTTH-GetReady-16 Mobily adds ACES as Open Access Service Provider	

Functional Block Diagram

Functional block diagrams considers the E2E design as an input that constitutes a series of user tasks, system activities, and the sequence of how the services are delivered.

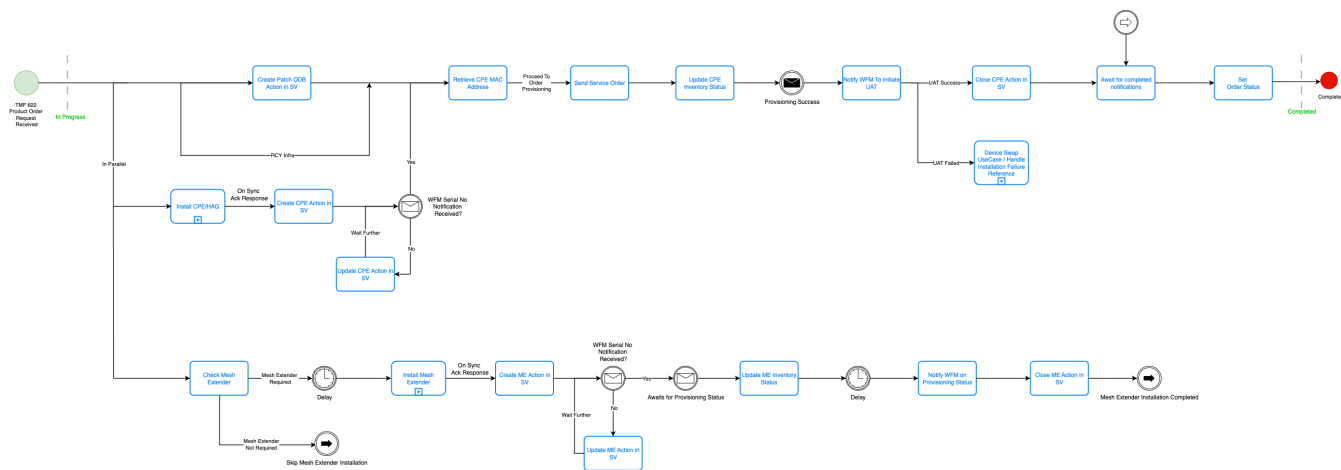
Mobily FTTH Service Activation

The below block diagram covers the following use cases,

1. Customer Buys New FTTH product in POS / Tafeely- Mobily Infrastructure
2. Customer Buys New FTTH product via Digital - Mobily Infrastructure
3. Staff buys new FTTH Product
4. RCY FTTH Activation
5. Customer buys new prepaid FTTH Package using GCC ID or Border Control Number using Digital and POS

for the above use cases, below sub-process has been used that can be referred in their respective sections:

1. Handle work order



S. No	Activity	Assigned Team /System	Description
1	Create Patch ODB Action in SV	SingleView	Telflow initiates ODB patching action in SingleView and awaits for completion notification. Email notification shall be triggered to patching team. A patching task shall be initiated irrespective of whether the ODB is old or new to attach the customer device with the ODB. This is applicable only for non-RCY customers.
2	Install CPE /HAG	WFM	This sub-process invokes 'Handle work order' sub-process. Note: - Patch ODB, installing CPE/HAG and ME are parallel activities - Upon receiving the synchronous response, Install CPE/HAG follows step 5.
3	Create CPE Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
4	WFM Serial No Notification Received?	Telflow	Telflow awaits for WFM serial number notification. Possible outcomes: - Yes Follow step 14: Retrieve CPE MAC Address - No Follow step 7: Update CPE Action in SV
5	Update CPE Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 6. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details - Comments, if any.
6	Check Mesh Extender	Telflow	Check whether the customer had opted for mesh extender. Possible outcomes: - Mesh Extender Required Follow step 10: Create ME Action in SV - Mesh Extender Not Required Follow step 20: Await for completed notifications
7	Install Mesh Extender	WFM	This sub-process invokes 'Handle work order' sub-process. Note: Upon receiving the synchronous response, this step follows step 10.

8	Create ME Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id (If required) 3. SingleView Order Id Note: • This activity shall be triggered with a delay of 5 seconds post step 4.
9	WFM Serial No Notification Received?	Telflow	Telflow awaits for WFM serial number notification. Possible outcomes: 1. Yes Follow step 13: Awaits for Provisioning Status 2. No Follow step 12: Update ME Action in SV
10	Update ME Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 11. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id (If required) 3. SingleView Order Id 4. Task status 5. Engineer details (Id, Name, etc) 6. Comments, if any.
11	Awaits for Provisioning Status	Telflow	Upon receiving the provisioning success message, follows step 21.
12	Retrieve CPE MAC Address	SingleView	Based on CPE Serial Number, Telflow retrieves CPE MAC Address from SingleView via TMF639 and proceeds the order further.
13	Send Service Order	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
14	Update CPE Inventory Status	MapleWave	Telflow updates the inventory status of the CPE allocated to the customer in MapleWave.
15	Provisioning Success	Telflow	Sends the message event which shall be consumed by step 13.
16	Notify WFM to initiate UAT	WFM	Telflow notifies WFM that the provisioning is success (for CPE) to initiate User Acceptance. Below outcomes are possible: 1. UAT Success Follow step 19: Close CPE Action in SV 2. UAT Failed (Device Swap UseCase / Handle Installation Failure to be referred)
17	Close CPE Action in SV	SingleView	Telflow closes the CPE action in SV.
18	Await for completed notifications	Telflow	Awaits for completion notification from WFM for both CPE and Mesh Extender. Upon receiving the notifications, the process follows step 24.
19	Update ME Inventory Status	MapleWave	Telflow updates the inventory status of the ME allocated to the customer in MapleWave.
20	Notify WFM on Provisioning Status	WFM	Telflow notifies WFM that the provisioning is success for Mesh Extender. Note: • This activity shall be triggered with a delay of 5 seconds post step 18 and awaits for completion notification from WFM. Upon receiving the notification, follows step 23.
21	Close ME Action in SV	SingleView	Telflow closes the ME action in SV and follows step 20.
22	Set Order Status	Telflow	Order status is set to complete.

Open Access FTTH Service Activation

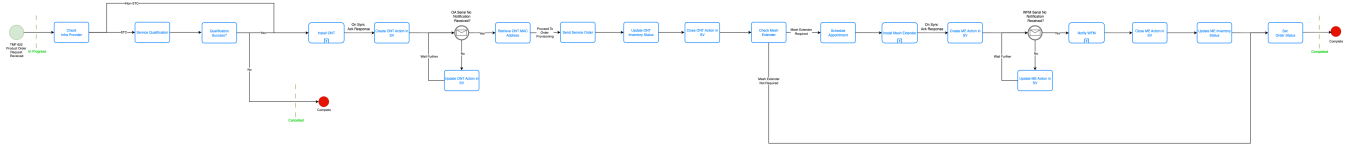
The below block diagram covers the following use cases,

1. Customer Buys New FTTH product in POS / Tafeely - Open Access
2. Customer Buys New FTTH product via Digital - Open Access
3. Staff buys new FTTH Product

4. Customer buys new prepaid FTTH Package using GCC ID or Border Control Number using Digital and POS
5. Mobily adds ACES as Open Access service provider

for the above use cases, below sub-process has been used that can be referred in their respective sections:

1. Handle work order



S. No	Item	Assigned Team	Detail
1	Check Infra Provider	Telflow	Identify the Open Access Infra Provider. The order is already in 'In Progress' state.
2	Service Qualification	Open Access	If the Infra Provider is STC, then Telflow shall initiate service qualification towards Open Access whereas for Non-STC operators, step 4 (Install ONT) shall follow.
3	Qualification Success?	Telflow	Checks whether the service is qualified. Possible outcomes: 1. Yes Follow step 4: Install ONT 2. No Completes the order and set the status to 'Cancelled'
4	Install ONT	Open Access	This sub-process invokes 'Handle work order' sub-process.
5	Create ONT Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id 3. SingleView Order Id
6	OA Serial No Notification Received?	Telflow	Telflow awaits for OA serial number notification. Possible outcomes: 1. Yes Follow step 8: Retrieve ONT MAC Address 2. No Follow step 7: Update ONT Action in SV
7	Update ONT Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 6. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id 3. SingleView Order Id 4. Task status 5. Comments, if any.
8	Retrieve ONT MAC Address	SingleView	Based on ONT Serial Number, Telflow retrieves ONT MAC Address from SingleView via TMF639 and proceeds the order further.
9	Send Service Order	Open Access	Telflow performs order decomposition and sends service order to Open Access provider.
10	Update ONT Inventory Status	MapleWave	Telflow updates the inventory status of the ONT allocated to the customer in MapleWave.
11	Close ONT Action in SV	SingleView	Telflow closes the ONT action in SV.
12	Check Mesh Extender	Telflow	On successful notification to Open Access, checks whether the customer had opted for mesh extender. Possible outcomes: 1. Mesh Extender Required Follow step 13: Schedule Appointment 2. Mesh Extender Not Required -> Set the order status to completed.
13	Schedule Appointment	SingleView	Telflow shall initiate SingleView to schedule an appointment with the customer to install Mesh Extender.

14	Install Mesh Extender	WFM	This sub-process invokes 'Handle work order' sub-process. Note: - This is a standalone Mesh Installation. - Upon receiving the synchronous response, this step follows step 15.
15	Create ME Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
16	WFM Serial No Notification Received?	Telflow	Telflow awaits for WFM serial number notification. Possible outcomes: - Yes Follow step 18: Notify WFM - No Follow step 17: Update ME Action in SV
17	Update ME Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 17. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details (Id, Name, etc) - Comments, if any.
18	Notify WFM	WFM	Telflow notifies WFM that the provisioning is success for Mesh Extender and awaits for completion notification.
19	Close ME Action in SV	SingleView	Telflow closes the ME action in SV.
20	Update ME Inventory Status	MapleWave	Telflow updates the inventory status of the ME allocated to the customer in MapleWave.
21	Set Order Status	Telflow	Order status is set to completed.

Auto Cancellation of FTTH Order

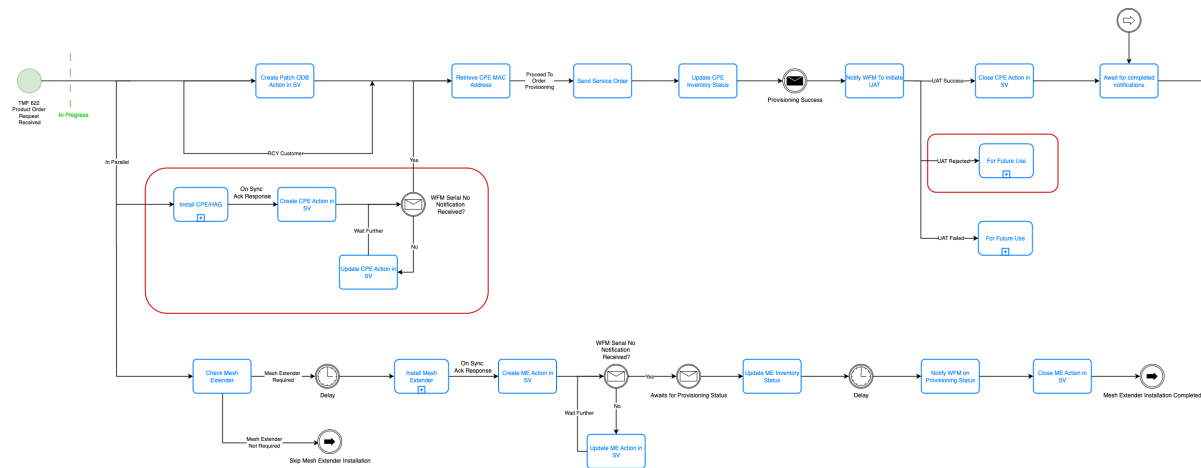
Note:

- Auto-cancellation timer to be configured in Telflow 15 days.
- During WFM installation stage, configure the 'X' days to maximum so that auto-cancellation shall not be triggered. This is to avoid the auto cancellation when the number of days left for appointment since the work order creation is > configured 'X' days.
- Open Access refers to STC / ITC / DAWIYAT / ACES

Auto Cancellation is applicable for the below use cases,

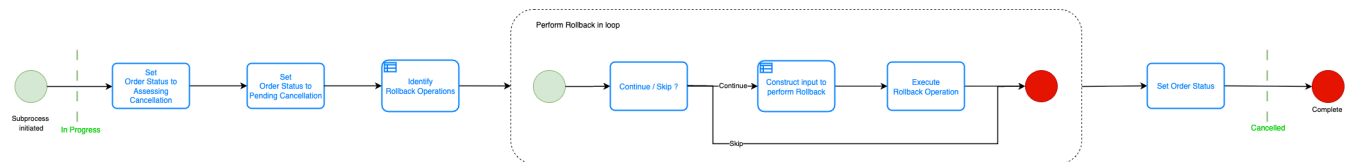
S. No	Use Case	Mobily Infra	Open Access	Scenario
1	FTTH Activation	Yes	Yes	When WFM/Open Access doesn't send CPE/ONT serial number notification within 'X' days. - When WFM/Open Access returns the error response during CPE/ONT installation, the order shall be transitioned to 'In Progress / On Hold' status and the treatment shall be initiated. If the treatment doesn't complete within 'X' days, then auto-cancellation triggers. - When the customer rejects UAT for 3 times or do not want to continue the UAT. Note: - UAT is not applicable for Open Access.
2	Relocation	Yes	Yes	When WFM/Open Access doesn't send CPE/ONT serial number notification within 'X' days. - When WFM/Open Access returns the error response during CPE/ONT installation, the order shall be transitioned to 'In Progress / On Hold' status and the treatment shall be initiated. If the treatment doesn't complete within 'X' days, then auto-cancellation triggers. - When the customer rejects UAT for 3 times or do not want to continue the UAT. Note: - UAT is not applicable for Open Access.

3	Rewiring	Yes	Yes	1. When WFM/Open Access doesn't send completion notification within 'X' days. 2. When WFM/Open Access returns the error response during CPE/ONT installation, the order shall be transitioned to 'In Progress / On Hold' status and the treatment shall be initiated. If the treatment doesn't complete within 'X' days, then auto-cancellation triggers. Note: 1. UAT is not applicable.
4	Device Swap	Yes	Yes	1. When WFM/Open Access doesn't send CPE/ONT serial number notification within 'X' days. 2. When WFM/Open Access returns the error response during CPE/ONT installation, the order shall be transitioned to 'In Progress / On Hold' status and the treatment shall be initiated. If the treatment doesn't complete within 'X' days, then auto-cancellation triggers. Note: 1. UAT is not applicable.



The below block diagram covers the following use cases,

1. Auto Cancellation of FTTH Order



S. No	Activity	Assigned Team /System	Description
1	Set Order Status to Assessing Cancellation	Telflow	Set order to 'Assessing Cancellation' state.
2	Set Order Status to Pending Cancellation	Telflow	Set order to 'Pending Cancellation' state.
3	Identify Rollback Operations	Telflow	Telflow identifies the rollback operations to be performed.
4	Continue / Skip?	Telflow	Possible outcomes: 1. Continue continue the rollback activities 2. Skip skip the rollback activities
5	Construct Input to perform Rollback	Telflow	Based on the rollback operation identified in Step 4, construct the request to be sent to the target system to initiate rollback.
6	Execute Rollback Operation	Target Systems	Telflow executes the rollback operations in sequence.
7	Set Order Status	Telflow	Set order to 'Cancelled' state.

Rollback Operations

During auto cancel, the activities that were initiated/completed in the existing order shall be identified and rolled back by the auto cancel process.

Note:

1. If both CPE and Mesh Extender exists in an order, cancelling CPE work order in WFM implicitly closes the Mesh Extender work order. Telflow doesn't initiate separate cancel work order for Mesh Extender.

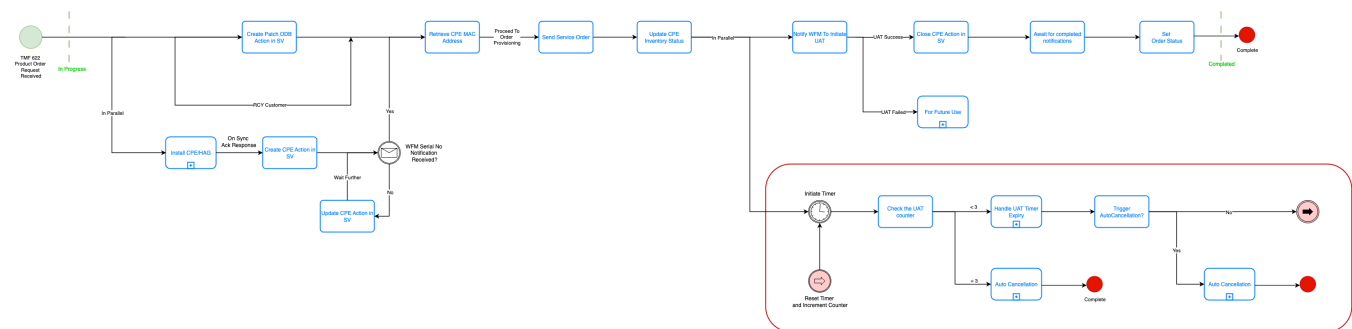
S.No	Stage	Infra Provider	System	Operation	Rollback Operation
1	CPE Installation	Mobily	WFM	Task Creation Request	Task Cancellation Request
			SingleView	Create CPE Action	Close CPE Action Note: Close the action with comments. No need to update.
				Create ME Action	Close ME Action Note: Close the action with comments. No need to update.
2	ONT Installation	Open Access	Open Access	Service Installation Request	Cancel Service Installation Request Send mail to Venkat asking for Cancel API details.
			SingleView	Create ONT Action	Close ONT Action
				Create ME Action (If applicable)	Close ME Action (If applicable)
3	UAT	Mobily	Nokia FlowOne	Activate Service	Deactivate Service
			MapleWave Supply	Update Inventory (SALE)	Update Inventory (RETURN)
			WFM	Task Creation Request	Task Cancellation Request

Customer Rejects UAT - Mobily Infra

The below block diagram (Mobily FTTH Service Activation) highlights the activity to handle UAT rejection.

Below sub-process has been used that can be referred in their respective sections:

1. Handle UAT Timer Expiry



S.No	Activity	Assigned Team/System	Description
1	Initiate Timer	Telflow	Initiate the UAT timer of 'X' days. Note: 1. UAT timer to be configured 10 days.
2	Check the UAT Counter	Telflow	Possible outcomes: 1. < 3 Follow step 3: Handle UAT Timer Expiry 2. = 3 Follow step 6: Auto Cancellation
3	Handle UAT Timer Expiry	Telflow	Invokes the subprocess 'Handle UAT Timer Expiry'. Post the completion of subprocess, follows step 4.
4	Trigger AutoCancellation?	Telflow	Possible outcomes: 1. Yes Follow step 6: Auto Cancellation 2. No Follow step 5: Reset Timer and Increment Counter

5	Reset Timer and Increment Counter	Telflow	Resets the UAT timer, increment the counter by 1, and follows step 1.
6	Auto Cancellation	Telflow	Initiates Auto Cancellation to perform rollback and cancels the process.
7	Set Order Status	Telflow	Set order to 'Cancelled' state.

Customer buys FTTH AddOn

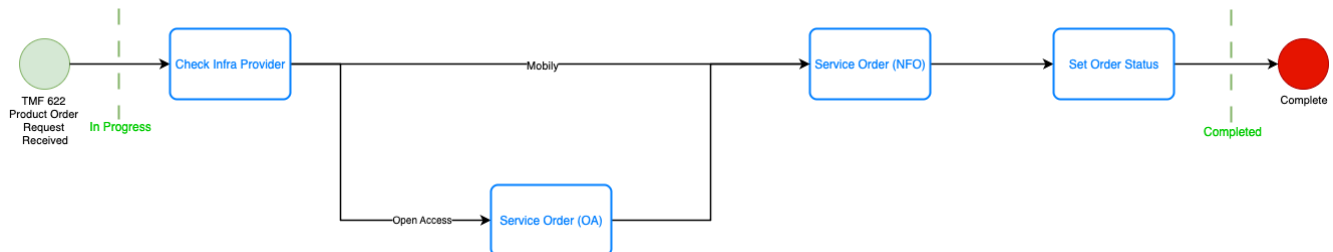
The below block diagram covers the following use case for FTTH AddOn in Mobily/Open Access Infrastructure.

1. Customer buys FTTH Add-ons in Store, Digital and CC

In Mobily/Open Access, customer purchases FTTH AddOn to boost the speed of the FTTH service for limited period of time, i.e., the bandwidth (QoS Profile) of the FTTH service shall be upgraded for certain duration and then downgraded post the duration ends. In Telflow and from network perspective, this is a bandwidth change. Hence, SingleView sends the bandwidth upgrade/downgrade request to Telflow and the same shall be sent to Nokia FlowOne or Open Access to provision FTTH AddOn.

Note:

1. This use case is currently not in production, hence Mobily business team must validate this design and share requisite details to implement.



S.No	Activity	Assigned Team/System	Description
1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily' or 'Open Access'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access system.
2	Service Order OA	Open Access	Telflow performs order decomposition and sends service order to Open Access provider.
3	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
4	Set Order Status	Telflow	Order status is set to complete.

Installation Failures

Installation failure scenarios are applicable when WFM is unable to successfully complete the work order. During such circumstances, WFM shall send the error response to Telflow and based on the error response, Telflow shall initiate and handle it via treatment. As part of the treatment outcome, either the order has to be resumed or cancelled.

During treatment handling, the comments entered by the technician/agent shall be cascaded between systems by Telflow.

In case of Mobily Infra,

1. When WFM sends the error response to Telflow, the comments entered by the technician in WFM work order shall be cascaded to SingleView via actions.
2. When the agent completes the action, any comments captured by the agent on the action shall be cascaded from SingleView to Telflow and the same shall be cascaded in the subsequent actions initiated by Telflow towards SingleView as part of treatment handling.
3. Post treatment, when Telflow creates a new work order, the comments shall be passed in the WFM request.

In case of Open Access,

1. When Open Access sends the error response to Telflow, the comments entered by the technician in the Open Access installation order shall be cascaded to SingleView via actions.
2. When the agent completes the action, any comments captured by the agent on the action shall be cascaded from SingleView to Telflow and the same shall be cascaded in the subsequent actions initiated by Telflow towards SingleView as part of treatment handling.
3. Post treatment, when Telflow creates a new installation order, the comments shall be passed in the installation request (if applicable, based on the API support).

Note:

Telflow doesn't persist the history of the comments, rather it will be overwritten.

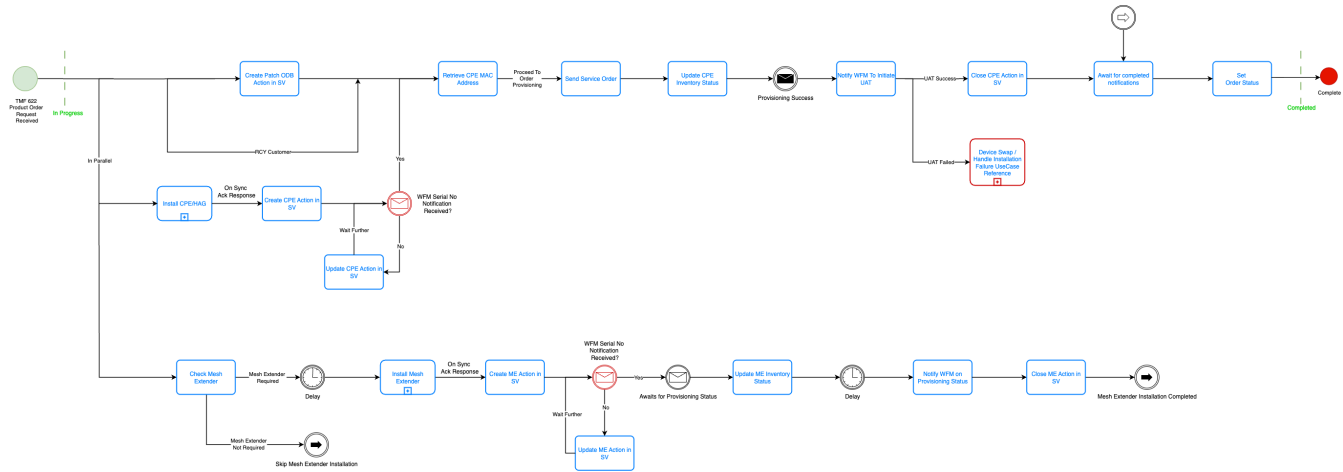
The installation failure scenarios are to be handled for the below use cases,

1. Mobily FTTH Service Activation
2. Open Access FTTH Service Activation
3. FTTH Relocation

4. FTTH Rewiring
5. FTTH Device Swap
6. Mobily adds ACES as Open Access Service Provider

The below block diagram (Mobily FTTH Service Activation) highlights the WFM activity to be considered to handle installation failures. Below are the sub-processes that shall be invoked by the WFM activity,

1. Handle Work Order
2. Handle Installation Failure



Mobily FTTH Rewiring

The below block diagram covers the following use case for FTTH Rewiring in Mobily/Open Access Infrastructure.

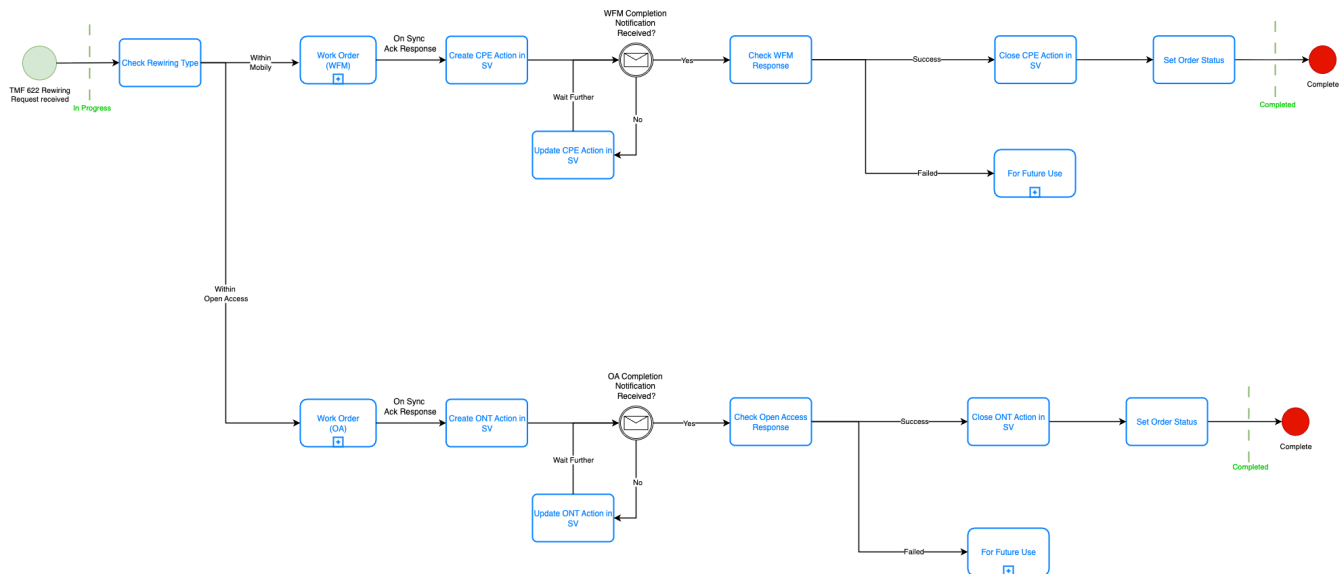
1. Customer requests FTTH Rewiring from POS
2. FTTH Rewiring from WFM
3. Customer requests FTTH Rewiring using CC

Note:

1. UAT is not applicable for Rewiring.
2. Open Access refers to DAWIYAT / ACES

for the above use cases, below sub-process has been used that can be referred in their respective sections:

1. Handle work order



S. No	Activity	Assigned Team /System	Description
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1	Check Rewiring Type	Telflow	Identify the Rewiring Type ' <i>Within Mobily</i> ' or ' <i>Within OpenAccess</i> '. The order is already in ' <i>In Progress</i> ' state.
Mobily			
2	Work Order (WFM)	WFM	This sub-process invokes '<i>Handle work order</i>' sub-process when the Rewiring Type '<i>Within Mobily</i>'. Note: Upon receiving the synchronous response, follows step 3.
3	Create CPE Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type 'Rewiring' - Appointment Id - SingleView Order Id
4	WFM Completion Notification Received?	Telflow	Telflow awaits for WFM Completion notification. Possible outcomes: - Yes Follow step 6: Check WFM Response - No Follow step 5: Update CPE Action in SV
5	Update CPE Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 4. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details - Comments, if any.
6	Check WFM Response	Telflow	Below outcomes are possible: - Success Follow step 7: Close CPE Action in SV - Failed
7	Close CPE Action in SV	SingleView	Telflow closes the CPE action in SV.
8	Set Order Status	Telflow	Order status is set to complete.
Open Access			
9	Work Order (OA)	Open Access	This sub-process invokes '<i>Handle work order</i>' sub-process when the Rewiring Type '<i>Within OpenAccess</i>'. Note: Upon receiving the synchronous response, follows step 10.
10	Create ONT Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type 'Rewiring' - Appointment Id (If required) - SingleView Order Id
11	OA Completion Notification Received?	Telflow	Telflow awaits for OA Completion notification. Possible outcomes: - Yes Follow step 13: Check Open Access Response - No Follow step 12: Update ONT Action in SV
12	Update ONT Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 12. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id (If required) - SingleView Order Id - Task status - Engineer details (Id, Name, etc) - Comments, if any.

13	Check Open Access Response	Telflow	Below outcomes are possible: 1. Success Follow step 14: Close ONT Action in SV 2. Failed
14	Close ONT Action in SV	SingleView	Telflow closes the ONT action in SV.
15	Set Order Status	Telflow	Order status is set to complete.

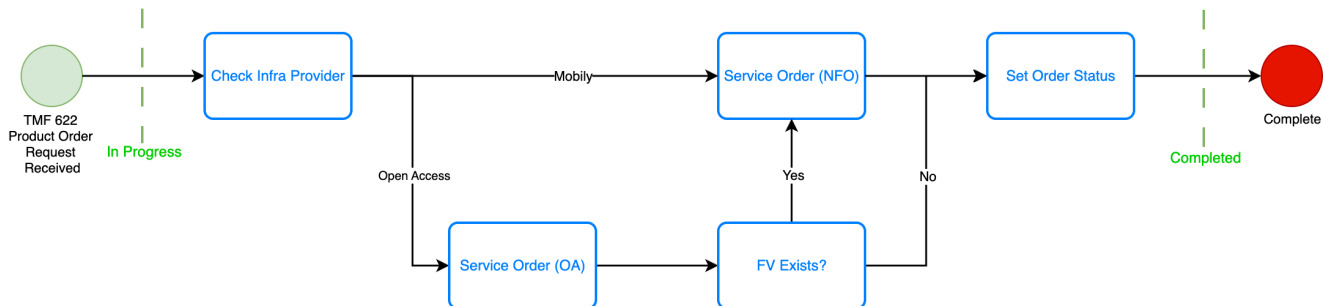
Mobily FTTH Suspend/Resume

Note:

- In case of Mobily Infra, SingleView shall send the suspend/resume request to Telflow if the customer has Fixed Voice. Telflow shall sent the barring/unbarring the Fixed Voice service request to Nokia FlowOne.
- In case of Open Access, irrespective of whether the customer has FTTH or FTTH/FV service, SingleView shall send the suspend/resume request to Telflow.
 - If the customer has FTTH + FV, then FTTH request shall be sent to Open Access and FV request to Nokia FlowOne.
 - If the customer has only FTTH, then FTTH request shall be sent to Open Access and then Telflow shall closes the order.
- Open Access refers to STC / ITC / DAWIYAT / ACES

The below block diagram covers the following use case for FTTH Suspension in Mobily/Open Access Infrastructure. ~~SingleView shall send the TMF 622 Product Order to Telflow, only if the customer has Fixed Voice service to either suspend or resume.~~

- Customer voluntary suspends FTTH services in POS, CC and Digital
- Customer resumes FTTH service via CC/POS/Digital
- Customer fails to pay FTTH invoice and gets treated in Collections Treatment Flow (FV Suspend/Resume - Phase 4B - Not Applicable currently)
- Impact of dunning life cycle in case customer has Fixed Voice (FV Suspend/Resume - Phase 4B - Not Applicable currently)
- Customer agrees Promise-to-Pay to maintain FTTH service while in Treatment (FV Suspend/Resume)
- Customer Prepaid FTTH service recharge with auto renewal using Best Fit Method
- Customer with Prepaid FTTH service receives Grace Period



S.No	Activity	Assigned Team/System	Description
1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily' or 'Open Access'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access system.
2	Service Order OA	Open Access	Telflow performs order decomposition and sends service order to Open Access provider.
3	FV Exists?	Telflow	Checks whether Fixed Voice service exists. Possible outcomes: 1. Yes Follow step 4: Service Order NFO 2. No Follow step 5: Set Order Status
4	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
5	Set Order Status	Telflow	Order status is set to complete.

Cancel FTTH Order

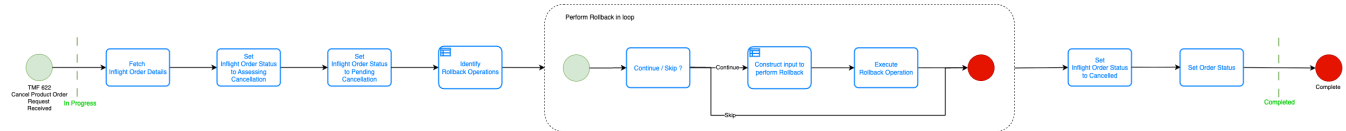
The below block diagram covers the following use cases,

- Customer requests to cancel FTTH order from POS and CC

Note:

- PoNR validation shall be performed by SingleView and not by Telflow. Once the cancel order reaches Telflow, it's assumed that the existing inflight order had not reached PoNR and can be cancelled.
- It is assumed that the existing inflight order shall be in the WFM installation stage or Open Access installation stage when the order cancellation had been initiated.
- TMF 622 Cancel Product Order shall be used by SingleView to initiate cancel order in Telflow.

- The SingleView order Id to be cancelled shall be sent in the TMF 622 Cancel Product Order request by SingleView.
- Open Access refers to STC / ITC / DAWIYAT / ACES



S. No	Activity	Assigned Team /System	Description
1	Fetch inflight Order Details	Telflow	Telflow fetches inflight order details to be cancelled. Order is set to ' <i>In Progress</i> ' state.
2	Set Inflight Order Status to Assessing Cancellation	Telflow	Set inflight order to ' <i>Assessing Cancellation</i> ' state.
3	Set Inflight Order Status to Pending Cancellation	Telflow	Set inflight order to ' <i>Pending Cancellation</i> ' state.
4	Identify Rollback Operations	Telflow	Telflow identifies the rollback operations to be performed.
5	Continue / Skip?	Telflow	Possible outcomes: 1. Continue continue the rollback activities 2. Skip skip the rollback activities
6	Construct Input to perform Rollback	Telflow	Based on the rollback operation identified in Step 4, construct the request to be sent to the target system to initiate rollback.
7	Execute Rollback Operation	Target Systems	Telflow executes the rollback operations in sequence.
8	Set Inflight Order Status to Cancelled	Telflow	Set inflight order to ' <i>Cancelled</i> ' state.
9	Set Order Status	Telflow	Order status is set to complete.

Rollback Operations

During cancel order processing, the activities that were initiated/completed in the existing inflight order shall be identified and rolled back by the cancel process.

Note:

- If both CPE and Mesh Extender exists in an order, cancelling CPE work order in WFM implicitly closes the Mesh Extender work order. Telflow doesn't initiate separate cancel work order for Mesh Extender.

S.No	Stage	Infra Provider	System	Operation	Rollback Operation
1	CPE Installation	Mobily	WFM	Task Creation Request	Task Cancellation Request
			SingleView	Create CPE Action	Close CPE Action Note: Close the action with comments. No need to update.
				Create ME Action	Close ME Action Note: Close the action with comments. No need to update.
2	ONT Installation	Open Access	Open Access	Service Installation Request	Cancel Service Installation Request
			SingleView	Create ONT Action	Close ONT Action Note: Close the action with comments. No need to update.
				Create ME Action (If applicable)	Close ME Action (If applicable) Note: Close the action with comments. No need to update.

Update FTTH Order

During order execution, the customer can request Mobily to update the below information in the existing inflight order.

- Appointment details
- Secondary contact details (to be addressed in future - Phase 4B - not considered currently)

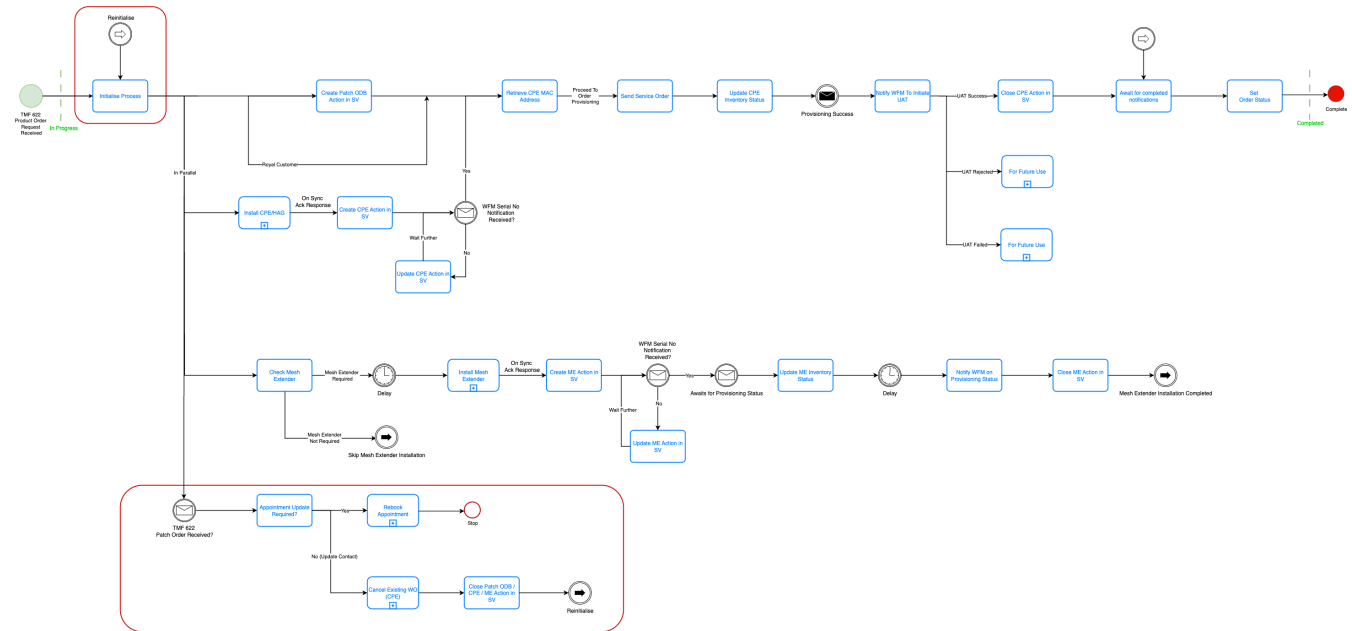
3. Service information (to be addressed in future - Phase 4B - not considered currently)

The order update has to be handled for the below use cases,

1. Mobily FTTH Service Activation
2. Open Access FTTH Service Activation
3. FTTH Relocation
4. FTTH Rewiring
5. FTTH Device Swap
6. Mobily adds ACES as Open Access service provider
7. Customer receives call for appointment confirmation

The below block diagram (Mobily FTTH Service Activation) highlights the activity to be considered to handle order update.

1. Customer request update on open order from POS and CC



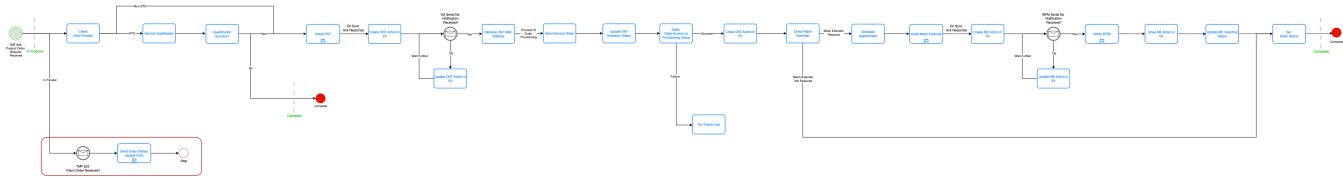
S. No	Activity	Assigned Team /System	Description
1	TMF 622 Patch Order Received?	Telflow	Telflow awaits for TMF 622 Patch Order request from SingleView.
2	Appointment Update Required?	Telflow	Telflow validates whether the order update request is for appointment update (validate by flag sent by SingleView). Possible outcomes: 1. Yes Follow step 3: Rebook Appointment 2. No (Update Contact) Follow step 4: Cancel Existing WO
3	Rebook Appointment	WFM	Based on the old appointment Id and slot Id received from the SingleView, Telflow sends the request to WFM to reschedule the appointment for the existing work order. Post receiving the appointment Id from WFM, Telflow stalls the token. Telflow shall notify SingleView, the new appointment Id details. The new appointment Id shall be notified to SV via further TMF 622 notifications and the CPE installation action update from Telflow. How to send the appointment Id to SV, via 622 notifications or case action? Check with Conor. Note: 1. The activity 'WFM Serial No Notification Received?' continues to wait for WFM notifications.
4	Cancel Existing WO (CPE)	WFM	If the order update is for contact, then Telflow initiates request to WFM to cancel the existing work order. Note: 1. Cancelling the CPE work order implicitly cancels the Mesh Extender work order in WFM.
5	Close Patch ODB / CPE / ME Action in SV	SingleView	Telflow closes the actions that was created in SingleView and then reinitialise the process from the beginning.

6	Initialise Process	Telflow	Initialise the process from the beginning with the latest contact data to be updated in the order. Note: 1. The activities 'Install CPE/HAG' and 'Install Mesh Extender' shall create a new work order in WFM. 2. The activity 'Create Patch ODB Action in SV' creates a new action in SingleView.
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Note:

1. For Open Access, SingleView sends the appointment Id (along with the existing order details) and Telflow shall perform order details update in Open Access. The existing flow is not disturbed in Telflow and it still listens for the service installation request notifications from Open Access.
2. Open Access shall send synchronous 'success' response ~~'completed' notification~~ for 'order details update' API to Telflow. Telflow shall ignore this status and the original order status or sub-status remain intact.
3. The open order update is not applicable for STC operator since the operation is not supported currently. Hence, SingleView should restrict the open order update for STC from its end.

The below block diagram (Open Access FTTH Service Activation) highlights the activity to be considered to handle order update.



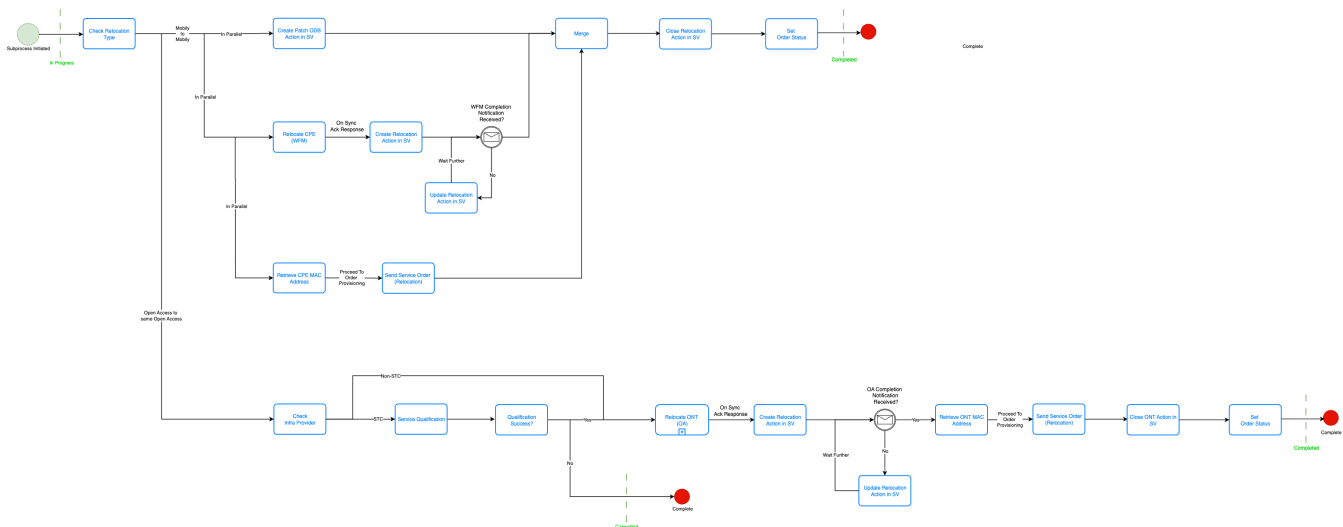
Mobily FTTH Relocation

The below block diagram covers the following use cases,

1. Customer requests FTTH Relocation from POS and CC
2. Fixed Voice Disconnection due to FTTH relocation (Relocation within the same fixed voice geographical zone)

Note:

1. Open Access refers to STC / ITC / DAWIYAT / ACES



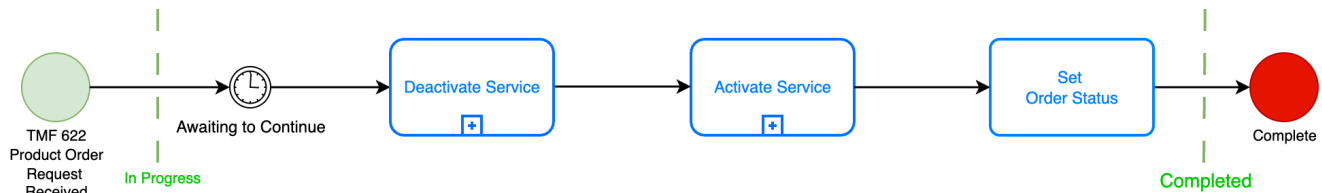
S. No	Activity	Assigned Team /System	Description
1	Check Relocation Type	Telflow	Identify the relocation type 'Within Mobily' or 'Within OpenAccess'. The order is already in 'In Progress' state.
Within Mobily			
2	Create Patch ODB Action in SV	SingleView	Telflow initiates ODB patching action in SingleView and awaits for completion notification. Email notification shall be triggered to patching team. A patching task shall be initiated irrespective of whether the ODB is old or new to attach the customer device with the ODB.

3	Relocate CPE (WFM)	WFM	This sub-process invokes 'Handle work order' sub-process. Note: Upon receiving the synchronous response, Relocate CPE follows step 5.
4	Retrieve CPE MAC Address	SingleView	Based on CPE Serial Number, Telflow retrieves CPE MAC Address from SingleView via TMF639 and proceeds the order further to step 8. Note: Patch ODB, Relocate CPE and Retrieve CPE MAC Address are parallel activities.
5	Create Relocation Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
6	WFM Completion Notification Received?	Telflow	Telflow awaits for WFM Completion notification. Possible outcomes: - Yes Follow step 9: Close Relocation Action in SV - No Follow step 7: Update Relocation Action in SV
7	Update Relocation Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 6. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details - Comments, if any.
8	Send Service Order (Relocation)	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
9	Close Relocation Action in SV	SingleView	Telflow closes the CPE action in SV.
10	Set Order Status	Telflow	Order status is set to complete.
Within Open Access			
11	Check Infra Provider	Telflow	Identify the Open Access Infra Provider. The order is already in 'In Progress' state.
12	Service Qualification	Open Access	If the Infra Provider is STC, then Telflow shall initiate service qualification towards Open Access whereas for Non-STC operators, step 14 (Relocate ONT (OA)) shall follow.
13	Qualification Success?	Telflow	Checks whether the service is qualified. Possible outcomes: - Yes Follow step 14: Relocate ONT - No Completes the order and set the status to 'Cancelled'
14	Relocate ONT (OA)	Open Access	This sub-process invokes 'Handle work order' sub-process.
15	Create Relocation Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
16	OA Completion Notification Received?	Telflow	Telflow awaits for OA completion notification. Possible outcomes: - Yes Follow step 18: Retrieve ONT MAC Address - No Follow step 17: Update Relocation Action in SV

17	Update Relocation Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 16. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id 3. SingleView Order Id 4. Task status 5. Comments, if any.
18	Retrieve ONT MAC Address	SingleView	Based on ONT Serial Number, Telflow retrieves ONT MAC Address from SingleView via TMF639 and proceeds the order further.
19	Send Service Order (Relocation)	Open Access	Telflow performs order decomposition and sends service order to Open Access provider.
20	Close ONT Action in SV	SingleView	Telflow closes the ONT action in SV.
21	Set Order Status	Telflow	Order status is set to complete.

<to be reused in phase 4B>

1. Relocation Deactivate
2. Relocation Activate



S. No	Activity	Assigned Team /System	Description
1	Awaiting to Continue	Telflow	Awaits until X hrs before the appointment time to initiate service deactivation in the customer's existing location.
2	Deactivate Service	Nokia FlowOne	This sub-process invokes 'Relocation Deactivate' sub-process.
3	Activate Service	WFM / Nokia FlowOne	This sub-process invokes 'Relocation Activate' sub-process.
4	Set Order Status	Telflow	Order status is set to complete.

Mobily FTTH Device Swap

On-Demand Device Swap

The below block diagram covers the following use case for FTTH Device Swap in Mobily/Open Access Infrastructure.

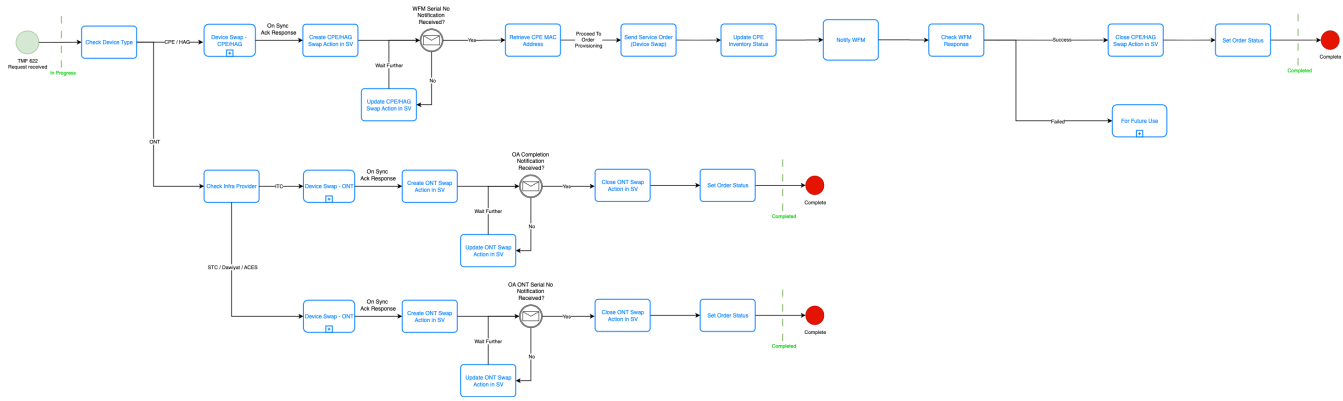
1. Customer requests FTTH Device Swap from POS
2. FTTH Device Swap from WFM

Note:

1. UAT is not applicable for Device Swap.
2. The on-demand device swap shall also be triggered as an outcome of maintenance work order Customer raises issue in POS/Digital use case.
3. Open Access refers to STC / ITC / DAWIYAT / ACES

for the above use cases, below sub-process has been used that can be referred in their respective sections:

1. Handle work order



S. No	Activity	Assigned Team /System	Description
1	Check Device Type	Telflow	Identify the Device Type 'CPE / HAG' or 'ONT'. The order is already in 'In Progress' state.
CPE / HAG			
2	Device Swap - CPE/HAG	WFM	This sub-process invokes 'Handle work order' sub-process. Note: Upon receiving the synchronous response, follows step 3.
3	Create CPE/HAG Swap Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
4	WFM Serial No Notification Received?	Telflow	Telflow awaits for WFM serial number notification. Possible outcomes: - Yes Follow step 6: Retrieve CPE MAC Address - No Follow step 5: Update CPE/HAG Swap Action in SV
5	Update CPE/HAG Swap Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 4. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details - Comments, if any.
6	Retrieve CPE MAC Address	SingleView	Based on CPE Serial Number, Telflow retrieves CPE MAC Address from SingleView via TMF639 and proceeds the order further.
7	Send Service Order (Device Swap)	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
8	Update CPE Inventory Status	MapleWave	Telflow updates the inventory status of the CPE allocated to the customer in MapleWave.
9	Notify WFM	WFM	Telflow notifies WFM that the provisioning is success.
10	Check WFM Response	Telflow	Below outcomes are possible: - Success Follow step 11: Close CPE/HAG Swap Action in SV - Failed
11	Close CPE/HAG Swap Action in SV	SingleView	Telflow closes the CPE/HAG Swap action in SV.
12	Set Order Status	Telflow	Order status is set to complete.
ONT			

13	Check Infra Provider	Telflow	Identify the Infra Provider 'ITC' or 'STC/Dawiyat/ACES'. The below steps are applicable for both the provider types.
14	Device Swap - ONT	Open Access	This sub-process invokes 'Handle work order' sub-process. Note: Upon receiving the synchronous response, follows step 13.
15	Create ONT Swap Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id (If required) 3. SingleView Order Id
16	OA Completion Notification Received? (or) OA ONT Serial No Notification Received?	Telflow	Telflow awaits for OA completion notification in case of ITC or serial number notification in case of STC/Dawiyat/ACES. Possible outcomes: 1. Yes Follow step 16: Close ONT Swap Action in SV 2. No Follow step 15: Update ONT Swap Action in SV
17	Update ONT Swap Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 14. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id (If required) 3. SingleView Order Id 4. Task status 5. Engineer details (Id, Name, etc) 6. Comments, if any.
18	Close ONT Swap Action in SV	SingleView	Telflow closes the ONT Swap action in SV.
19	Set Order Status	Telflow	Order status is set to complete.

WFM - Device Swap

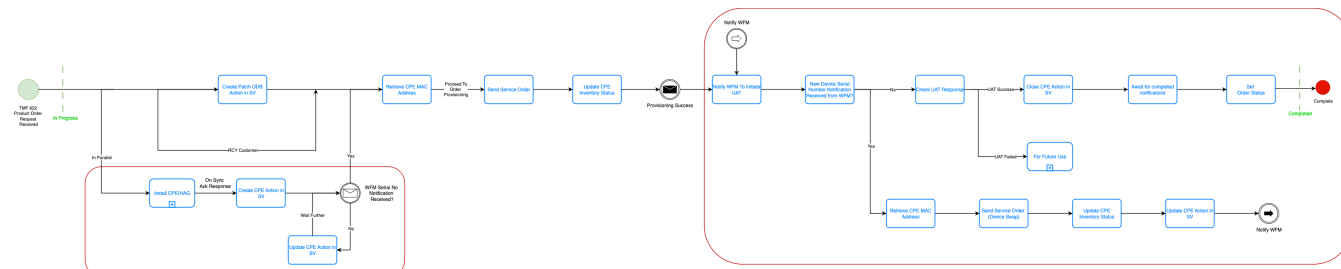
Scenario - 1 - FTTH Device Swap

This design had been covered in the previous section 'On-Demand Device Swap'.

Scenario - 2 - FTTH Device Swap during service installation

During service installation, there are two scenarios to be considered for device swap,

- When Telflow awaits for serial number notification from WFM. (positive scenario during service installation)
 - In this situation, when Telflow awaits for serial number notification, WFM in the backend performs stock check or OSS check and it returns the serial number to Telflow only when the serial number is valid. Unless the device serial number is valid, WFM doesn't send the serial number notification to Telflow. Hence, there is no impact to Telflow.
- When Telflow awaits for completion notification from WFM.
 - In this situation, if the WFM technician finds that the device is faulty, he/she will scan the new device serial number followed by stock check or OSS check. If the serial number is valid, this results in WFM re-sending the serial number notification to Telflow whereas Telflow awaits for completion notification. Upon receiving this serial number notification again, Telflow must perform device swap in the NFO and update the Maplewave inventory with the new serial number. Post this, Telflow sends 'provisioning complete' message to WFM and starts awaiting for completion notification.
 - WFM sends both the old and new serial number in the notification and Telflow shall process the notification accordingly. Note, there are two different notifications for device serial number from WFM and Telflow should recognise that this notification is for device swap and handle it in the process. (Refer to the sample sent by Abdalla in the mail)



Scenario - 3 - FTTH Device Swap - Complaint Management

During complaint management, if the resolution demands device replacement, then the 'On-Demand' device swap process shall be invoked to resolve the complaint. This design had been covered in the use case 'Customer raises issue'.

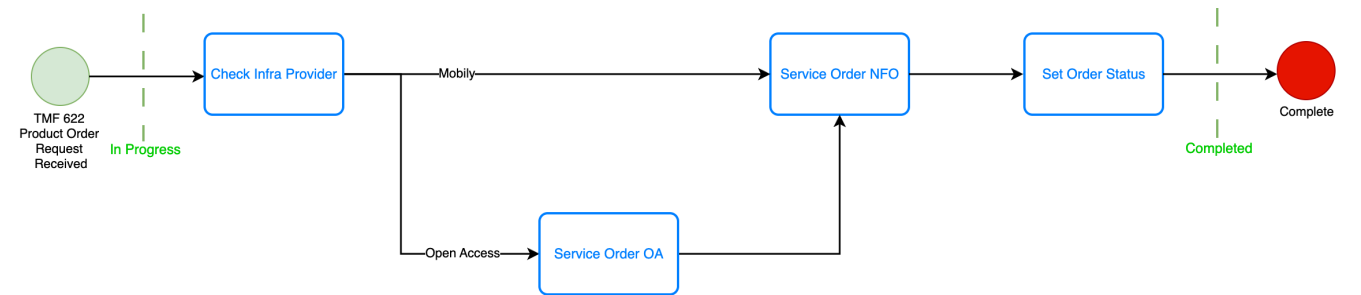
Mobily FTTH Bandwidth Upgrade/Downgrade

The below block diagram covers the following use cases,

- 1. Customer requests FTTH bandwidth upgrade/downgrade from POS, Digital and CC

Note:

- 1. Open Access refers to STC / ITC / DAWIYAT / ACES



S.No	Activity	Assigned Team/System	Description
1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily' or 'Open Access'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access system.
2	Service Order OA	Open Access	Telflow performs order decomposition and sends service order to Open Access provider.
3	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
4	Set Order Status	Telflow	Order status is set to complete.

Customer Raises Issue

Note:

- 1. This use case shall be handled as a product order (technical) in Telflow and as SR (Service Request) in SingleView and Open Access systems and as task in WFM system.
- 2. In Mobily Infra, as an outcome of customer visit, if WFM responds back to Telflow that the resolution needs device swap to be performed, then Telflow must close the order successfully. Based on the resolution that's been sent by Telflow to SingleView, it initiates device swap order to Telflow. This is covered as part of 'Customer requests device swap from POS' use case.
 - a. resolution code 2064
 - b. resolution text Wrong Info - Wrong Package
- 3. In Mobily Infra, re-opening an issue in SingleView shall create a new maintenance order in Telflow. No need to create maintenance order in Telflow while closing the issue.
- 4. Handling of negative scenarios is not applicable for Open Access.
 - a. If the SR been resolved/rejected by Open Access, Telflow should notify SingleView and place the order in 'Pending' state (i.e., the order is still open in Telflow). At this stage, SingleView can perform below SR operations based on the interaction with the customer,
 - i. Re-Open SR
 - ii. Close SR
 - b. To Re-Open / Close SR, SingleView shall patch the existing product order in Telflow and, Telflow continue processing the order.
- 5. Telflow updates the existing action that was created in SingleView. The action Id shall be passed by SingleView to Telflow.
- 6. ONT Inquiry API invocation towards Open Access is not applicable for Telflow. This API shall be triggered directly from SingleView to Open Access.
- 7. Rebook Appointment is applicable during maintenance work order (for Mobily Infra).
- 8. Open Access refers to STC / ITC / DAWIYAT / ACES

Additional Note: (to be removed later)

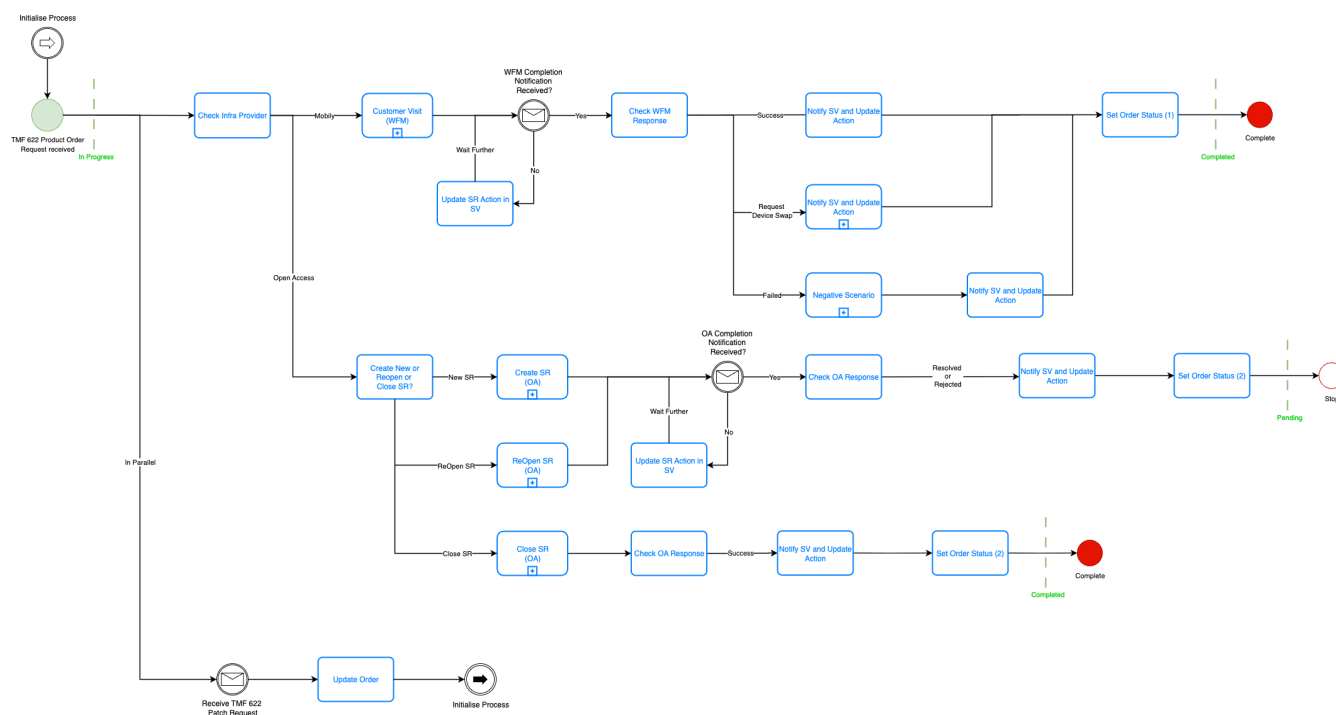
- 1. Data Mapping required for the Open Access systems shall be performed in Telflow as per the supporting mail sent by Venkat (Subject Create SR Mapping for All Three Providers).

The below block diagram covers the following use case for Mobily/Open Access Infrastructure.

- 1. Customer raises issue in POS, Digital and CC

for the above use cases, below sub-process has been used that can be referred in their respective sections:

- 1. Handle work order



S. No	Activity	Assigned Team /System	Description
1	Check Infra Provider	Telflow	Identify the Infra Provider 'Mobily' or 'OpenAccess'. The order is already in 'In Progress' state.
Mobily			
2	Customer Visit (WFM)	WFM	This sub-process invokes 'Handle work order' sub-process. Note: Upon receiving the synchronous response, follows step 3.
3	Create CPE Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
4	WFM Completion Notification Received?	Telflow	Telflow awaits for WFM completion notification. Possible outcomes: - Yes Follow step 6: Check WFM Response - No Follow step 5: Update CPE Action in SV
5	Update CPE Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 4. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details - Comments, if any.
6	Check WFM Response	Telflow	Possible Outcomes: - Success Follow step 7: Close CPE Action in SV - Failed

7	Close CPE Action in SV	SingleView	Telflow closes the CPE action in SV.
8	Set Order Status (1)	Telflow	Order status is set to complete.
Open Access			
9	Create New SR or ReOpen SR?	Telflow	Telflow validates whether new SR has to be created or reopen the existing SR. Possible outcomes: 1. New SR Follow step 10: Create SR (OA) 2. ReOpen SR Follow step 11: ReOpen SR (OA)
10	Create SR (OA)	Open Access	This sub-process invokes 'Handle work order' sub-process. Note: • Upon receiving the synchronous response, follows step 12.
11	ReOpen SR (OA)	Open Access	This sub-process invokes 'Handle work order' sub-process. Note: • Upon receiving the synchronous response, follows step 12.
12	Create ONT Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id (If required) 3. SingleView Order Id
13	OA Completion Notification Received?	Telflow	Telflow awaits for OA completion notification. Possible outcomes: 1. Yes Follow step 15: Check OA Response 2. No Follow step 14: Update ONT Action in SV
14	Update ONT Action in SV	SingleView	Upon receiving the OA notification, Telflow updates the action in SV and follows step 13. Below are the information that has to be cascaded to SV from Telflow, 1. Type 2. Appointment Id (If required) 3. SingleView Order Id 4. Task status 5. Engineer details (Id, Name, etc) 6. Comments, if any.
15	Check OA Response	Telflow	Possible Outcomes: 1. Success Follow step 16: Close ONT Action in SV 2. Failed
16	Close ONT Action in SV	SingleView	Telflow closes the ONT action in SV.
17	Set Order Status (2)	Telflow	Order status is set to complete.

FTTH Complaints - Negative Scenarios

Note:

1. Negative scenario is not applicable for Open Access. (to be decided for ACES)
2. In case of Mobily Infra, negative scenario is not applicable when WFM sends the resolution code stating, the device has to be swapped.
3. During treatment handling, the comments entered by the technician/agent shall be cascaded between systems by Telflow.
 - a. In case of Mobily Infra,
 - i. When WFM sends the error response to Telflow, the comments entered by the technician in WFM work order shall be cascaded to SingleView via actions.
 - ii. When the agent completes the action, any comments captured by the agent on the action shall be cascaded from SingleView to Telflow and the same shall be cascaded in the subsequent actions initiated by Telflow towards SingleView as part of treatment handling.
 - iii. Post treatment, when Telflow creates a new work order, the comments shall be passed in the WFM request.
4. Telflow doesn't persist the history of the comments, rather it will be overwritten.

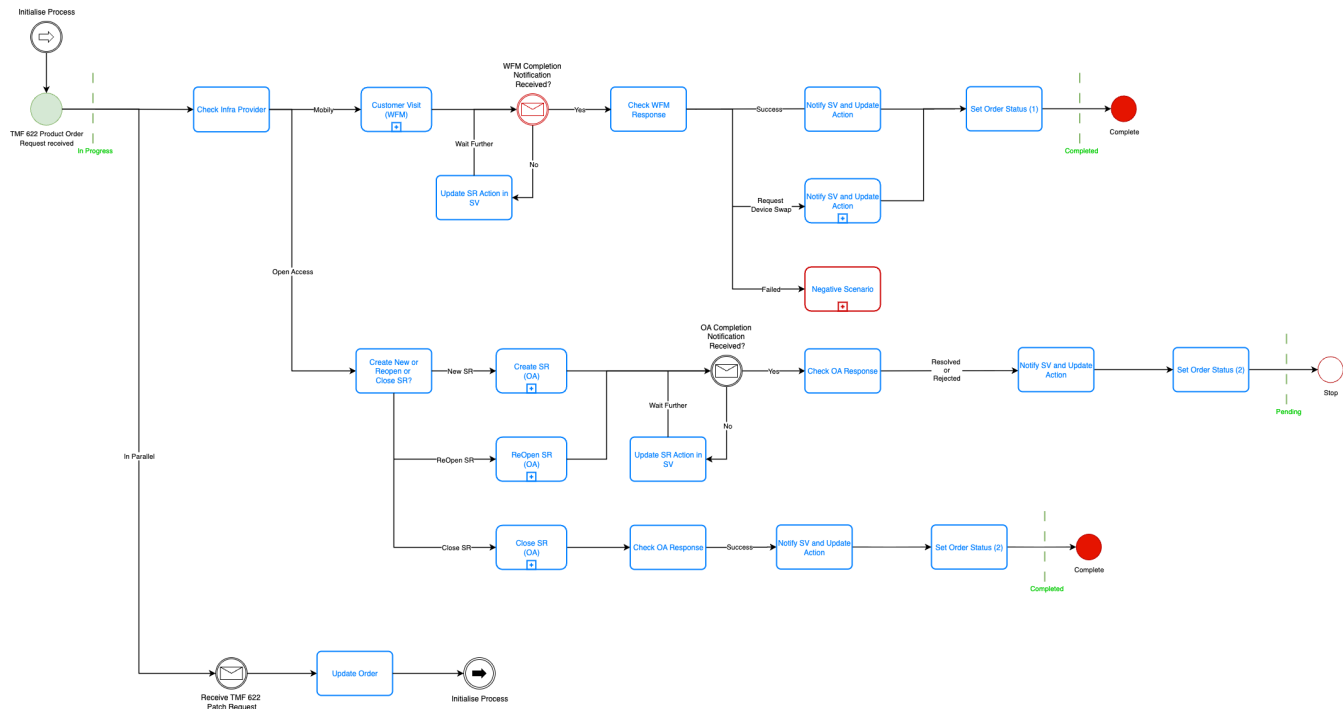
Negative scenarios are applicable when WFM is unable to successfully complete the maintenance customer visit task. During such circumstances, WFM shall send the error response to Telflow and based on the error response, Telflow shall initiate and handle it via treatment. As part of the treatment outcome, either the order has to be resumed or cancelled.

The negative scenarios are to be handled for the below use cases,

- 1. Customer raises issue

The below block diagram (Customer raises issue) highlights the WFM / Open Access activity to be considered to handle negative scenarios. Below are the sub-processes that shall be invoked,

- 1. Handle Maintenance Work Order
- 2. Handle Negative Scenarios



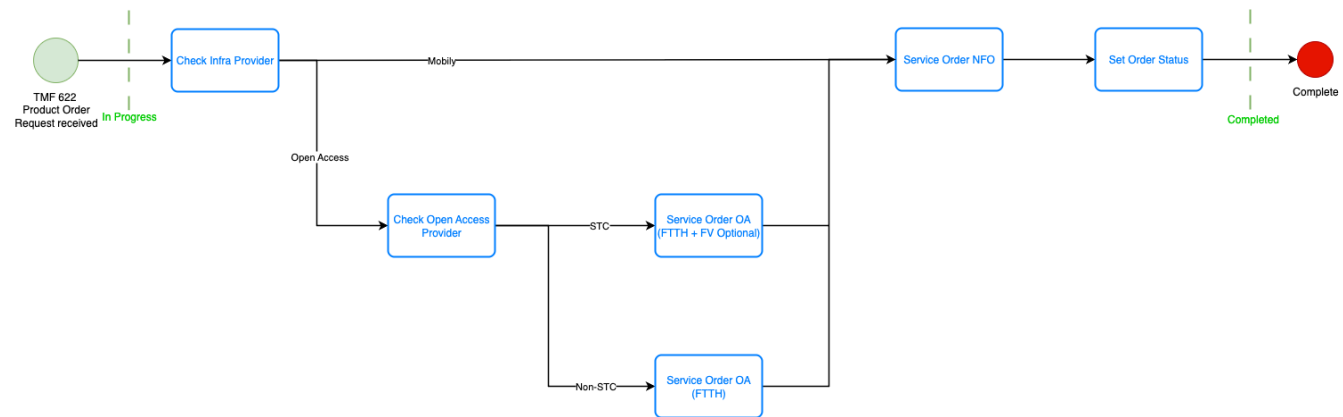
Mobily FTTH Service Termination

The below block diagram covers the following use cases,

- 1. Customer terminates FTTH service in POS, Digital and CC
- 2. Customer request FTTH service and Fixed Voice disconnection (Phase 4B - Not considered currently)
- 3. Customer fails to pay FTTH invoice and gets treated in Collections Treatment Flow (FTTH Deactivation)

Note:

- 1. Open Access refers to STC / ITC / DAWIYAT / ACES



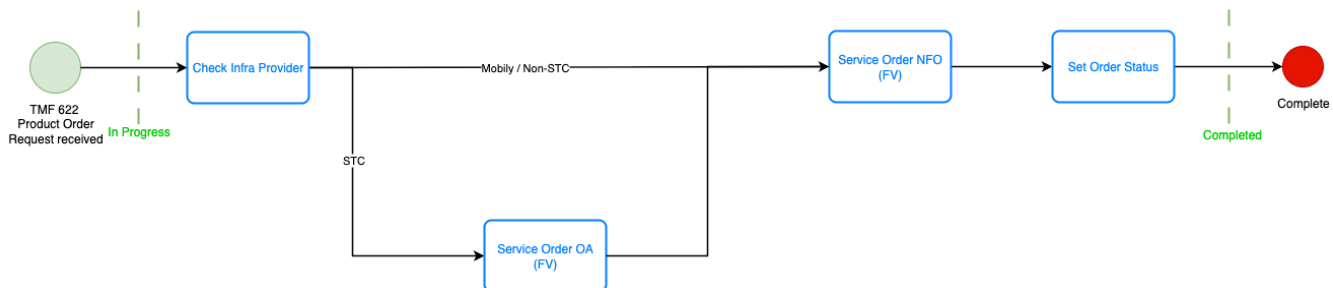
S. No	Activity	Assigned Team /System	Description

1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily' or 'Open Access'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access system.
2	Check Open Access Provider	Telflow	Identify the Open Access Provider ('STC' or 'Non-STC'). Possible outcomes: 1. STC Follow step 3: Service Order OA (FTTH + FV Optional) 2. Non-STC Follow step 4: Service Order OA (FTTH)
3	Service Order OA (FTTH + FV Optional)	Open Access	Telflow performs order decomposition and sends service order to Open Access provider. Note: - If the customer has Fixed Voice, Telflow shall send both FTTH and Fixed Voice details to the Open Access provider. - Abdalla confirmed that if the customer has Fixed Voice, then for STC provider, the request for FTTH and FV shall be sent in the separate request. This shall be covered in Phase 4B (to be decided - not considered currently).
4	Service Order OA (FTTH)	Open Access	Telflow performs order decomposition and sends service order to Open Access provider. Note: If the customer has Fixed Voice, Telflow shall send only FTTH details to the Open Access provider.
5	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
6	Set Order Status	Telflow	Order status is set to complete.

Mobily Fixed Voice Service Termination

The below block diagram covers the following use cases,

- Customer requests Fixed Voice Disconnection in POS
- Fixed voice disconnection due to FTTH Postpaid to Prepaid package change
- Fixed Voice Disconnection due to FTTH relocation (Relocation outside the fixed voice geographical zone)



S. No	Activity	Assigned Team /System	Description
1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily / Non-STC' or 'STC'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access system.
2	Service Order OA (FV)	Open Access	Telflow performs order decomposition and sends service order to Open Access provider. Note: If the Open Access provider is STC, then Telflow shall send FV disconnection request to Open Access system.
3	Service Order NFO (FV)	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
4	Set Order Status	Telflow	Order status is set to complete.

Mobily FTTH Package Conversion

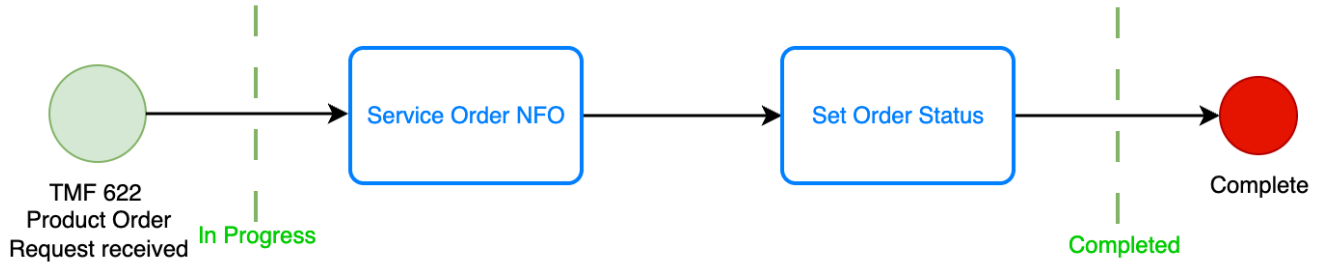
Postpaid to Prepaid

The below block diagram covers the following use cases,

- Customer changes from Postpaid to Prepaid FTTH package in POS

Note:

1. Open Access is not applicable.
2. Prepaid payment type doesn't have support for fixed voice. If the customer has fixed voice, then the service must be deactivated in the network via Nokia FlowOne. Both package conversion and fixed voice deactivation must be performed via a single request in Nokia FlowOne.



S. No	Activity	Assigned Team /System	Description
1	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface. Note: 1. This step performs FTTH package conversion (Postpaid to Prepaid) and optionally FV deactivation in the same request.
2	Set Order Status	Telflow	Order status is set to complete.

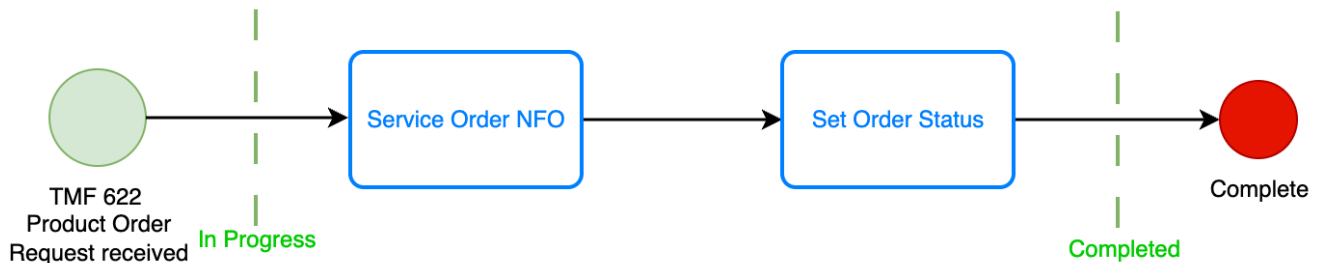
Postpaid to Prepaid

The below block diagram covers the following use cases,

1. Customer changes from Prepaid to Postpaid FTTH package in POS

Note:

1. Open Access is not applicable.
2. If the customer opts for fixed voice, then the FV service must be activated in the network via separate request/order post package conversion completes successfully.

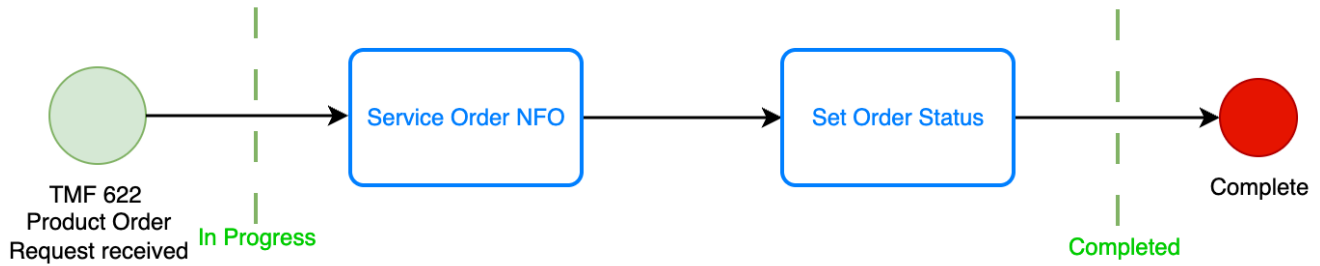


S.No	Activity	Assigned Team/System	Description
1	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
2	Set Order Status	Telflow	Order status is set to complete.

Fixed Voice Number Change

The below block diagram covers the following use cases,

1. Customer changes Fixed Voice number in POS

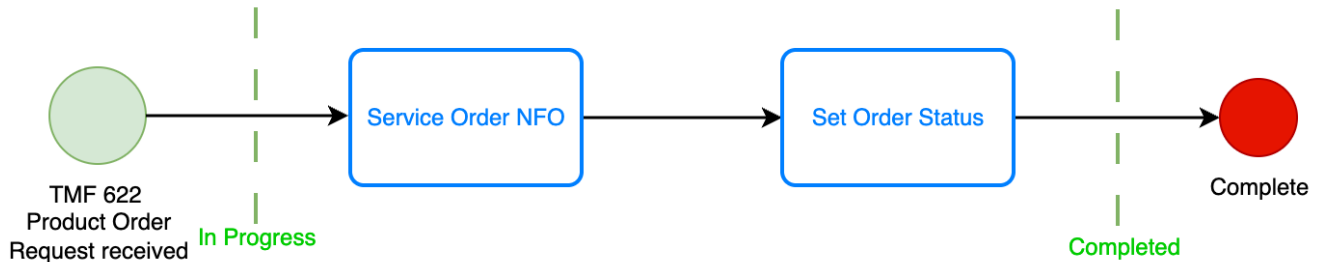


S.No	Activity	Assigned Team/System	Description
1	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
2	Set Order Status	Telflow	Order status is set to complete.

FTTH Account Ownership Change

The below block diagram covers the following use cases,

1. Customer changes FTTH Account Ownership in POS



S.No	Activity	Assigned Team/System	Description
1	Service Order NFO	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
2	Set Order Status	Telflow	Order status is set to complete.

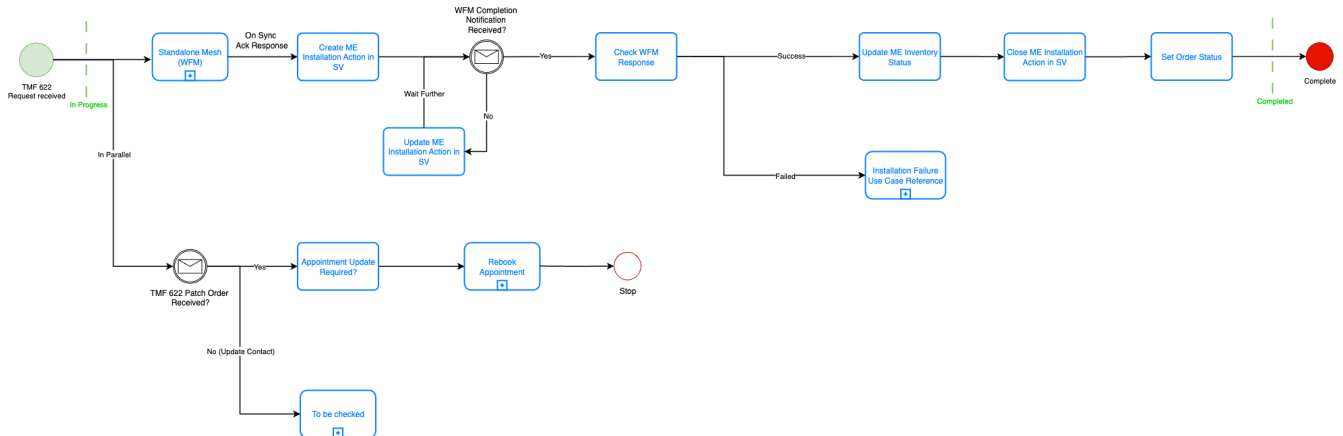
Customer buys Mesh Extender (Standalone)

Note:

The WFM and SingleView APIs that were developed in 4A shall be reused for this use case, whereas the BPMN process is newly developed.

The below block diagram covers the following use cases,

1. Customer buys Mesh Extender in POS



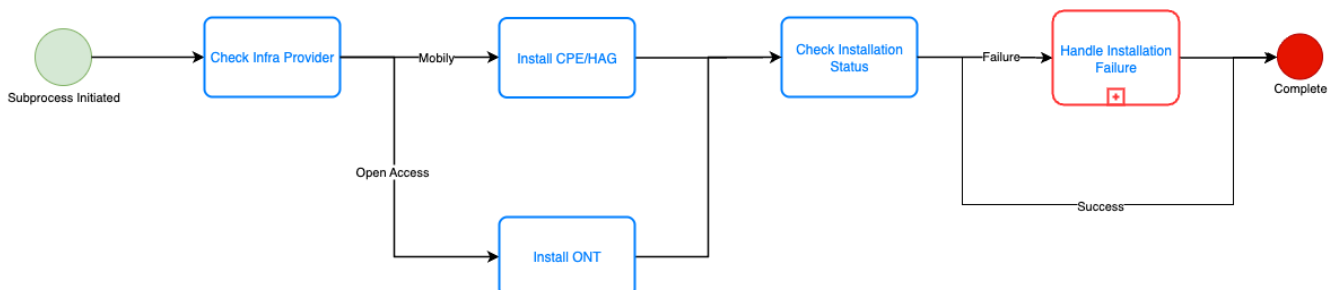
S. No	Activity	Assigned Team /System	Description
1	Standalone Mesh (WFM)	WFM	This sub-process invokes 'Handle work order' sub-process. Note: Upon receiving the synchronous response, follows step 2.
2	Create ME Installation Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id
3	WFM Completion Notification Received?	Telflow	Telflow awaits for WFM completion notification. Possible outcomes: - Yes Follow step 5: Check WFM Response - No Follow step 4: Update ME Installation Action in SV
4	Update ME Installation Action in SV	SingleView	Upon receiving the WFM notification, Telflow updates the action in SV and follows step 3. Below are the information that has to be cascaded to SV from Telflow, - Type - Appointment Id - SingleView Order Id - Task status - Engineer details - Comments, if any.
5	Check WFM Response	Telflow	Telflow validates the WFM response. Possible outcomes: - Success Follow step 6: Update ME Inventory Status - Failed Follow: Installation failure use case reference Note: Installation failure scenario is applicable for the Mesh Extender except the below Mesh Extender specific reason codes, 2101 - Mesh Device is not available 2089 - SANAD is rejected by customer 2102 - Customer refused to sign PN
6	Update ME Inventory Status	MapleWave	Telflow updates the inventory status of the ME allocated to the customer in MapleWave.
7	Close ME Installation Action in SV	SingleView	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
8	Set Order Status	Telflow	Order status is set to complete.

Sub-processes

Handling Work Order

The below block diagram covers the following subprocess,

1. Handle work Order

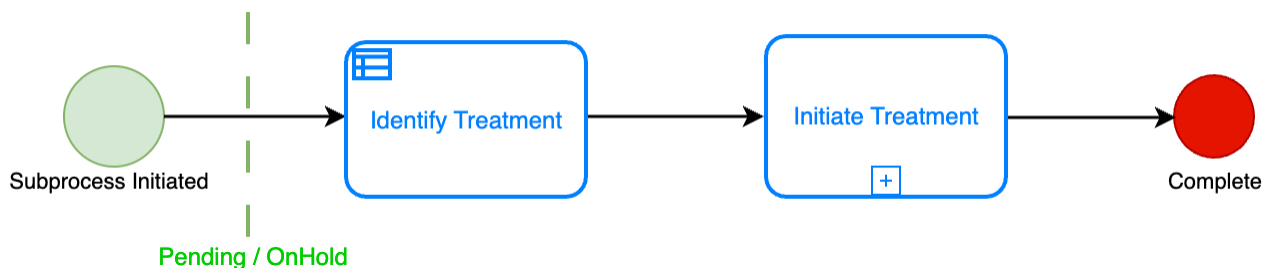


S.No	Item	Assigned Team	Detail
1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily' or 'Open Access'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access WFM.
2	Install CPE/HAG/ME	WFM	1. Work order is created in WFM for the technician to perform CPE/HAG installation. 2. Receives CPE serial number sent by WFM in the CPE/HAG installation response.
3	Install ONT	Open Access	1. Work order is created in Open Access for the technician to perform ONT installation. 2. Receives ONT serial number sent by Open Access in the CPE/HAG installation response.
4	Check Installation Status	Telflow	Possible outcomes: 1. Success Completes the process. 2. Failure Follow step 5: Handle Installation Failure
5	Handle Installation Failure	Telflow	Invokes Handle Installation Failure sub-process.

Handling Installation Failure

The below block diagram covers the following sub-process

1. Handle Installation Failure



S.No	Item	Assigned Team	Detail
1	Identify Treatment	Telflow	Based on the Infra Provider and the rejection reason sent by WFM, the treatment to be performed shall be identified.
2	Initiate Treatment	Telflow	Based on the treatment identified, the respective treatment sub-process shall be initiated.

Following DMN shall be configured to identify the treatment pertaining to the service provider based on the rejection reason,

Input				Output	
S. No	Reason Code	Reason Text	Infrastructure Provider	Treatment	Action Type
1	2010	NS-No Signal	Mobily	1	FTTH Installation Failure - Network Problem
2	2011	NS-ODB Upgrade required	Mobily	1	FTTH Installation Failure - Network Problem
3	2012	NS-Damaged ODB	Mobily	1	FTTH Installation Failure - Network Problem
4	2013	NS-High Loss	Mobily	1	FTTH Installation Failure - Network Problem
5	2014	NS-No ODB	Mobily	1	FTTH Installation Failure - Network Problem
6	2015	NS-Duplicate ODB ID	Mobily	1	FTTH Installation Failure - Network Problem
7	2016	NS-ODB relocation required	Mobily	1	FTTH Installation Failure - Network Problem
8	2017	NS-Wrong ODB ID	Mobily	1	FTTH Installation Failure - Network Problem
9	2018	NS-NO ODB on GIS	Mobily	1	FTTH Installation Failure - Network Problem

10	2067	NC-CPE not visible	Mobily	1	FTTH Installation Failure - Network Problem
11	2068	NC-CPE not taking IP	Mobily	1	FTTH Installation Failure - Network Problem
12	2069	NC-Speed Problem	Mobily	1	FTTH Installation Failure - Network Problem
13	2025	Duplicate WO	Mobily	1	FTTH Installation Failure - Network Problem
14	2020	SYS-CPE/STB Swap Failure	Mobily	2	FTTH Installation Failure - System Issue
15	2021	SYS-CPE not visible	Mobily	2	FTTH Installation Failure - System Issue
16	2022	SYS-CPE not taking IP	Mobily	2	FTTH Installation Failure - System Issue
17	2023	SYS-Speed Problem	Mobily	2	FTTH Installation Failure - System Issue
18	2024	SYS-OSS Discovered Problems	Mobily	2	FTTH Installation Failure - System Issue
19	2031	SYS-Provisioning Unsuccessful	Mobily	2	FTTH Installation Failure - System Issue
20	2060	Wrong Info - Wrong contact number	Mobily	3	FTTH Installation Failure - Wrong Information
21	2061	Wrong Info - Wrong ODB ID	Mobily	3	FTTH Installation Failure - Wrong Information
22	2062	Wrong Info - Wrong CPE/STB	Mobily	3	FTTH Installation Failure - Wrong Information
23	2063	Wrong Info - Wrong Location	Mobily	3	FTTH Installation Failure - Wrong Information
24	2064	Wrong Info - Wrong Package	Mobily	3	FTTH Installation Failure - Wrong Information
25	2035	Device Faulty - refused to pay	Mobily	4	FTTH Installation Failure - Mobily Customer Rejection
26	2040	Customer Not Reachable	Mobily	4	FTTH Installation Failure - Mobily Customer Rejection
27	2041	Customer Want to Cancel	Mobily	4	FTTH Installation Failure - Mobily Customer Rejection
28	2042	Reschedule later	Mobily	4	FTTH Installation Failure - Mobily Customer Rejection
29	B09-OLO	B09-OLO - Loss Signal	STC		FTTH Installation Failure - Provider Rejection
30	B10-OLO	B10-OLO - TB Damage	STC		FTTH Installation Failure - Provider Rejection
31	B11-OLO	B11-OLO - Fiber Cut B/W TB & FDT	STC		FTTH Installation Failure - Provider Rejection
32	B01-OLO	B01-OLO - No Port Spare	STC		FTTH Installation Failure - Provider Rejection
33	B14-OLO	B14-OLO - No Fiber	STC		FTTH Installation Failure - Provider Rejection
34	B13-OLO	B13-OLO - Need Owner Permission	STC		FTTH Installation Failure - Customer Rejection
35	B26-OLO	B26-OLO - Customer Reject The Service	STC		FTTH Installation Failure - Customer Rejection
36	B31-OLO	B31-OLO - Customer not Answering	STC		FTTH Installation Failure - Customer Rejection
37	B24-OLO	Internal wiring	STC		FTTH Installation Failure - Customer Rejection
38	B167-OLO	B167-OLO - No HAG Available	STC		FTTH Installation Failure - Logistic Rejection
39	B23-OLO	B23-OLO - Wrong contact Number	STC		FTTH Installation Failure - Wrong Customer Information

40	B15-OLO	wrong plate ID.	STC		FTTH Installation Failure - Wrong Service Information
41	B166-OLO	Commercial site/area	STC		FTTH Installation Failure - Wrong Service Information
42	No Details	No Details	STC		FTTH Installation Failure - Unknown Rejection
43		Wrong ODB	ITC		FTTH Installation Failure - Wrong Service Information
44		Commercial area	ITC		FTTH Installation Failure - Wrong Service Information
45		Customer requested for reappointment	ITC		FTTH Installation Failure - Customer Rejection
46		Internal wiring	ITC		FTTH Installation Failure - Customer Rejection
47		Customer requested to cancel	ITC		FTTH Installation Failure - Customer Rejection
48		Customer is not reachable	ITC		FTTH Installation Failure - Customer Rejection
49		No HAG available	ITC		FTTH Installation Failure - Logistic Rejection
50		Wrong customer contact	ITC		FTTH Installation Failure - Wrong Customer Information
51		No Fiber	ITC		FTTH Installation Failure - Provider Rejection
52		No Signal	ITC		FTTH Installation Failure - Provider Rejection
53		ODB Damage	ITC		FTTH Installation Failure - Provider Rejection
54		Fiber Cut	ITC		FTTH Installation Failure - Provider Rejection
55		Cancelled as Order can't be fulfilled	ITC		FTTH Installation Failure - Provider Rejection
56		Customer cancelation	Dawiyat		FTTH Installation Failure - Customer Rejection
57		Internal wiring	Dawiyat		FTTH Installation Failure - Customer Rejection
58		Customer appointment reschedule	Dawiyat		FTTH Installation Failure - Customer Rejection
59		Customer is not reachable	Dawiyat		FTTH Installation Failure - Customer Rejection
60		Wiring issue	Dawiyat		FTTH Installation Failure - Customer Rejection
61		Customer request Reschedule > 5 Days	Dawiyat		FTTH Installation Failure - Customer Rejection
62		Customer request Re-appointment	Dawiyat		FTTH Installation Failure - Customer Rejection
63		Canceled/has another service	Dawiyat		FTTH Installation Failure - Customer Rejection
64		Canceled By Customer	Dawiyat		FTTH Installation Failure - Customer Rejection
65		customer unavailable	Dawiyat		FTTH Installation Failure - Customer Rejection
66		Customer No answer	Dawiyat		FTTH Installation Failure - Customer Rejection
67		Customer Mobile Off	Dawiyat		FTTH Installation Failure - Customer Rejection
68		Customer appointment reschedule:No answer	Dawiyat		FTTH Installation Failure - Customer Rejection
69		Customer appointment reschedule:Mobile Off	Dawiyat		FTTH Installation Failure - Customer Rejection

70		Customer appointment reschedule:customer unavailable	Dawiyat		FTTH Installation Failure - Customer Rejection
71		Customer appointment reschedule:Canceled By Customer	Dawiyat		FTTH Installation Failure - Customer Rejection
72		Customer appointment reschedule:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Customer Rejection
73		Customer appointment reschedule:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Customer Rejection
74		Customer appointment reschedule:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Customer Rejection
75		Customer appointment reschedule:Wiring issue	Dawiyat		FTTH Installation Failure - Customer Rejection
76		Customer appointment reschedule:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Customer Rejection
77		Customer appointment reschedule:Duplicate Order	Dawiyat		FTTH Installation Failure - Customer Rejection
78		Customer appointment reschedule:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
79		Customer appointment reschedule:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
80		Customer appointment reschedule:No Signal	Dawiyat		FTTH Installation Failure - Customer Rejection
81		Customer appointment reschedule:No Port Available	Dawiyat		FTTH Installation Failure - Customer Rejection
82		Customer appointment reschedule:No Plate	Dawiyat		FTTH Installation Failure - Customer Rejection
83		Customer appointment reschedule:No ODB	Dawiyat		FTTH Installation Failure - Customer Rejection
84		Customer appointment reschedule:ODB Damage	Dawiyat		FTTH Installation Failure - Customer Rejection
85		Customer appointment reschedule:Wrong Contact	Dawiyat		FTTH Installation Failure - Customer Rejection
86		Customer appointment reschedule:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Customer Rejection
87		Customer appointment reschedule:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Customer Rejection
88		Customer appointment reschedule:Successful	Dawiyat		FTTH Installation Failure - Customer Rejection
89		Customer appointment reschedule:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Customer Rejection
90		Internal wiring:No answer	Dawiyat		FTTH Installation Failure - Customer Rejection
91		Internal wiring:Mobile Off	Dawiyat		FTTH Installation Failure - Customer Rejection
92		Internal wiring:customer unavailable	Dawiyat		FTTH Installation Failure - Customer Rejection
93		Internal wiring:Canceled By Customer	Dawiyat		FTTH Installation Failure - Customer Rejection
94		Internal wiring:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Customer Rejection
95		Internal wiring:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Customer Rejection
96		Internal wiring:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Customer Rejection
97		Internal wiring:Wiring issue	Dawiyat		FTTH Installation Failure - Customer Rejection
98		Internal wiring:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Customer Rejection
99		Internal wiring:Duplicate Order	Dawiyat		FTTH Installation Failure - Customer Rejection

100		Internal wiring:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
101		Internal wiring:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
102		Internal wiring:No Signal	Dawiyat		FTTH Installation Failure - Customer Rejection
103		Internal wiring:No Port Available	Dawiyat		FTTH Installation Failure - Customer Rejection
104		Internal wiring:No Plate	Dawiyat		FTTH Installation Failure - Customer Rejection
105		Internal wiring:No ODB	Dawiyat		FTTH Installation Failure - Customer Rejection
106		Internal wiring:ODB Damage	Dawiyat		FTTH Installation Failure - Customer Rejection
107		Internal wiring:Wrong Contact	Dawiyat		FTTH Installation Failure - Customer Rejection
108		Internal wiring:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Customer Rejection
109		Internal wiring:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Customer Rejection
110		Internal wiring:Successful	Dawiyat		FTTH Installation Failure - Customer Rejection
111		Internal wiring:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Customer Rejection
112		Customer cancelation:No answer	Dawiyat		FTTH Installation Failure - Customer Rejection
113		Customer cancelation:Mobile Off	Dawiyat		FTTH Installation Failure - Customer Rejection
114		Customer cancelation:customer unavailable	Dawiyat		FTTH Installation Failure - Customer Rejection
115		Customer cancelation:Canceled By Customer	Dawiyat		FTTH Installation Failure - Customer Rejection
116		Customer cancelation:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Customer Rejection
117		Customer cancelation:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Customer Rejection
118		Customer cancelation:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Customer Rejection
119		Customer cancelation:Wiring issue	Dawiyat		FTTH Installation Failure - Customer Rejection
120		Customer cancelation:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Customer Rejection
121		Customer cancelation:Duplicate Order	Dawiyat		FTTH Installation Failure - Customer Rejection
122		Customer cancelation:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
123		Customer cancelation:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
124		Customer cancelation:No Signal	Dawiyat		FTTH Installation Failure - Customer Rejection
125		Customer cancelation:No Port Available	Dawiyat		FTTH Installation Failure - Customer Rejection
126		Customer cancelation:No Plate	Dawiyat		FTTH Installation Failure - Customer Rejection
127		Customer cancelation:No ODB	Dawiyat		FTTH Installation Failure - Customer Rejection
128		Customer cancelation:ODB Damage	Dawiyat		FTTH Installation Failure - Customer Rejection
129		Customer cancelation:Wrong Contact	Dawiyat		FTTH Installation Failure - Customer Rejection

130		Customer cancelation:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Customer Rejection
131		Customer cancelation:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Customer Rejection
132		Customer cancelation:Successful	Dawiyat		FTTH Installation Failure - Customer Rejection
133		Customer cancelation:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Customer Rejection
134		Customer is not reachable:No answer	Dawiyat		FTTH Installation Failure - Customer Rejection
135		Customer is not reachable:Mobile Off	Dawiyat		FTTH Installation Failure - Customer Rejection
136		Customer is not reachable:customer unavailable	Dawiyat		FTTH Installation Failure - Customer Rejection
137		Customer is not reachable:Canceled By Customer	Dawiyat		FTTH Installation Failure - Customer Rejection
138		Customer is not reachable:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Customer Rejection
139		Customer is not reachable:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Customer Rejection
140		Customer is not reachable:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Customer Rejection
141		Customer is not reachable:Wiring issue	Dawiyat		FTTH Installation Failure - Customer Rejection
142		Customer is not reachable:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Customer Rejection
143		Customer is not reachable:Duplicate Order	Dawiyat		FTTH Installation Failure - Customer Rejection
144		Customer is not reachable:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
145		Customer is not reachable:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Customer Rejection
146		Customer is not reachable:No Signal	Dawiyat		FTTH Installation Failure - Customer Rejection
147		Customer is not reachable:No Port Available	Dawiyat		FTTH Installation Failure - Customer Rejection
148		Customer is not reachable:No Plate	Dawiyat		FTTH Installation Failure - Customer Rejection
149		Customer is not reachable:No ODB	Dawiyat		FTTH Installation Failure - Customer Rejection
150		Customer is not reachable:ODB Damage	Dawiyat		FTTH Installation Failure - Customer Rejection
151		Customer is not reachable:Wrong Contact	Dawiyat		FTTH Installation Failure - Customer Rejection
152		Customer is not reachable:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Customer Rejection
153		Customer is not reachable:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Customer Rejection
154		Customer is not reachable:Successful	Dawiyat		FTTH Installation Failure - Customer Rejection
155		Customer is not reachable:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Customer Rejection
156		No HAG	Dawiyat		FTTH Installation Failure - Logistic Rejection
157		HAG Issue	Dawiyat		FTTH Installation Failure - Logistic Rejection
158		No HAG available	Dawiyat		FTTH Installation Failure - Logistic Rejection
159		No HAG available:No answer	Dawiyat		FTTH Installation Failure - Logistic Rejection

160		No HAG available:Mobile Off	Dawiyat		FTTH Installation Failure - Logistic Rejection
161		No HAG available:customer unavailable	Dawiyat		FTTH Installation Failure - Logistic Rejection
162		No HAG available:Canceled By Customer	Dawiyat		FTTH Installation Failure - Logistic Rejection
163		No HAG available:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Logistic Rejection
164		No HAG available:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Logistic Rejection
165		No HAG available:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Logistic Rejection
166		No HAG available:Wiring issue	Dawiyat		FTTH Installation Failure - Logistic Rejection
167		No HAG available:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Logistic Rejection
168		No HAG available:Duplicate Order	Dawiyat		FTTH Installation Failure - Logistic Rejection
169		No HAG available:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Logistic Rejection
170		No HAG available:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Logistic Rejection
171		No HAG available:No Signal	Dawiyat		FTTH Installation Failure - Logistic Rejection
172		No HAG available:No Port Available	Dawiyat		FTTH Installation Failure - Logistic Rejection
173		No HAG available:No Plate	Dawiyat		FTTH Installation Failure - Logistic Rejection
174		No HAG available:No ODB	Dawiyat		FTTH Installation Failure - Logistic Rejection
175		No HAG available:ODB Damage	Dawiyat		FTTH Installation Failure - Logistic Rejection
176		No HAG available:Wrong Contact	Dawiyat		FTTH Installation Failure - Logistic Rejection
177		No HAG available:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Logistic Rejection
178		No HAG available:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Logistic Rejection
179		No HAG available:Successful	Dawiyat		FTTH Installation Failure - Logistic Rejection
180		No HAG available:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Logistic Rejection
181		Fiber cut	Dawiyat		FTTH Installation Failure - Provider Rejection
182		The customer can't be served due to facility	Dawiyat		FTTH Installation Failure - Provider Rejection
183		ODB damage	Dawiyat		FTTH Installation Failure - Provider Rejection
184		No ODB fiber signal	Dawiyat		FTTH Installation Failure - Provider Rejection
185		The customer can't be served due to facility:No answer	Dawiyat		FTTH Installation Failure - Provider Rejection
186		The customer can't be served due to facility:Mobile Off	Dawiyat		FTTH Installation Failure - Provider Rejection
187		The customer can't be served due to facility:customer unavailable	Dawiyat		FTTH Installation Failure - Provider Rejection
188		The customer can't be served due to facility:Canceled By Customer	Dawiyat		FTTH Installation Failure - Provider Rejection
189		The customer can't be served due to facility:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Provider Rejection

190		The customer can't be served due to facility:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Provider Rejection
191		The customer can't be served due to facility:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Provider Rejection
192		The customer can't be served due to facility:Wiring issue	Dawiyat		FTTH Installation Failure - Provider Rejection
193		The customer can't be served due to facility:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Provider Rejection
194		The customer can't be served due to facility:Duplicate Order	Dawiyat		FTTH Installation Failure - Provider Rejection
195		The customer can't be served due to facility:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
196		The customer can't be served due to facility:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
197		The customer can't be served due to facility:No Signal	Dawiyat		FTTH Installation Failure - Provider Rejection
198		The customer can't be served due to facility:No Port Available	Dawiyat		FTTH Installation Failure - Provider Rejection
199		The customer can't be served due to facility:No Plate	Dawiyat		FTTH Installation Failure - Provider Rejection
200		The customer can't be served due to facility:No ODB	Dawiyat		FTTH Installation Failure - Provider Rejection
201		The customer can't be served due to facility:ODB Damage	Dawiyat		FTTH Installation Failure - Provider Rejection
202		The customer can't be served due to facility:Wrong Contact	Dawiyat		FTTH Installation Failure - Provider Rejection
203		The customer can't be served due to facility:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Provider Rejection
204		The customer can't be served due to facility:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Provider Rejection
205		The customer can't be served due to facility:Successful	Dawiyat		FTTH Installation Failure - Provider Rejection
206		The customer can't be served due to facility:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Provider Rejection
207		ODB damage:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Provider Rejection
208		ODB damage:Wiring issue	Dawiyat		FTTH Installation Failure - Provider Rejection
209		ODB damage:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Provider Rejection
210		ODB damage:Duplicate Order	Dawiyat		FTTH Installation Failure - Provider Rejection
211		ODB damage:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
212		ODB damage:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
213		ODB damage:No Signal	Dawiyat		FTTH Installation Failure - Provider Rejection
214		ODB damage:No Port Available	Dawiyat		FTTH Installation Failure - Provider Rejection
215		ODB damage:No Plate	Dawiyat		FTTH Installation Failure - Provider Rejection
216		ODB damage:No ODB	Dawiyat		FTTH Installation Failure - Provider Rejection
217		ODB damage:ODB damage	Dawiyat		FTTH Installation Failure - Provider Rejection
218		ODB damage:Wrong Contact	Dawiyat		FTTH Installation Failure - Provider Rejection
219		ODB damage:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Provider Rejection

220		ODB damage:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Provider Rejection
221		ODB damage:Successful	Dawiyat		FTTH Installation Failure - Provider Rejection
222		ODB damage:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Provider Rejection
223		Fiber cut:No answer	Dawiyat		FTTH Installation Failure - Provider Rejection
224		Fiber cut:Mobile Off	Dawiyat		FTTH Installation Failure - Provider Rejection
225		Fiber cut:customer unavailable	Dawiyat		FTTH Installation Failure - Provider Rejection
226		Fiber cut:Canceled By Customer	Dawiyat		FTTH Installation Failure - Provider Rejection
227		Fiber cut:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Provider Rejection
228		Fiber cut:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Provider Rejection
229		Fiber cut:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Provider Rejection
230		Fiber cut:Wiring issue	Dawiyat		FTTH Installation Failure - Provider Rejection
231		Fiber cut:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Provider Rejection
232		Fiber cut:Duplicate Order	Dawiyat		FTTH Installation Failure - Provider Rejection
233		Fiber cut:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
234		Fiber cut:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
235		Fiber cut:No Signal	Dawiyat		FTTH Installation Failure - Provider Rejection
236		Fiber cut:No Port Available	Dawiyat		FTTH Installation Failure - Provider Rejection
237		Fiber cut:No Plate	Dawiyat		FTTH Installation Failure - Provider Rejection
238		Fiber cut:No ODB	Dawiyat		FTTH Installation Failure - Provider Rejection
239		Fiber cut:ODB Damage	Dawiyat		FTTH Installation Failure - Provider Rejection
240		Fiber cut:Wrong Contact	Dawiyat		FTTH Installation Failure - Provider Rejection
241		Fiber cut:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Provider Rejection
242		Fiber cut:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Provider Rejection
243		Fiber cut:Successful	Dawiyat		FTTH Installation Failure - Provider Rejection
244		Fiber cut:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Provider Rejection
245		No ODB fiber signal:No answer	Dawiyat		FTTH Installation Failure - Provider Rejection
246		No ODB fiber signal:Mobile Off	Dawiyat		FTTH Installation Failure - Provider Rejection
247		No ODB fiber signal:customer unavailable	Dawiyat		FTTH Installation Failure - Provider Rejection
248		No ODB fiber signal:Canceled By Customer	Dawiyat		FTTH Installation Failure - Provider Rejection
249		No ODB fiber signal:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Provider Rejection

250		No ODB fiber signal:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Provider Rejection
251		No ODB fiber signal:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Provider Rejection
252		No ODB fiber signal:Wiring issue	Dawiyat		FTTH Installation Failure - Provider Rejection
253		No ODB fiber signal:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Provider Rejection
254		No ODB fiber signal:Duplicate Order	Dawiyat		FTTH Installation Failure - Provider Rejection
255		No ODB fiber signal:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
256		No ODB fiber signal:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Provider Rejection
257		No ODB fiber signal:No Signal	Dawiyat		FTTH Installation Failure - Provider Rejection
258		No ODB fiber signal:No Port Available	Dawiyat		FTTH Installation Failure - Provider Rejection
259		No ODB fiber signal:No Plate	Dawiyat		FTTH Installation Failure - Provider Rejection
260		No ODB fiber signal:No ODB	Dawiyat		FTTH Installation Failure - Provider Rejection
261		No ODB fiber signal:ODB Damage	Dawiyat		FTTH Installation Failure - Provider Rejection
262		No ODB fiber signal:Wrong Contact	Dawiyat		FTTH Installation Failure - Provider Rejection
263		No ODB fiber signal:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Provider Rejection
264		No ODB fiber signal:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Provider Rejection
265		No ODB fiber signal:Successful	Dawiyat		FTTH Installation Failure - Provider Rejection
266		No ODB fiber signal:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Provider Rejection
267		ODB damage:No answer	Dawiyat		FTTH Installation Failure - Provider Rejection
268		ODB damage:Mobile Off	Dawiyat		FTTH Installation Failure - Provider Rejection
269		ODB damage:customer unavailable	Dawiyat		FTTH Installation Failure - Provider Rejection
270		ODB damage:Canceled By Customer	Dawiyat		FTTH Installation Failure - Provider Rejection
271		ODB damage:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Provider Rejection
272		ODB damage:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Provider Rejection
273		Duplicate Order	Dawiyat		FTTH Installation Failure - Systematic Rejection
274		Others	Dawiyat		FTTH Installation Failure - Unknown Rejection
275		Wrong customer contact	Dawiyat		FTTH Installation Failure - Wrong Customer Information
276		Wrong Contact	Dawiyat		FTTH Installation Failure - Wrong Customer Information
277		Wrong Customer Contact:No answer	Dawiyat		FTTH Installation Failure - Wrong Customer Information
278		Wrong Customer Contact:Mobile Off	Dawiyat		FTTH Installation Failure - Wrong Customer Information
279		Wrong Customer Contact:customer unavailable	Dawiyat		FTTH Installation Failure - Wrong Customer Information

280		Wrong Customer Contact:Canceled By Customer	Dawiyat		FTTH Installation Failure - Wrong Customer Information
281		Wrong Customer Contact:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Wrong Customer Information
282		Wrong Customer Contact:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Wrong Customer Information
283		Wrong Customer Contact:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Wrong Customer Information
284		Wrong Customer Contact:Wiring issue	Dawiyat		FTTH Installation Failure - Wrong Customer Information
285		Wrong Customer Contact:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Wrong Customer Information
286		Wrong Customer Contact:Duplicate Order	Dawiyat		FTTH Installation Failure - Wrong Customer Information
287		Wrong Customer Contact:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Wrong Customer Information
288		Wrong Customer Contact:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Wrong Customer Information
289		Wrong Customer Contact:No Signal	Dawiyat		FTTH Installation Failure - Wrong Customer Information
290		Wrong Customer Contact:No Port Available	Dawiyat		FTTH Installation Failure - Wrong Customer Information
291		Wrong Customer Contact:No Plate	Dawiyat		FTTH Installation Failure - Wrong Customer Information
292		Wrong Customer Contact:No ODB	Dawiyat		FTTH Installation Failure - Wrong Customer Information
293		Wrong Customer Contact:ODB Damage	Dawiyat		FTTH Installation Failure - Wrong Customer Information
294		Wrong Customer Contact:Wrong Contact	Dawiyat		FTTH Installation Failure - Wrong Customer Information
295		Wrong Customer Contact:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Wrong Customer Information
296		Wrong Customer Contact:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Wrong Customer Information
297		Wrong Customer Contact:Successful	Dawiyat		FTTH Installation Failure - Wrong Customer Information
298		Wrong Customer Contact:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Wrong Customer Information
299		Wrong ODB ID	Dawiyat		FTTH Installation Failure - Wrong Service Information
300		commercial site	Dawiyat		FTTH Installation Failure - Wrong Service Information
301		Wrong Location	Dawiyat		FTTH Installation Failure - Wrong Service Information
302		Commercial area	Dawiyat		FTTH Installation Failure - Wrong Service Information
303		Wrong ODB ID:No answer	Dawiyat		FTTH Installation Failure - Wrong Service Information
304		Wrong ODB ID:Mobile Off	Dawiyat		FTTH Installation Failure - Wrong Service Information
305		Wrong ODB ID:customer unavailable	Dawiyat		FTTH Installation Failure - Wrong Service Information
306		Wrong ODB ID:Canceled By Customer	Dawiyat		FTTH Installation Failure - Wrong Service Information
307		Wrong ODB ID:has another service by other Seeker	Dawiyat		FTTH Installation Failure - Wrong Service Information
308		Wrong ODB ID:Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Wrong Service Information
309		Wrong ODB ID:Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Wrong Service Information

310		Wrong ODB ID:Wiring issue	Dawiyat		FTTH Installation Failure - Wrong Service Information
311		Wrong ODB ID:commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Wrong Service Information
312		Wrong ODB ID:Duplicate Order	Dawiyat		FTTH Installation Failure - Wrong Service Information
313		Wrong ODB ID:Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Wrong Service Information
314		Wrong ODB ID:Pending/No HAGs	Dawiyat		FTTH Installation Failure - Wrong Service Information
315		Wrong ODB ID:No Signal	Dawiyat		FTTH Installation Failure - Wrong Service Information
316		Wrong ODB ID:No Port Available	Dawiyat		FTTH Installation Failure - Wrong Service Information
317		Wrong ODB ID:No Plate	Dawiyat		FTTH Installation Failure - Wrong Service Information
318		Wrong ODB ID:No ODB	Dawiyat		FTTH Installation Failure - Wrong Service Information
319		Wrong ODB ID:ODB Damage	Dawiyat		FTTH Installation Failure - Wrong Service Information
320		Wrong ODB ID:Wrong Contact	Dawiyat		FTTH Installation Failure - Wrong Service Information
321		Wrong ODB ID:Wrong ODB ID	Dawiyat		FTTH Installation Failure - Wrong Service Information
322		Wrong ODB ID:Wrong Customer Location	Dawiyat		FTTH Installation Failure - Wrong Service Information
323		Wrong ODB ID:Successful	Dawiyat		FTTH Installation Failure - Wrong Service Information
324		Wrong ODB ID:ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Wrong Service Information
325		Commercial area(for FTTH):No answer	Dawiyat		FTTH Installation Failure - Wrong Service Information
326		Commercial area(for FTTH):Mobile Off	Dawiyat		FTTH Installation Failure - Wrong Service Information
327		Commercial area(for FTTH):customer unavailable	Dawiyat		FTTH Installation Failure - Wrong Service Information
328		Commercial area(for FTTH):Canceled By Customer	Dawiyat		FTTH Installation Failure - Wrong Service Information
329		Commercial area(for FTTH):has another service by other Seeker	Dawiyat		FTTH Installation Failure - Wrong Service Information
330		Commercial area(for FTTH):Re-appointment within SLA	Dawiyat		FTTH Installation Failure - Wrong Service Information
331		Commercial area(for FTTH):Reschedule > 4 Days	Dawiyat		FTTH Installation Failure - Wrong Service Information
332		Commercial area(for FTTH):Wiring issue	Dawiyat		FTTH Installation Failure - Wrong Service Information
333		Commercial area(for FTTH):commercial site (for FTTH)	Dawiyat		FTTH Installation Failure - Wrong Service Information
334		Commercial area(for FTTH):Duplicate Order	Dawiyat		FTTH Installation Failure - Wrong Service Information
335		Commercial area(for FTTH):Canceled/No HAGs	Dawiyat		FTTH Installation Failure - Wrong Service Information
336		Commercial area(for FTTH):Pending/No HAGs	Dawiyat		FTTH Installation Failure - Wrong Service Information
337		Commercial area(for FTTH):No Signal	Dawiyat		FTTH Installation Failure - Wrong Service Information
338		Commercial area(for FTTH):No Port Available	Dawiyat		FTTH Installation Failure - Wrong Service Information
339		Commercial area(for FTTH):No Plate	Dawiyat		FTTH Installation Failure - Wrong Service Information

340		Commercial area(for FTTH):No ODB	Dawiyat		FTTH Installation Failure - Wrong Service Information
341		Commercial area(for FTTH):ODB Damage	Dawiyat		FTTH Installation Failure - Wrong Service Information
342		Commercial area(for FTTH):Wrong Contact	Dawiyat		FTTH Installation Failure - Wrong Service Information
343		Commercial area(for FTTH):Wrong ODB ID	Dawiyat		FTTH Installation Failure - Wrong Service Information
344		Commercial area(for FTTH):Wrong Customer Location	Dawiyat		FTTH Installation Failure - Wrong Service Information
345		Commercial area(for FTTH):Successful	Dawiyat		FTTH Installation Failure - Wrong Service Information
346		Commercial area(for FTTH):ODB-Signal issue/OSP	Dawiyat		FTTH Installation Failure - Wrong Service Information
347	R-OM-001	Validation of customer order is not passed	ACES		
348	R-OM-002	Validation of product order is not passed	ACES		
349	R-OM-003	Cannot contact the end-customer	ACES		
350	R-OM-004	Network is planning or no idle resource for service	ACES		
351	R-OM-005	Network is available for service	ACES		
352	R-OM-006	The relationship cannot be circulated between product A [prodInstId] product B [prodInstId]	ACES		
353	R-OM-007	This product [prodSpecCode] cannot be ordered as the agreement	ACES		
354	R-OM-008	End-customer require to cancel the product order	ACES		
355	R-OM-010	The product [prodSpecCode] cannot be deleted, because it is depended by the active product instance [prodInstId]	ACES		
356	R-OM-012	No available address can be found with address Id [addressId]	ACES		
357	R-OM-014	Address Unknown	ACES		
358	R-OM-015	No available network resource for service fulfillment	ACES		
359	R-OM-016	Customer Order Cancelled	ACES		

Treatment Handling processes (Installation Failures)

Mobily Infrastructure

Following are the treatments that has to be considered for the WFM rejection reasons,

Mobily WFM Rejection Reasons Categorization

Treatment#1 Category:

Network Problem

2010 NS-No Signal
2011 NS-ODB Upgrade required
2012 NS-Damaged ODB
2013 NS-High loss
2014 NS-No ODB
2015 NS-Duplicate ODB ID
2016 NS-ODB relocation required
2017 NS-Wrong ODB ID
2018 NS-NO ODB on GIS
2067 NC-CPE not visible
2068 NC-CPE not taking IP
2069 NC-Speed Problem
2025 Duplicate WO

Treatment#2 Category:

System Issues

2024 SYS - OSS Discovered Problems
2020 SYS - CPE/STB Swap Failure
2021 SYS - CPE not visible
2022 SYS - CPE not taking IP
2023 SYS - Speed Problem
2031 SYS - Provisioning Unsuccessful
2101 Mesh Device is not available

Treatment#3 Category:

Wrong Information

2060 Wrong Info - Wrong contact No.
2061 Wrong Info - Wrong ODB ID
2062 Wrong Info - Wrong CPE/STB
2063 Wrong Info - Wrong Location
2064 Wrong Info - Wrong Package

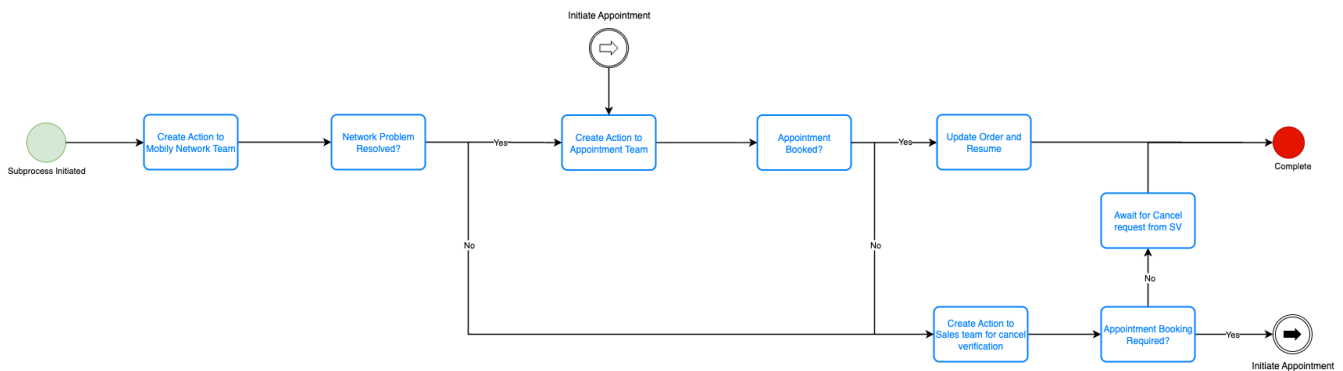
Treatment#4 Category:

Mobily Customer Rejection

2035 Device Faulty - refused to pay
2040 Customer Not Reachable
2041 Customer Want to Cancel
2042 Reschedule later
2089 SANAD is rejected by customer
2102 Customer refused to sign PN

The rejection reasons are categorised into groups and each group shall be handled by distinct treatment. In Telflow, four sub-processes are created to handle each treatment independently such that any business process changes to the treatment in future can be handled with no impact to the other treatments.

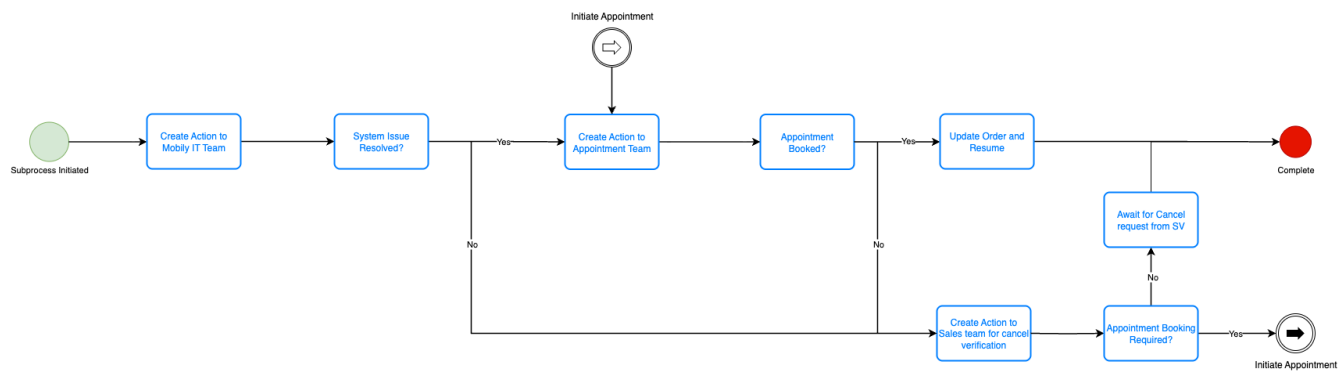
Treatment 1 - Network Problem



S. No	Item	Assigned Team	Detail
1	Create Action to Mobily Network Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Mobily network team and awaits for the outcome of the problem resolution.
2	Network Problem Resolved?	Telflow	Telflow validates whether the problem is resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.

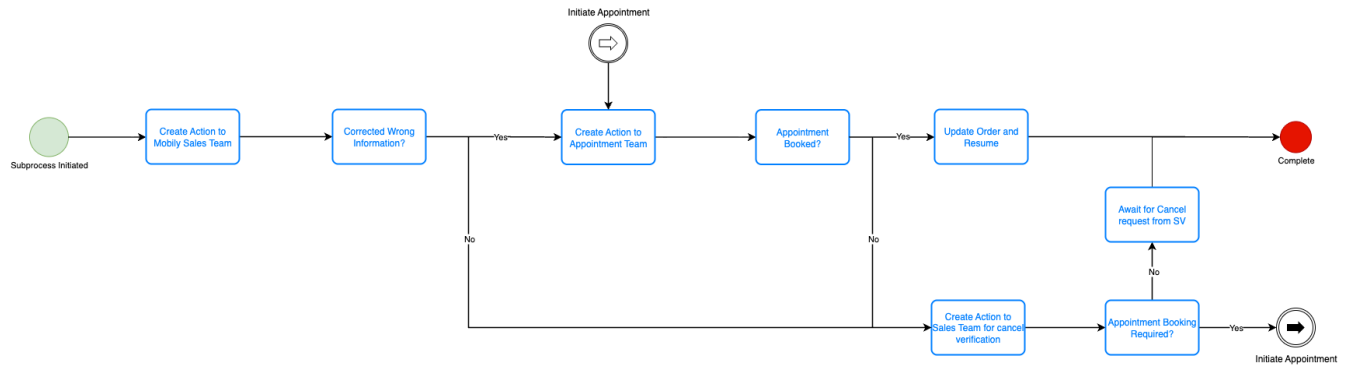
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Treatment 2 - System Issues



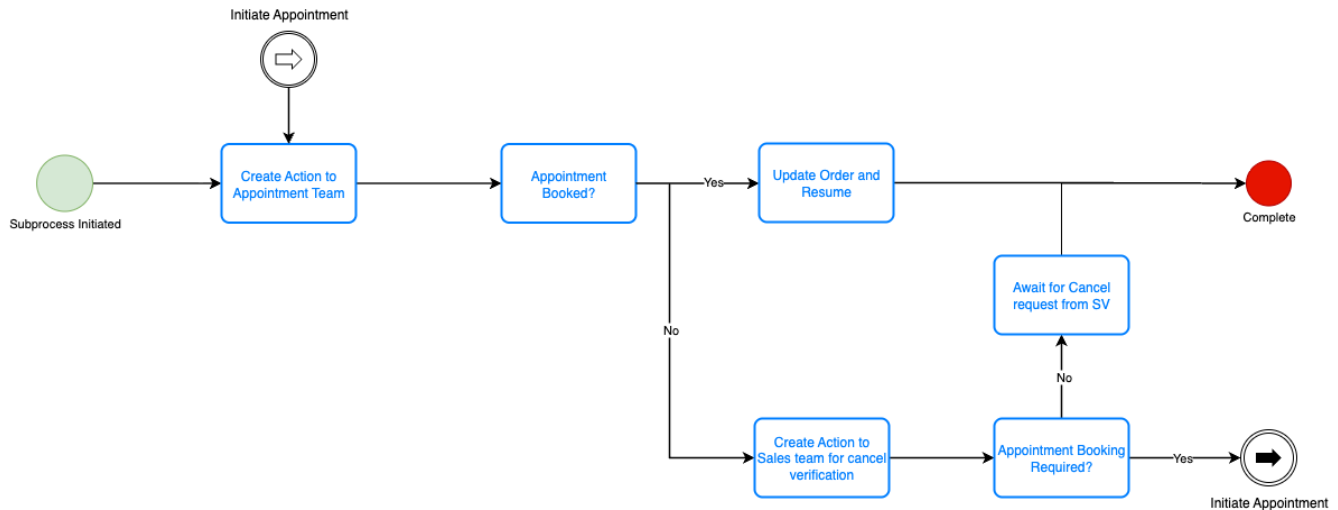
S. No	Item	Assigned Team	Detail
1	Create Action to Mobily IT Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Mobily IT team and awaits for the outcome of the issue resolution.
2	System Issue Resolved?	Telflow	Telflow validates whether the issue is resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Treatment 3 - Wrong Information



S. No	Item	Assigned Team	Detail
1	Create Action to Mobily Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Mobily network team and awaits for the outcome of the information update.
2	Corrected Wrong Information?	Telflow	Telflow validates whether the information had been corrected based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Treatment 4 - Customer Rejection



S. No	Item	Assigned Team	Detail
1	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment.
2	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Update Order and Resume 2. No Follow step 4: Create Action to Sales team for cancel verification
3	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
4	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
5	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
6	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Open Access Infrastructure

Following are the treatments that has to be considered for the Open Access rejection reasons,

Open Access Rejection Reasons Categorization

Treatment#1 Category:

Provider Rejection

- Loss Signal
- Fiber Cut
- ODB Damage
- No Port
- No Fiber
- ...etc.

Wrong Service Information

- Wrong Plate ID
- Wrong ODB ID
- Wrong Location
- ...etc.

Systematic Rejection

- Duplicate Order
- ...etc.

Treatment#2 Category:

Customer Rejection

- Customer Not Answering
- Customer Asked for Reappointment
- Customer Request to Cancel
- Customer not Reachable
- Customer Mobil Off
- ...etc.

Treatment#4 Category:

Logistic Rejection

- No HAG Available
- HAG Issue
- ...etc.

Treatment#3 Category:

Wrong Customer Information

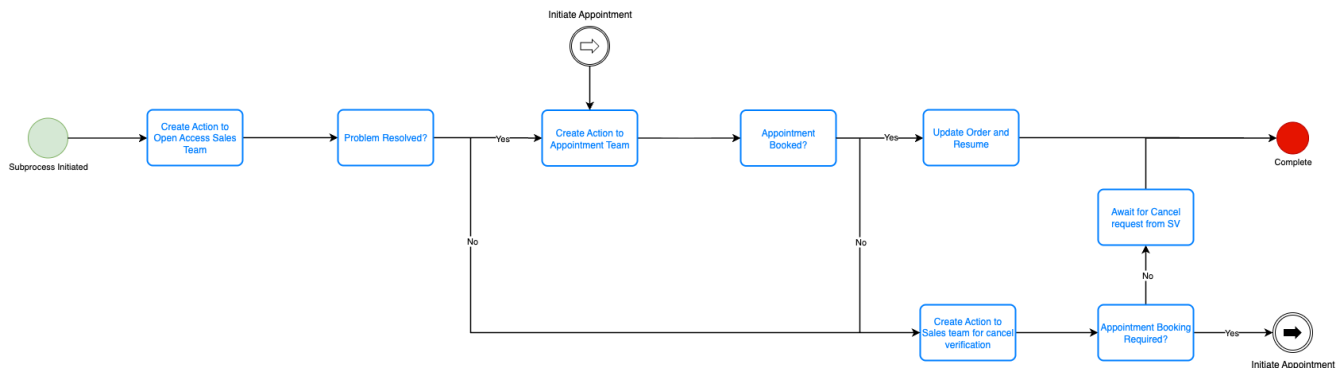
- Wrong Contact Number
- ...etc.

Treatment#5 Category:

Unknown Rejection

- No Details
- Others

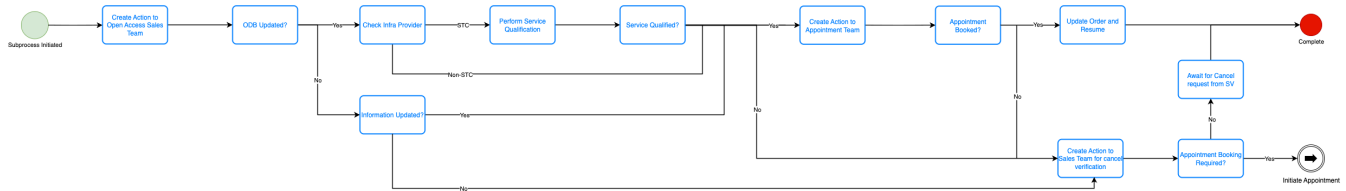
Treatment 1 - Variant 1 - Provider Rejection



S. No	Item	Assigned Team	Detail
1	Create Action to Open Access Sales Team	SingleView	Telfow shall initiate an action in SingleView which assigns to Open Access sales team and awaits for the outcome of the problem resolution.
2	Problem Resolved?	Telflow	Telfow validates whether the problem is resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 7: Create Action to Appointment Team 2. No Follow step 10: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telfow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telfow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 9: Update Order and Resume 2. No Follow step 10: Create Action to Sales team for cancel verification

5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

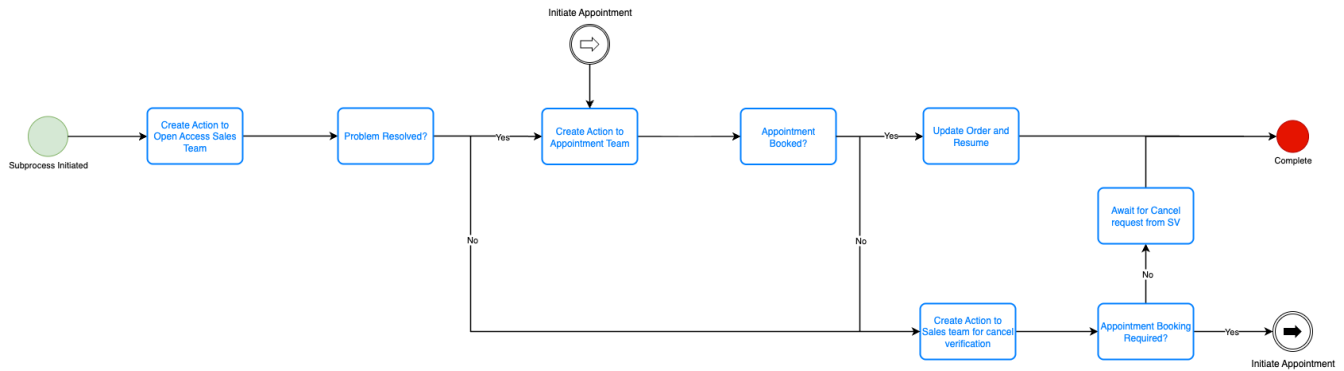
Treatment 1 - Variant 2 - Wrong Service Information



S. No	Item	Assigned Team	Detail
1	Create Action to Open Access Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Open Access sales team and awaits for the outcome of the problem resolution.
2	ODB Updated?	Telflow	Based on the asynchronous response from SingleView, Telflow checks whether the ODB info had been updated. 1. Yes Follow step 3: Check Infra Provider 2. No Follow step 6: Information Updated?
3	Check Infra Provider	Telflow	Checks the infra provider. 1. STC Follow step 4: Perform Service Qualification 2. Non-STC Follow step 7: Create Action to Appointment Team
4	Perform Service Qualification	Open Access	If the Infra Provider is STC, then Telflow shall initiate service qualification towards Open Access whereas for Non-STC operators, step 7 shall follow.
5	Service Qualified?	Telflow	Checks whether the service is qualified. Possible outcomes: 1. Yes Follow step 7: Create Action to Appointment Team 2. No Follow step 10: Create Action to Sales team for cancel verification
6	Information Updated?	Telflow	Telflow validates whether the information had been updated based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 7: Create Action to Appointment Team 2. No Follow step 10: Create Action to Sales team for cancel verification
7	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
8	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 9: Update Order and Resume 2. No Follow step 10: Create Action to Sales team for cancel verification
9	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
10	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.

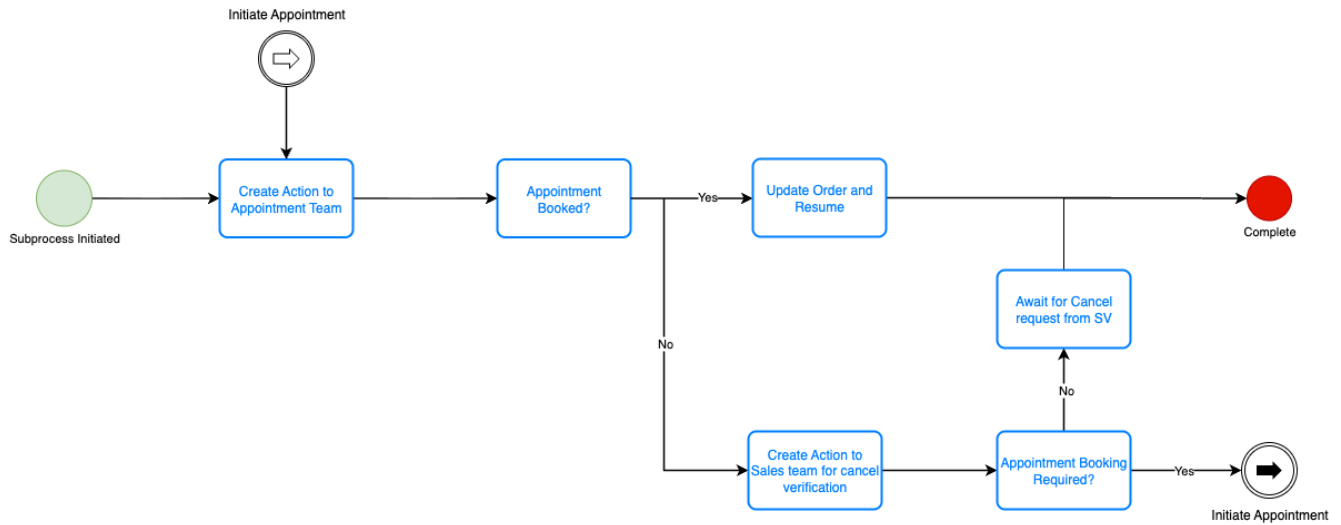
11	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 7: Create Action to Appointment Team 2. No Follow step 12: Await for Cancel request from SV
12	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Treatment 1 - Variant 3 - Systematic Rejection



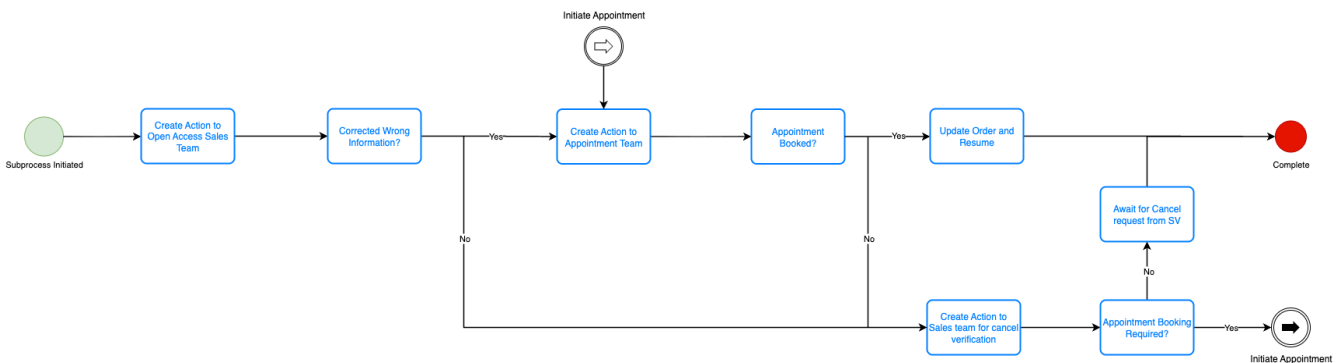
S. No	Item	Assigned Team	Detail
1	Create Action to Open Access Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Open Access sales team and awaits for the outcome of the problem resolution.
2	Problem Resolved?	Telflow	Telflow validates whether the problem is resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 7: Create Action to Appointment Team 2. No Follow step 10: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 9: Update Order and Resume 2. No Follow step 10: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Treatment 2 - Customer Rejection



S. No	Item	Assigned Team	Detail
1	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment.
2	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Update Order and Resume 2. No Follow step 4: Create Action to Sales team for cancel verification
3	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
4	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
5	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
6	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

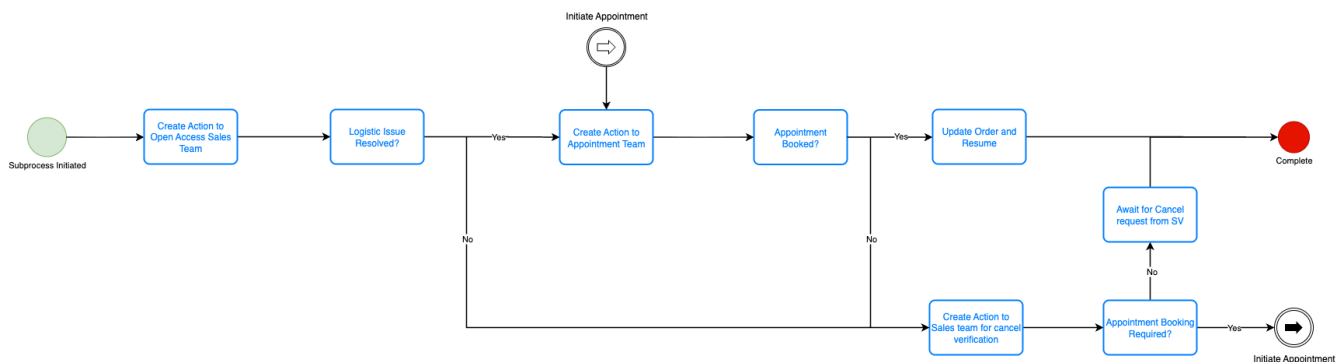
Treatment 3 - Wrong Customer Information



S. No	Item	Assigned Team	Detail
1	Create Action to Open Access Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Open Access sales team and awaits for the outcome of the information update.

2	Corrected Wrong Information?	Telflow	Telflow validates whether the information had been corrected based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

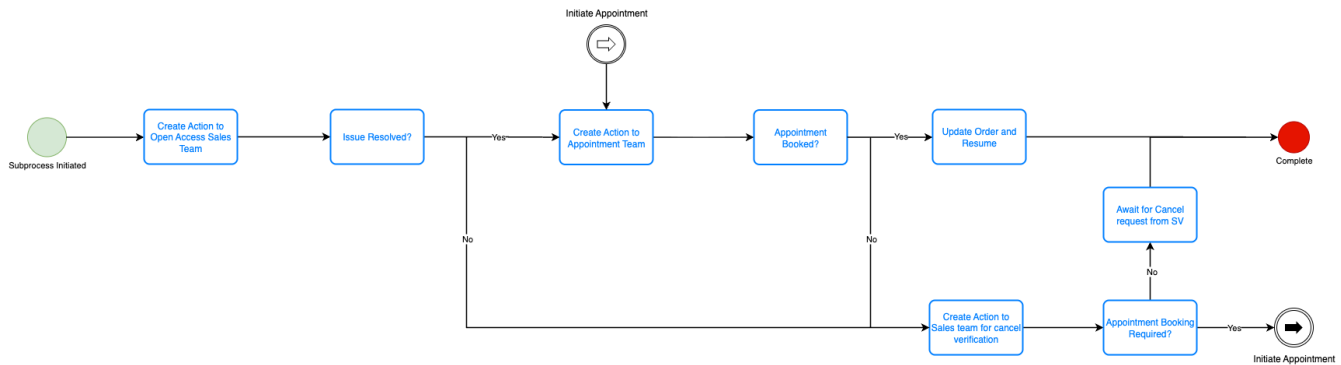
Treatment 4 - Logistic Rejection



S. No	Item	Assigned Team	Detail
1	Create Action to Open Access Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Open Access sales team and awaits for the outcome of the issue resolution.
2	Logistic Issue Resolved?	Telflow	Telflow validates whether the issue had been resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.

7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Treatment 5 - Unknown Rejection

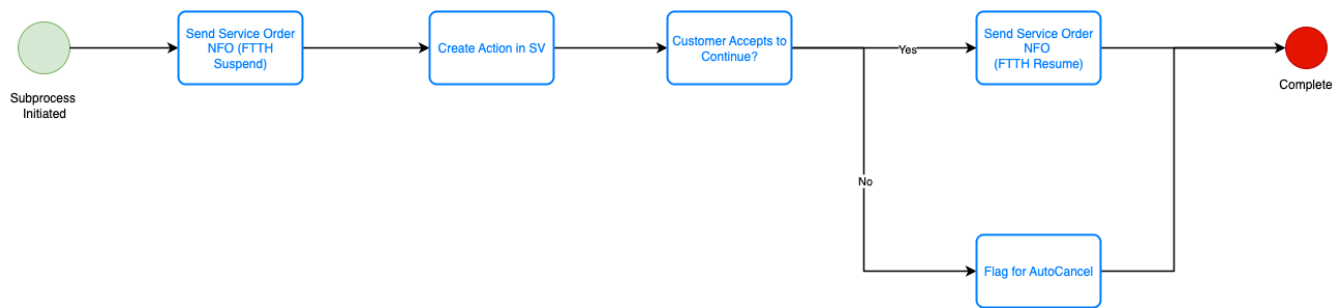


S. No	Item	Assigned Team	Detail
1	Create Action to Open Access Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Open Access sales team and awaits for the outcome of the issue resolution.
2	Issue Resolved?	Telflow	Telflow validates whether the issue had been resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Create Action to Sales team for cancel verification
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Create Action to Sales team for cancel verification
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Create Action to Sales team for cancel verification	SingleView	Telflow requests SingleView to create action and assign to Mobily sales team. Sales team shall reach out to customer and based on that, either new appointment shall be scheduled or order shall be cancelled.
7	Appointment Booking Required?	Telflow	Telflow validates whether appointment action has to be created in SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 8: Await for Cancel request from SV
8	Await for Cancel request from SV	Telflow	Telflow awaits for cancel order request from SV.

Handle UAT Timer

The below block diagram covers the following subprocess,

1. Handle UAT Timer Expiry



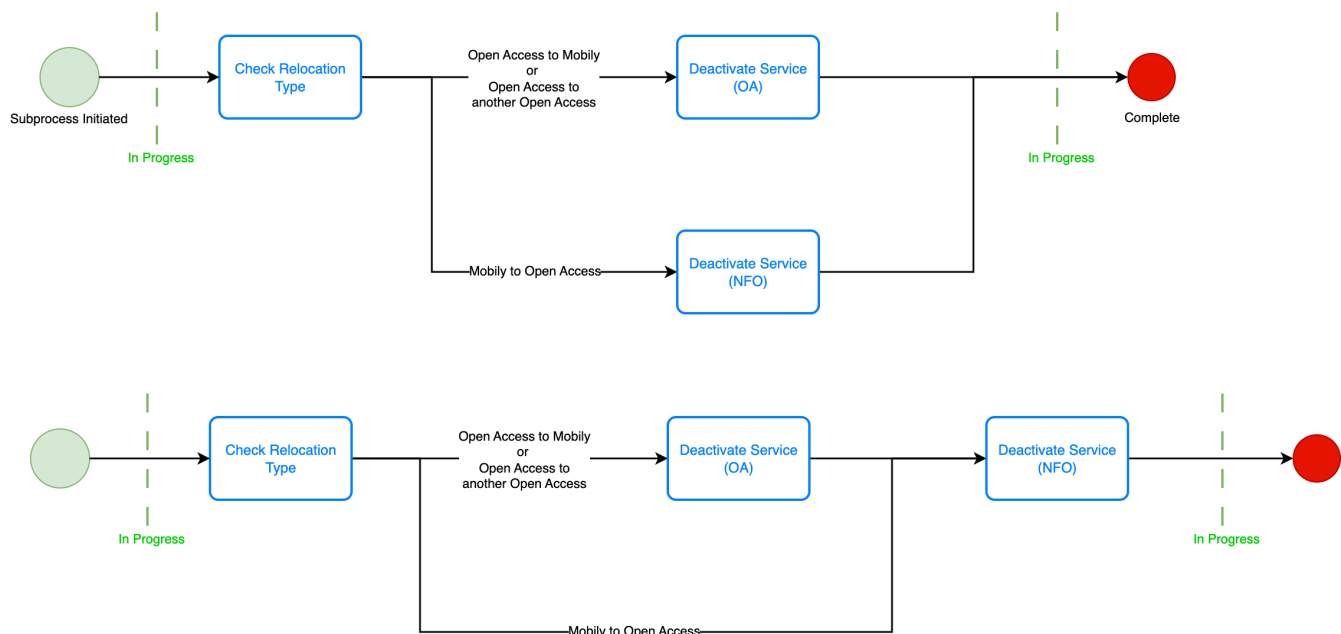
S. No	Item	Assigned Team	Detail
1	Send Service Order NFO (FTTH Suspend)	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface to suspend FTTH service.
2	Create Action in SV	SingleView	Telflow creates an action in SV and expects action Id in the response. Below are the information that has to be cascaded to SV from Telflow, 1. Type 'Contact Customer for UAT' 2. SingleView Order Id
3	Customer Accepts to Continue?	Telflow	Based on the asynchronous response from SingleView, Telflow validates whether the customer had accepted UAT. Possible outcomes: 1. Yes Follow step 4: Send Service Order NFO (FTTH Resume) 2. No Follow step 5: Flag for AutoCancel
4	Send Service Order NFO (FTTH Resume)	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface to resume FTTH service.
5	Flag for AutoCancel	Telflow	Flag to identify that the AutoCancellation is required.

Relocation Deactivate

<for future reference in 4B>

The below block diagram covers the following subprocess

1. Relocation Deactivate



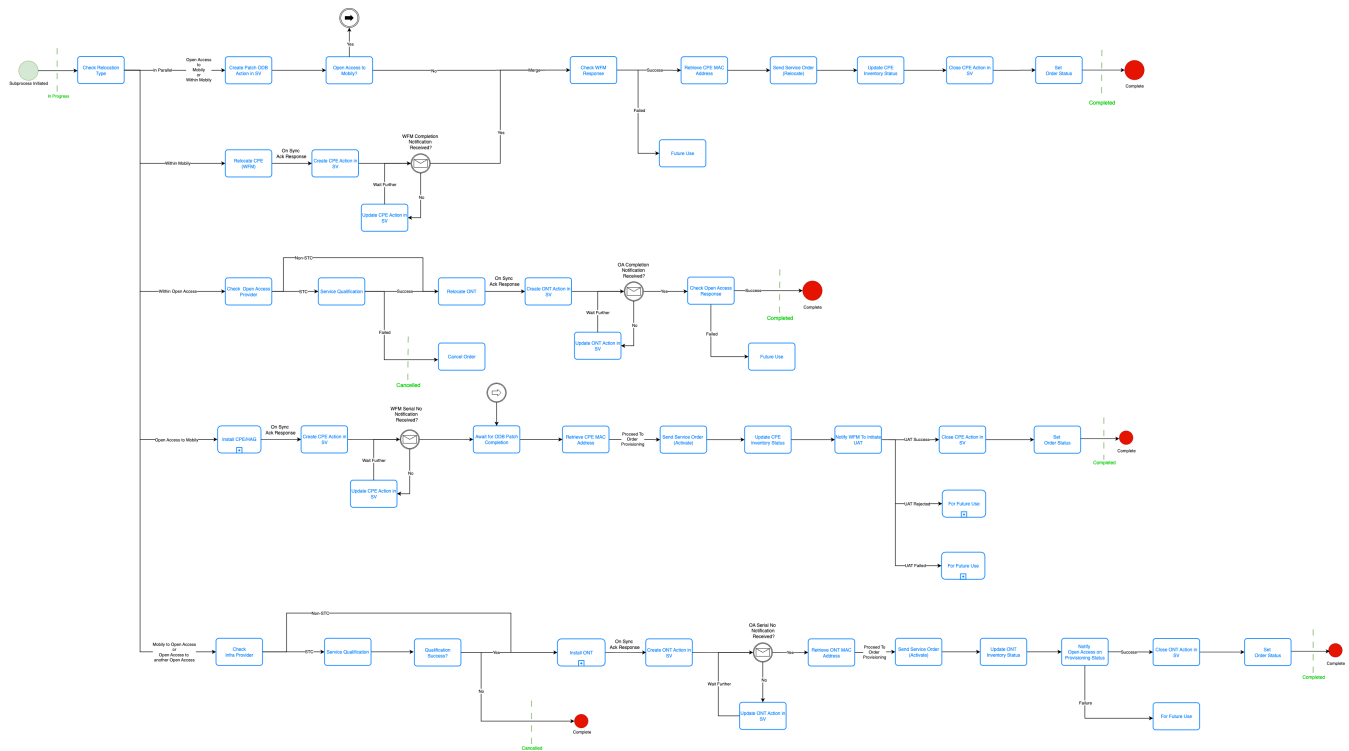
S. No	Item	Assigned Team	Detail
1	Check Relocation Type	Telflow	Identify the relocation type. Possible outcomes: 1. If Open Access to Mobily Infra or Open Access to another Open Access Infra -> Follow step 2: Deactivate Service (OA) 2. If Mobily to Open Access Infra -> Follow step 3: Deactivate Service (NFO) Order is set to In Progress status.
2	Deactivate Service (OA)	Open Access	Telflow sends service order to Open Access to deactivate the service.
3	Deactivate Service (NFO)	Nokia FlowOne	Telflow sends service order to Nokia FlowOne via TMF641 interface to deactivate the service.

Relocation Activate

<for future reference in 4B>

The below block diagram covers the following subprocess

1. Relocation Activate



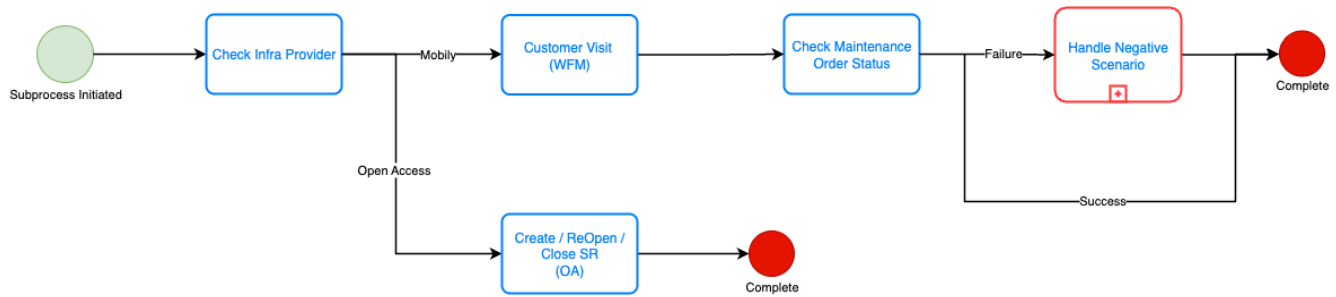
S. No	Activity	Assigned Team /System	Description
1	Check Relocation Type	Telflow	Identify the relocation type. Possible outcomes: 1. If Open Access to Mobily Infra -> Follow step 2: Patch ODB and step 3: Install CPE/HAG 2. If Mobily to Open Access Infra or Open Access to another Open Access Infra -> Follow step 4: Check Open Access Provider
2	Patch ODB	WFM	A patching task is created in Telflow irrespective of whether the ODB is old or new to attach the customer device with the ODB. The Patch ODB task is applicable for regular customer only and the WFM team shall be notified of this task. The WFM technician must complete this task in Telflow for the order to proceed further.

3	Install CPE /HAG	WFM	This sub-process invokes 'Handle work order' sub-process. Note: • Patch ODB and Install CPE/HAG are parallel tasks.
4	Check Open Access Provider	Telflow	Identify the Open Access Provider. Possible outcomes: 1. If STC Follow step 5: Service Qualification 2. If Non-STC Follow step 6: Install ONT
5	Service Qualification	SingleView	Telflow shall inform SingleView to initiate service qualification.
6	Install ONT	Open Access	This sub-process invokes 'Handle work order' sub-process.
7	Check Infra Provider	Telflow	Identify the Infra Provider (Mobily or Open Access). Possible outcomes: 1. If Mobily Follow step 8: Retrieve CPE MAC Address 2. If Open Access Follow step 9: Send Service Order
8	Retrieve CPE MAC Address	SingleView	Based on CPE Serial Number, Telflow retrieves CPE MAC Address from SingleView via TMF639 and proceeds the order further.
9	Send Service Order	Nokia FlowOne	Telflow performs order decomposition and sends service order to Nokia FlowOne via TMF641 interface.
10	Check Infra Provider	Telflow	Identify the Infra Provider (Mobily or Open Access). Possible outcomes: 1. If Mobily Follow step 11: Update CPE Inventory Status 2. If Open Access Follow step 14: Service Provisioned in OA
11	Update CPE Inventory Status	MapleWave	Telflow updates the inventory status of the CPE allocated to the customer in MapleWave.
12	Notify WFM to initiate UAT	WFM	Telflow notifies WFM to initiate User Acceptance.
13	Check UAT Response		Check the UAT result. Possible outcomes: 1. UAT Success Follow step 16: Set Order Status 2. UAT Rejected 3. UAT Failed
14	Service Provisioned in OA	Open Access	Service order creation in Open Access.
15	Check Provisioning Status	Telflow	Check the provisioning result. Possible outcomes: 1. Success Follow step 16: Set Order Status 2. Failure
16	Set Order Status	Telflow	Order status is set to complete.

Handling Maintenance Work Order

The below block diagram covers the following subprocess,

1. Handle Maintenance Work order

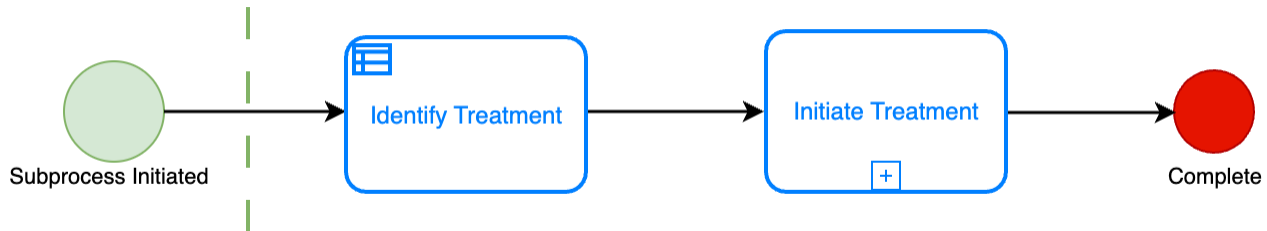


S. No	Item	Assigned Team	Detail
1	Check Infra Provider	Telflow	Identify the Infra Provider ('Mobily' or 'Open Access'). The order is already in 'In Progress' state. Based on the Infra Provider, initiates the request towards Mobily/Open Access WFM.
2	Customer Visit (WFM)	WFM	1. Maintenance Work order is created in WFM for the technician to visit the customer premise and fix the issue.
3	Create / ReOpen SR (OA)	Open Access	1. Service Request is created in Open Access for the technician to visit the customer premise and fix the issue.
4	Check Maintenance Order Status	Telflow	Possible outcomes: 1. Success Completes the process. 2. Failure Follow step 5: Handle Negative Scenario
5	Handle Negative Scenario	Telflow	Invokes Handle Negative Scenario sub-process.

Handling Negative Scenario

The below block diagram covers the following sub-process

1. Handle Negative Scenario



S. No	Item	Assigned Team	Detail
1	Identify Treatment	Telflow	Based on the Infra Provider and the rejection reason sent by WFM or Open Access, the treatment to be performed shall be identified.
2	Initiate Treatment	Telflow	Based on the treatment identified, the respective treatment sub-process shall be initiated.

Following DMN shall be configured to identify the treatment pertaining to the service provider based on the rejection reason,

Note:

1. Send mail to Sharanya to extract the reason codes for SR. Below list shall vary based on the data shared by WFM team.
2. As per Mobily confirmation, the reason code and reason text for negative scenario are same as that of installation failure. Below is for reference only, not actuals.

	Input			Output	
S. No	Reason Code	Reason Text	Infrastructure Provider	Treatment	Action Type

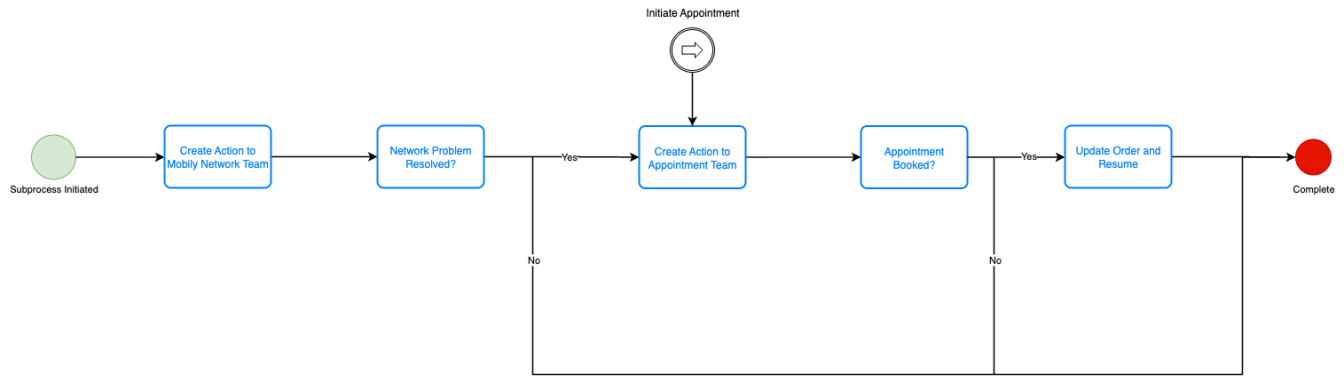
1	2010	NS-No Signal	Mobily	1	FTTH Negative Scenario - Network Problem
2	2011	NS-ODB Upgrade required	Mobily	1	FTTH Negative Scenario - Network Problem
3	2012	NS-Damaged ODB	Mobily	1	FTTH Negative Scenario - Network Problem
4	2013	NS-High Loss	Mobily	1	FTTH Negative Scenario - Network Problem
5	2014	NS-No ODB	Mobily	1	FTTH Negative Scenario - Network Problem
6	2015	NS-Duplicate ODB ID	Mobily	1	FTTH Negative Scenario - Network Problem
7	2016	NS-ODB relocation required	Mobily	1	FTTH Negative Scenario - Network Problem
8	2017	NS-Wrong ODB ID	Mobily	1	FTTH Negative Scenario - Network Problem
9	2018	NS-NO ODB on GIS	Mobily	1	FTTH Negative Scenario - Network Problem
10	2067	NC-CPE not visible	Mobily	1	FTTH Negative Scenario - Network Problem
11	2068	NC-CPE not taking IP	Mobily	1	FTTH Negative Scenario - Network Problem
12	2069	NC-Speed Problem	Mobily	1	FTTH Negative Scenario - Network Problem
13	2025	Duplicate WO	Mobily	1	FTTH Negative Scenario - Network Problem
14		OSS Discovered Problems	Mobily	1	FTTH Negative Scenario - Network Problem
15		NC - CPE/STB Swap Failure	Mobily	1	FTTH Negative Scenario - Network Problem
16		Network Issue : Others	Mobily	1	FTTH Negative Scenario - Network Problem
17		System Issues (NOC, IT)	Mobily	2	FTTH Negative Scenario - System Issues
18		System Issues - CP Not taking IP	Mobily	2	FTTH Negative Scenario - System Issues
19		System Issues - Speed Problems	Mobily	2	FTTH Negative Scenario - System Issues
20		System Issues - CPE not Visible	Mobily	2	FTTH Negative Scenario - System Issues
21		System Issues - OSS Discovered problems	Mobily	2	FTTH Negative Scenario - System Issues
22		System Issues - Duplicate WO	Mobily	2	FTTH Negative Scenario - System Issues
23		System Issues - Provisioning Unsuccessful	Mobily	2	FTTH Negative Scenario - System Issues
24		System Issues - Other	Mobily	2	FTTH Negative Scenario - System Issues
25		Wrong Info - Contact Number	Mobily	3	FTTH Negative Scenario - Wrong Information
26		Wrong Info - Wrong ODB	Mobily	3	FTTH Negative Scenario - Wrong Information
27		Wrong Info - Wrong CPE / STB	Mobily	3	FTTH Negative Scenario - Wrong Information
28		Wrong Info - Wrong Location	Mobily	3	FTTH Negative Scenario - Wrong Information
29		Wrong Info - Wrong Package	Mobily	3	FTTH Negative Scenario - Wrong Information
30		Customer Not available	Mobily	4	FTTH Negative Scenario - Mobily Customer Rejection
31		Customer Wants Cancel	Mobily	4	FTTH Negative Scenario - Mobily Customer Rejection
32		Appointment Reschedule	Mobily	4	FTTH Negative Scenario - Mobily Customer Rejection
33		Customer Not Ready	Mobily	4	FTTH Negative Scenario - Mobily Customer Rejection

Treatment Handling processes (Negative Scenarios)

Mobily Infrastructure

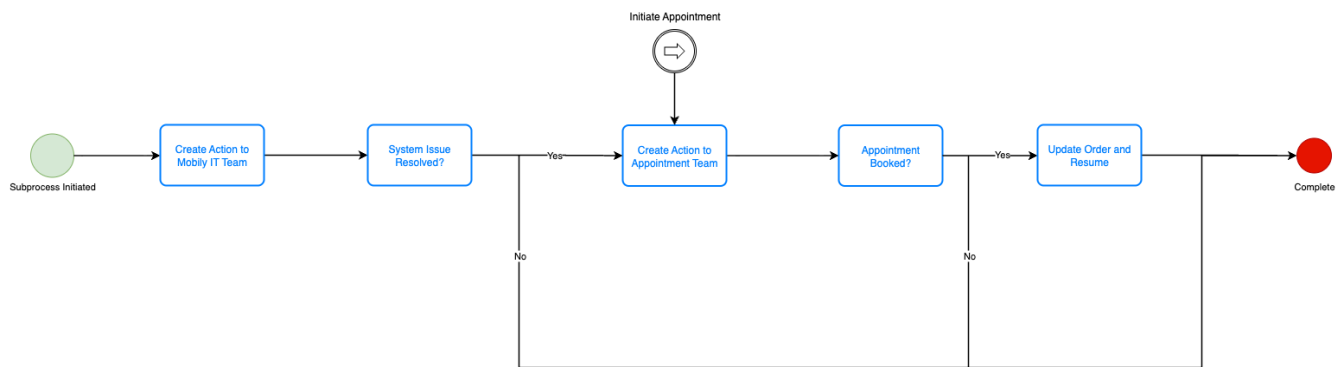
The rejection reasons are categorised into groups and each group shall be handled by distinct treatment. In Telflow, four sub-processes are created to handle each treatment independently such that any business process changes to the treatment in future can be handled with no impact to the other treatments.

Treatment 1 - Network Problem



S. No	Item	Assigned Team	Detail
1	Create Action to Mobily Network Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Mobily network team and awaits for the outcome of the problem resolution.
2	Network Problem Resolved?	Telflow	Telflow validates whether the problem is resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Complete
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Complete
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Complete	Telflow	Completes the process.

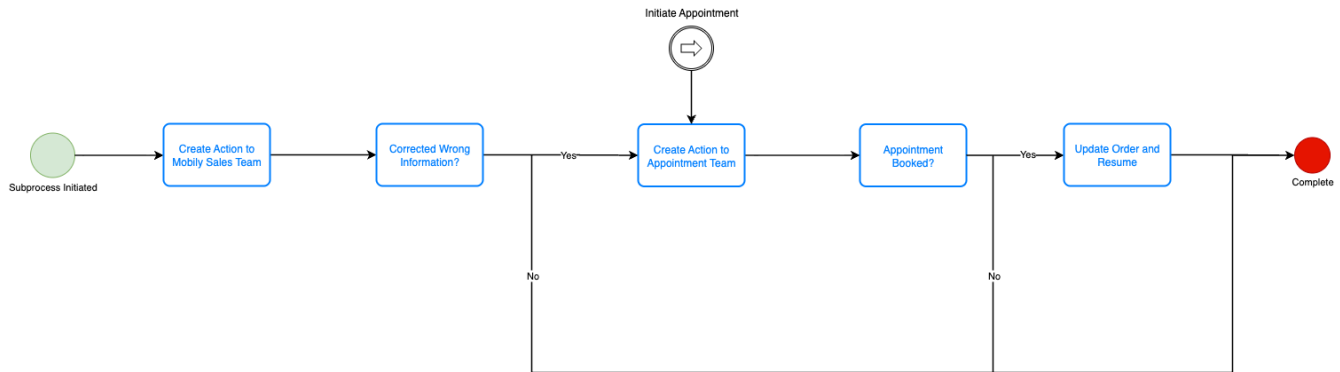
Treatment 2 - System Issues



S. No	Item	Assigned Team	Detail
1	Create Action to Mobily IT Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Mobily IT team and awaits for the outcome of the issue resolution.
2	System Issue Resolved?	Telflow	Telflow validates whether the issue is resolved based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Complete

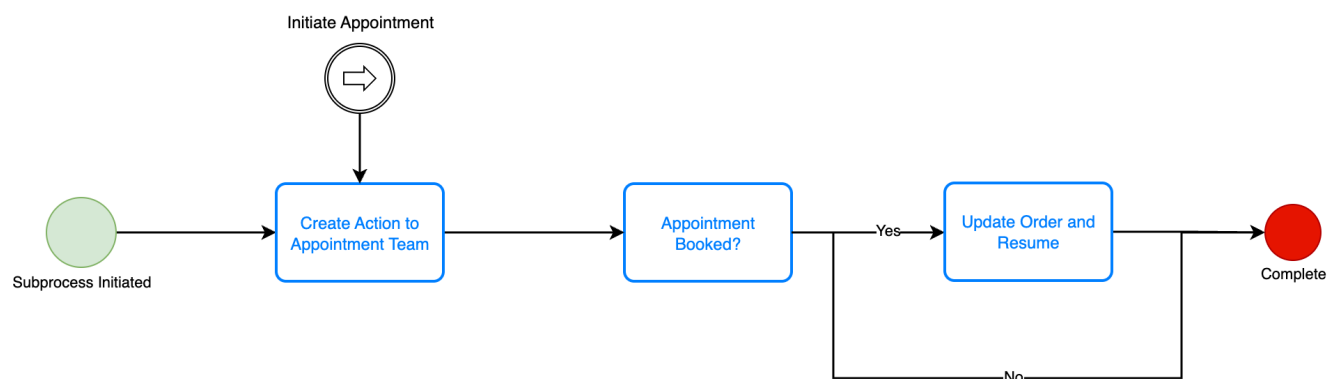
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Complete
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Complete	Telflow	Completes the process.

Treatment 3 - Wrong Information



S. No	Item	Assigned Team	Detail
1	Create Action to Mobily Sales Team	SingleView	Telflow shall initiate an action in SingleView which assigns to Mobily network team and awaits for the outcome of the information update.
2	Corrected Wrong Information?	Telflow	Telflow validates whether the information had been corrected based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Create Action to Appointment Team 2. No Follow step 6: Complete
3	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment. In addition, SingleView sends the updated order data that has to be captured in Telflow.
4	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 5: Update Order and Resume 2. No Follow step 6: Complete
5	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
6	Complete	Telflow	Completes the process.

Treatment 4 - Customer Rejection



S. No	Item	Assigned Team	Detail
1	Create Action to Appointment Team	SingleView	Telflow shall initiate an action in SingleView which assigns to appointment team and awaits for the outcome of the appointment.
2	Appointment Booked?	Telflow	Telflow validates whether the appointment is booked based on the asynchronous response from the SingleView. Possible outcomes: 1. Yes Follow step 3: Update Order and Resume 2. No Follow step 4: Complete
3	Update Order and Resume	Telflow	Telflow updates and resumes the order with the latest data.
4	Complete	Telflow	Completes the process.

Standard Integration Process

The standard integration process delivered by the product shall be used to enable the synchronous/asynchronous communication from Telflow to external systems.

Standard Exception Process

The standard exception process shall be initiated by the fulfilment process under certain circumstances as stated below,

1. When Telflow doesn't receive the response from the external system within the configured timeout duration. (Refer Note - point 2)
2. When Telflow receives the failure response from the external system. (Refer Note - point 3)

The exception process creates an exception task with the below options and the user must complete the task for the order to proceed further.

1. Skip Skips the current activity and proceeds the order further.
2. Retry Resend the request to the external system.
3. Continue Continue waiting for the external system notifications.

S.No	Scenario	Task Operations
1	Timeout occurred in Telflow while waiting for the notification response from the external system.	1. Skip 2. Continue
2	Telflow receives failure response from the external system for the initiated request.	1. Retry 2. Skip

The error code and the error description sent by the external systems shall be captured in the exception task for reference.

Note: (to be updated)

1. The standard exception process delivered by the product shall be reused/enhanced to handle the exception scenarios.
2. The timeout duration for the external system response to be configured based on the business and the external system requirements.
3. The external system failures to handle to be decided.
4. The due date and jeopardy calculation for the exception task to be decided based on the business requirement.

TMF 622 Product Order Notifications

During order processing, Telflow shall notify SingleView periodically the latest status of the order via TMF 622 Product Order notifications. Below are the stages/activities considered during which Telflow sends notifications to SingleView,

S. No	System	Activity	Telflow Order State	Order Sub-State Mapping		TMF 622 Notification Triggered (Y/N)	SingleView Action API		
				Telflow Order Sub-State	Mapped External System Order Status / message		Created	Updated	Closed
1	WFM	CPE Installation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	
			In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
			In Progress	Installation Completed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
		UAT	In Progress	Pending UAT	-	Y		Y	
			In Progress	UAT Completed Note: Telflow expects the async notification (for TMF 622) from SingleView to progress the order.	COMPLETED	Y			Y
		ME Installation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	✗ No Note: This notification shall not be triggered since Telflow already sends the 'Pending Visit' notification to SingleView when the CPE installation started. From Telflow perspective, there is no order sub-state change, hence notification shall not be triggered.	Y		
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	
			In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
			In Progress	-	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	Completed	COMPLETED	No			Y
		Relocation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	

		('X' to Mobily), Rewiring	In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
			In Progress	Completed	COMPLETED	Y			Y
		Customer Visit	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y			
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	
			In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
			In Progress	Completed	COMPLETED	Y		Y	
			In Progress	On Hold	COMPLETED (failed with reason code and text) Note: This is just for understanding purpose.	Y		Y	
		Device Swap	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	
			In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
			In Progress	Installation Completed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	On Hold	COMPLETED (with reason code and text) Note: This is just for understanding purpose.	Y			Y
			In Progress	Pending UAT	-	Y		Y	
			In Progress	UAT Completed	COMPLETED	Y			Y
		ITC Service Installation / Activation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Create Task	Create Task	No		Y	
			In Progress	Assigned	Assigned	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	Set Off	Set Off	No		Y	

2	Open Access		In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	Process	Process Note: Telflow can ignore this status.				
			In Progress	Installation Completed	Success	No			Y
			In Progress	Leave	Leave Note: Telflow can ignore this status.				
		Dawiyat Service Installation / Activation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	InProgress	InProgress	No		Y	
			In Progress	Acknowledged	Acknowledged	No		Y	
			In Progress	InProgress-Dispatch	InProgress-Dispatch	No		Y	
			In Progress	InProgress-Departure	InProgress-Departure	No		Y	
			In Progress	InProgress-Arrival	InProgress-Arrival	No		Y	
			In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	InProgress-HAG Activation-Successful	InProgress-HAG Activation-Successful	No		Y	
			In Progress	Installation Completed	Completed	No			Y
		STC Service Installation / Activation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	WO Created in WFMS	WO Created in WFMS	No		Y	
			In Progress	Technician Assignment	Technician Assignment	No		Y	
			In Progress	Technician Working	Technician Working	No		Y	
			In Progress	Activate ONT	Activate ONT	No		Y	
			In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	Installation Completed	Closed	No			Y
		ACES Service Installation / Activation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)	Y	Y		
			In Progress	Received	Received	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	In Progress	In Progress	No		Y	

			In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	Partial Completed	Partial Completed	No		Y	
			In Progress	Installation Completed	Completed	No			Y
		Service Qualification (Applicable only for STC)	In Progress	Feasibility Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Feasibility InProgress	Ordered	No			
			In Progress	Feasibility Completed	Completed (and FeasibilityStatus = "FEASIBLE") Note: Reservation Id shall be sent in this notification.	No			
			In Progress	Feasibility Closed	Closed (and FeasibilityStatus = "FEASIBLE")	No			
			Cancelled Note: 1. The order shall be transitioned to Pre-Cancellation and then the final state as Cancelled. 2. Telflow expects the async notification from SingleView for Pre-Cancellation to progress the order.	-	Completed / Closed (and FeasibilityStatus = "INFEASIBLE")	Y			
		ITC Service Modification, Order details update (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Note: For order details update, no need to update the order sub-state in Telflow.	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed Note: For order details update, no need to update the order sub-state in Telflow.	Success (Order Status and Provider Status)	Y			
		Dawiyat Service Modification, Order details update (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Note: For order details update, no need to update the order sub-state in Telflow.	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			

			In Progress	Completed	Completed	Y			
				Note: For order details update, no need to update the order sub-state in Telflow.	(Order Status, Provider Status and Provider Milestone)				
		STC Service Modification (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	-	No			
					Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.				
			In Progress	Completed	Closed	Y			
					(Order Status and Provider Status)				
		ACES Service Modification	In Progress	Pending	-	No			
					Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)				
			In Progress	Completed	Completed	Y			
		ITC Service Termination (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	-	No			
					Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.				
			In Progress	Completed	Success	Y			
					(Order Status and Provider Status)				
		Dawiyat Service Termination (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	-	No			
					Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.				
			In Progress	Completed	Completed	Y			
					(Order Status, Provider Status and Provider Milestone)				
		STC Service Termination (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	-	No			
					Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.				
			In Progress	Completed	Closed	Y			
					(Order Status and Provider Status)				
		ACES Service Termination	In Progress	Pending	-	No			
					Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)				
			In Progress	Completed	Completed	Y			

		ITC Suspend Service (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed	Success (Order Status and Provider Status)	Y			
		Dawiyat Suspend Service (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Completed	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y			
			In Progress	Completed	Completed (Order Status, Provider Status, and Provider Milestone)	No			
		STC Suspend Service (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed	Closed (Order Status and Provider Status)	Y			
		ACES Suspend Service	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)	No			
			In Progress	Completed	Completed	Y			
		ITC Resume Service (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed	Success (Order Status and Provider Status)	Y			
		Dawiyat Resume Service (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Completed	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y			
			In Progress	Completed	Completed (Order Status, Provider Status, and Provider Milestone)	No			
		STC Resume Service (Check with Abdalla	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			

		whether failed notification is applicable)	In Progress	Completed	Closed (Order Status and Provider Status)	Y			
		ACES Resume Service	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)	No			
			In Progress	Completed	Completed	Y			
		ITC AddOn Provisioning (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed	Success (Order Status and Provider Status)	Y			
		Dawiyat AddOn Provisioning (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed	Completed (Order Status, Provider Status and Provider Milestone)	Y			
		STC AddOn Provisioning (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	No			
			In Progress	Completed	Closed (Order Status and Provider Status)	Y			
		ACES AddOn Provisioning	In Progress	Pending	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)	No			
			In Progress	Completed	Completed	Y			
		ITC Device Swap Provisioning (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Create Task	Create Task	No		Y	
			In Progress	Assigned	Assigned	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	Set Off	Set Off	No		Y	

			In Progress	Installation Completed	Success	No			Y
		Dawiyat Device Swap Provisioning (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	InProgress	InProgress	No		Y	
			In Progress	Acknowledged	Acknowledged	No		Y	
			In Progress	InProgress-Dispatch	InProgress-Dispatch	No		Y	
			In Progress	InProgress-Departure	InProgress-Departure	No		Y	
			In Progress	InProgress-Arrival	InProgress-Arrival	No		Y	
			In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number. Dawiyat sends ONT serial number (and not HAG similar to installation) during Device Swap. If the implementation is complex, we can ignore this status post confirming with Abdalla.	No		Y	
			In Progress	InProgress-HAG Activation-Successful	InProgress-HAG Activation-Successful	No		Y	
			In Progress	Installation Completed	Completed	No			Y
		STC Device Swap Provisioning (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	WO Created in WFMS	WO Created in WFMS	No		Y	
			In Progress	Technician Assignment	Technician Assignment	No		Y	
			In Progress	Technician Working	Technician Working	No		Y	
			In Progress	Activate ONT	Activate ONT	No		Y	
			In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number. STC sends ONT serial number (and not HAG similar to installation) during Device Swap. If the implementation is complex, we can ignore this status post confirming with Abdalla.	No		Y	
			In Progress	Installation Completed	Closed	No			Y

		ACES Device Swap Provisioning	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)	Y	Y		
			In Progress	Received	Received	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	In Progress	In Progress	No		Y	
			In Progress	Device Installed	- Note: Sets the Telflow Order Sub-State on receiving the device serial number.	No		Y	
			In Progress	Partial Completed	Partial Completed	No		Y	
			In Progress	Installation Completed	Completed	No			Y
		ITC Relocation (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Create Task	Create Task	No		Y	
			In Progress	Assigned	Assigned	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	Set Off	Set Off	No		Y	
			In Progress	Installation Completed	Success	No			Y
		Dawiyat Relocation, Rewiring (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	InProgress	InProgress	No		Y	
			In Progress	Acknowledged	Acknowledged	No		Y	
			In Progress	InProgress-Dispatch	InProgress-Dispatch	No		Y	
			In Progress	InProgress-Departure	InProgress-Departure	No		Y	
			In Progress	InProgress-Arrival	InProgress-Arrival	No		Y	
			In Progress	InProgress-HAG Activation-Successful	InProgress-HAG Activation-Successful	No		Y	
			In Progress	Installation Completed	Completed	No			Y
		STC Relocation (Check with Abdalla whether failed notification is applicable)	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	WO Created in WFMS	WO Created in WFMS	No		Y	
			In Progress	Technician Assignment	Technician Assignment	No		Y	
			In Progress	Technician Working	Technician Working	No		Y	
			In Progress	Activate ONT	Activate ONT	No		Y	
			In Progress	Installation Completed	Closed	No			Y

		ACES Relocation / Rewiring	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub- State based on Synchronous acknowledgement. (Acknowledged)	Y	Y		
			In Progress	Received	Received	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	In Progress	In Progress	No		Y	
			In Progress	Partial Completed	Partial Completed	No		Y	
			In Progress	Installation Completed	Completed	No			Y
		ITC Create / Re-Open / Close SR (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub- State based on Synchronous acknowledgement.	No		Y	
			In Progress	Completed	Create / Re-Open Resolved / Rejected & Closed Close Closed This is a synchronous API. Notifications are not applicable.	Y		Y	
		Dawiyat Create / Re- Open / Close SR (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub- State based on Synchronous acknowledgement.	No		Y	
			In Progress	Completed	Create / Re-Open Resolved / Rejected & Closed Close Closed This is a synchronous API. Notifications are not applicable.	Y		Y	
		STC Create / Re-Open / Close SR (Check with Abdalla whether failed notification is applicable)	In Progress	Pending	- Note: Sets the Telflow Order Sub- State based on Synchronous acknowledgement.	No			
			In Progress	Pending	- Note: Sets the Telflow Order Sub- State based on Asynchronous acknowledgement. (On receiving SR number) Note: This notification might not be applicable. To be revisited during implementation.	No		Y	
			In Progress	Completed	Create / Re-Open Resolved / Rejected & Closed Close Closed This is a synchronous API. Notifications are not applicable.	Y		Y	
		ACES Create / Re- Open SR	In Progress	Pending	- Note: Sets the Telflow Order Sub- State based on Synchronous acknowledgement. (Acknowledged)	No		Y	

			In Progress	Completed	Create / Re-Open In Progress / Accepted / Rejected	Y		Y	
3	Nokia FlowOne	Service Order Provisioning	In Progress	Provisioning Started	- Note: Sets the Telflow Order Sub- State based on Synchronous acknowledgement.	No			
			In Progress	Provisioning Acknowledged	acknowledged	No			
			In Progress	Provisioning InProgress	inProgress (phaseNote.id NOTIF_01; phaseNote. message Phase VALIDATION has completed successfully.)	No			
			In Progress	Provisioning Completed Note: Telflow expects the async notification (for TMF 622) from SingleView to progress the order.	completed	Y (No double check for change scenarios)			
			In Progress	Provisioning Failed	failed	No			
4	MapleWave	Update Inventory	In Progress	-	Success	No			
5	SingleView	Schedule Appointment	In Progress	Scheduling Appointment		No			
			In Progress	Appointment Scheduled		No			

Note:

1. The failure order state and substates pertaining to WFM and Open Access shall be covered in their respective 'installation failure' use case HLD document.
2. The state and the substates depicted in the above table are specific to the systems involved and doesn't follow the sequence of order state /substate transitions during order progression.
3. In Telflow, the order substates pertaining to CPE and ME shall be differentiated, so that the users can identify whether the specific substate change is for either CPE/ME. When Mobily exports the data, they should be able to identify whether the substate change is for CPE/ME.
4. For short-lived orders, Telflow shall send the initial and the final order sub-states only to SingleView.
5. Telflow shall not process all the intermediate notifications sent by Open Access rather only the selected notifications that are mentioned in the above table. The intermediate notifications shall be logged in Telflow logs for reference purpose and for the operation team.
6. Telflow shall follow API based TMF 622 notifications (Async) to SingleView for the below order substates. The reason is based on these order substates, SingleView shall perform actions from its end (e.g., enable billing) and Telflow must not progress the order without SingleView's confirmation that the order had been updated. If there is an issue with the interface, Telflow shall retry for specified times and if the retries exhausted, then standard exception task shall be created. The operation team shall either retry/skip the request via the exception task in Telflow.
 - a. Provisioning Completed
 - b. UAT Completed
 - c. Pre-Completion
 - d. Pre-Cancellation
 - e. Pre-Failure

Below table depicts the order states / sub-states for the installation failures or negative use cases,

S. No	System	Activity	Telflow Order State	Order Sub-State Mapping		TMF 622 Notification Triggered (Y/N)	SingleView Action API		
				Telflow Order Sub- State	Mapped External System Order Status / message		Created	Updated	Closed
			In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	

1	WFM	CPE Installation, Relocation, Device Swap	In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
			In Progress	On Hold	COMPLETED (failed with reason code and text) Note: This is just for understanding purpose.	Y			Y
		ME Installation	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Ready	Ready	No		Y	
			In Progress	Acknowledged	ACKNOWLEDGED	No		Y	
			In Progress	Accepted	ACCEPTED	No		Y	
			In Progress	Trip Started	TRIP STARTED	No		Y	
			In Progress	Customer Premises	CUSTOMER PREMISES	No		Y	
			In Progress	In Work	IN WORK	No		Y	
		ME Installation (Mobily Infra)	-	-	COMPLETED (failed with reason code and text) Note: This is just for understanding purpose.	No			Y
		ME Installation (Open Access)	In Progress	On Hold	COMPLETED (failed with reason code and text) Note: This is just for understanding purpose.	Y			Y
2	Open Access	ONT Installation (ITC Service Installation), Relocation, Device Swap	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	Create Task	Create Task	No		Y	
			In Progress	Assigned	Assigned	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	Set Off	Set Off	No		Y	
			In Progress	On Hold	Failure (failed with rejection code and rejection description) Note: This is just for understanding purpose.	Y			Y
		ONT Installation (Dawiyat Service Installation), Relocation, Device Swap	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	InProgress	InProgress	No		Y	
			In Progress	Acknowledged	Acknowledged	No		Y	
			In Progress	InProgress-Dispatch	InProgress-Dispatch	No		Y	
			In Progress	InProgress-Departure	InProgress-Departure	No		Y	
			In Progress	InProgress-Arrival	InProgress-Arrival	No		Y	
			In Progress	On Hold	Pending (failed with rejection code and rejection description) Note: This is just for understanding purpose.	Y			Y

		ONT Installation (STC Service Installation), Relocation, Device Swap	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement.	Y	Y		
			In Progress	WO Created in WFMS	WO Created in WFMS	No		Y	
			In Progress	Technician Assignment	Technician Assignment	No		Y	
			In Progress	Technician Working	Technician Working	No		Y	
			In Progress	Activate ONT	Activate ONT	No		Y	
			In Progress	On Hold	ERROR (failed with rejection code and rejection description) Note: This is just for understanding purpose.	Y			Y
		ONT Installation (ACES Service Installation), Relocation, Device Swap	In Progress	Pending Visit	- Note: Sets the Telflow Order Sub-State based on Synchronous acknowledgement. (Acknowledged)	Y	Y		
			In Progress	Received	Received	No		Y	
			In Progress	Accepted	Accepted	No		Y	
			In Progress	In Progress	In Progress	No		Y	
			In Progress	On Hold	Pending (failed with rejection code and rejection description) Note: This is just for understanding purpose.	Y			Y

Below table depicts the order states / sub-states for the cancel order,

S. No	System	Activity	Telflow Order State	Order Sub-State Mapping		TMF 622 Notification Triggered (Y /N)	SingleView Action API		
				Telflow Order Sub-State	Mapped External System Order Status / message		Created	Updated	Closed
1	WFM	Cancel CPE Installation	In Progress	Cancelling Work Order	- Is there an acknowledgement sent by WFM ??? Send mail to Sharanya and Raqib.	No			
			In Progress	Aborted	ABORTED	No			
2	Open Access	Cancel Service Installation (STC)	In Progress						
		Cancel Service Installation (ITC)	In Progress						
		Cancel Service Installation (Dawiyat)	In Progress						
		Cancel Service Installation (ACES)	In Progress						

Below table depicts the order states / sub-states for the inflight order (to be cancelled),

S.No	Telflow Order State	TMF 622 Notification Triggered (Y/N)	SingleView Action API		
			Created	Updated	Closed
1	Assessing Cancellation	No			
2	Pending Cancellation	Y			
3	Cancelled	Y			

Email Notification

Email notification to be sent to Mobily teams/individuals when task had been created in Telflow. The organisation/teams/individuals to be associated with the correct email Id to use this functionality.

Following is the task to be associated with the Email notification preference,

1. Standard Exception Task

Email template to be configured for the aforementioned tasks. By default, Telflow sends email upon creating the task to teams email Id, whereas when the task completes it sends email to individuals email Id.

In addition, when the 'ODB Patching' action had been initiated by Telflow towards SingleView, Email notification shall be triggered to the respective team.

S.No	Organisation Name	Teams	Teams Email Id	Individuals	Individuals Email Id
1	Mobily	WFM Team			
		Provisioning Team			
		Exception Handling Team			

Note:

1. The email Id of the teams to be configured with the 'dummy' email Id. In future, Mobily shall configure with the correct email Id based on the requirement.

Tasks

Standard Exception Task

The standard exception process shall initiate this task when the external systems didn't respond or send failure response. Upon task creation, Telflow sends email notification to the configured team (Exception Handling).

Email Id	Email Template

Note:

1. The email Id of the teams to be configured with the 'dummy' email Id. In future, Mobily shall configure with the correct email Id based on the requirement.

Retry Mechanism

When Telflow sends the request to the external system, it expects success response to progress the order. In case of Telflow receiving failure response or exceptions from the external systems, instead of creating a manual task immediately to address the problem, it retries to send the request for the configured number of times in specific intervals. The manual task shall be created in Telflow once the retry attempts exhaust and the operation team shall decide on the progress of the order.

Integrations

Note:

1. Telflow shall implement and expose the callback URL to the external downstream systems to process the asynchronous notifications.
2. The parameters/data that are to be exchanged between the systems are specified in this document. Additional details to be exchanged if any, will be covered during EICD phase.

SingleView - Telflow (Direct)

TMF 622 (Asynchronous)

As part of FTTH order journey,

1. The order shall be captured in SingleView.
2. SingleView shall generate the access token in Telflow and send the authorisation information in the form of bearer token in the TMF 622 request.
3. Upon receiving the request, Telflow creates a product order and responds with the acknowledgement response. The key details to anticipate in the response are,
 - a. Telflow product order Id
 - b. Telflow Product inventory instance Id
 - c. Product Order state
4. Telflow manages the order and sends the intermediate TMF 622 notification to SingleView during order processing and the completed TMF 622 notification to SingleView post completion of the order. SingleView must register the callback URL through hub (listener) in Telflow to process the TMF 622 product order state change notifications received from Telflow during order provisioning.

The existing TMF 622 Integration Cartridge shall be reused for this integration with requisite enhancement based on the use case requirement. SingleView shall send the TMF 622 product order request as per the definition depicted in the Telflow TMF 622 EICD document.

S. No	API Name	Description
1	Create Product Order	Creates product order in Telflow. The product order request may contain multiple line items to be provisioned, say when the customer had opted for Mesh Extender along with the FTTH service. In that case, Telflow shall receive two line items, 1. FTTH along with CPE details 2. Mesh Extender Key Points: 1. The SingleView order Id shall be sent in the TMF 622 request as '<i>externalId</i>' and this will be captured and persisted against the Telflow product order. 2. The customer and the order item information shall be sent by SingleView in the TMF 622 request.

3. Following details are expected by Telflow from SingleView,
 - a. External ID (SingleView order action ID for correlation purposes)
 - b. Creator (Sales Agent ID, where applicable)
 - c. Ordered Item Details
 - FTTH
 - Service Type
 - This is the type of service to be installed. SingleView can provide the TLO name (e.g. FTTH Postpaid) or the Technical Product ID configured in reference data in SingleView, which can be aligned with the legacy stack names.
 - QoS profile
 - Name of the QoS profile as defined in SingleView reference data.
 - Service Qualification (Feasibility) ID
 - The ID returned by the service qualification check result, if available. This is usually available for open access providers, except STC.
 - ODB Details
 - ID
 - Provider
 - Latitude, Longitude
 - Region
 - Appointment ID, Slot ID, Appointment Start and Appointment End
 - Address Id and Reserve Id (for ACES)
 - Service Request Details
 - SR Number
 - Description
 - Type
 - Area
 - Sub Area
 - Network Category
 - Supported values are,
 - For normal customers FTTH Consumer
 - For RCY customers FTTH RCY
 - Router
 - Device Name (Offer Name or Technical Product ID of the device child offer, for the CPE device to be installed.)
 - Integration ID (ID of the router order item)
 - Mesh Extender
 - Number of mesh extenders which may be provided according to the SingleView product catalogue.
 - Integration ID (ID of the mesh extender order item)
 - d. Customer Details
 - Billing Account Number (Primary Account or SAN)
 - Customer Account Number
 - Customer Id
 - Customer Type
 - Customer Category
 - Contact Name
 - Contact Email, Number
 - Secondary Contact Number
 - Installation Address
 - ZipCode
 - Building Name
 - Street Name
 - VIP Flag
 - ~~SingleView will pass a null value for VIP Flag as there is no use case currently for setting this flag on the downstream work orders.~~
 - e. For RCY FTTH activation following are the considerations,
 - i. Infra provider value shall be Mobily.
 - ii. Customer category follows the existing values like Regular, Royal, Vip, etc.,
 - iii. Network category 'FTTH RCY' shall be passed from SV to Telflow and Telflow can follow the RCY infra path in the fulfilment process based on this value.
 - iv. Telflow shall send the 'isRcy' flag to Nokia FlowOne based on the Network Category -> FTTH RCY.

4. Following details must be specified in the TMF 622 request for FTTH and FV service.

Use Case	Parameter	Value
Activation	Action	add
	FTTH SubAction	Activate
Rewiring	Action	modify
	FTTH SubAction	Rewiring
Relocation	Action	modify
	FTTH SubAction	Relocation
Device Swap	Action	modify
	FTTH SubAction	CPESwap / HAGSwap / ONTSwap
Bandwidth Upgrade / Downgrade	Action	modify
	FTTH SubAction	Upgrade / Downgrade (ModifyQoS ???)
Negative Scenario	Action	add
Termination	Action	delete
	FTTH SubAction	Deactivate
Suspend	Action	modify
	FTTH SubAction	Suspend
Resume	Action	modify
	FTTH SubAction	Resume
Bar	Action	modify
	FV SubAction	Bar
UnBar	Action	modify
	FV SubAction	UnBar
FV Termination	Action	modify
	FTTH SubAction	Unchanged
	FV SubAction	Delete

2	Patch Product Order	Patches the existing inflight product order in Telflow. During patch, the following details shall be updated, 1. Appointment Id 2. Contact number (Not to be addressed in 4A) 3. Service information (Not to be addressed in 4A) SingleView must send the Telflow product inventory Id in the patch request to update the existing inflight order and their respective inventory.
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Telflow - ESB

Workforce Management

Bespoke Integration (Asynchronous)

During order provisioning, Telflow shall send the CPE/ME installation request to WFM via ESB for installing the CPE/ME in the customer premise. This integration is a REST based interface, asynchronous and of non-TMF standard (OAuth based custom integration). Based on the response from WFM, the order shall proceed further.



Following are the applicable APIs,

S. No	API Name	Expected WFM Notifications	Description																				
1	Task Creation Request	Following are the applicable notification status, 1. OK 2. Ready 3. Acknowledged 4. Accepted 5. Trip Started 6. Customer Premises 7. In Work	To install CPE/ME in the customer premises. Key Points: 1. Engineer details shall be received by Telflow in the 'Acknowledged' notification which shall be cascaded to SingleView. 2. The CPE/ME serial number shall be received by Telflow post 'In Work' notification and the process will proceed further to provision the FTTH service in the network. 3. The Telflow order Id (as Sales order Id) and the WFM work order Id for CPE/ME shall be sent in the 'Task Creation' request. If the customer has opted for Mesh Extender, then the 'parent' Id shall also be sent in the request. 4. In addition, appointment Id, ODB details, customer location, customer information and equipment information shall be sent in the 'Task Creation' request. 5. The standalone mesh request must be created towards WFM in case, a. If the customer opts for Mesh Extender during FTTH activation for Open Access provider. b. If the customer buys Mesh Extender post purchase of FTTH service. 6. WFM notifies Telflow the CPE/ME installation failure along with the 'reason code', reason text' and 'comments' if applicable. 7. Telflow initiates the treatment based on the 'reason code' and reason text' received from WFM. 8. For Device swap, old serial number shall be sent in the request. 9. Rewiring scenario is supported in WFM via 'Relocate' order type request. 10. Device serial number notification is not applicable for the Rewiring/Relocation request. Post 'In Work' notification, WFM sends the 'Completed' notification. 11. For maintenance work order, a. service request details shall be sent in the request. Check with Abdalla on the applicable notifications. b. Device serial number notification is not applicable. c. Sales order Id and SR number must be same in the request. (Check internally on how the responses can be co-related since Telflow order Id is not being passed in the request). d. If WFM respond to Telflow with the Device Swap related reason code/text, Telflow shall ignore this and completes the order successfully. i. resolution code 2064 ii. resolution text Wrong Info - Wrong Package 12. For the device swap during service installation, a. resolution code 2030 b. resolution text Device Provisioning Request Note: For Mobily Infra, Installation failure scenario is not applicable for the Mesh Extender, meaning Telflow shall continue processing the order even if the Mesh Extender installation fails. <table><tr><th>S. No</th><th colspan="2">Scenario</th><th>Outcome</th><th>Comments</th></tr><tr><th></th><th>CPE Installation</th><th>ME Installation</th><th></th><th></th></tr><tr><td>1</td><td>Success</td><td>Failed</td><td>Order can proceed further.</td><td> Following reason codes are expected from WFM for Mesh Extender which can be ignored, 2101 - Mesh Device is not available 2089 - SANAD is rejected by customer 2102 - Customer refused to sign PN 07-Apr-2026 - Based on Abdalla's feedback, in 4A following is the only error code that shall be handled in Telflow for Mesh Extender since the expected error codes 2101 / 2089 / 2102 are not configured in WFM system. In future, when WFM configure these missing error codes, Telflow can handle it accordingly. 2041 - Customer Want to Cancel Telflow, upon receiving this error code from WFM, ignore this and completes the order successfully. </td></tr><tr><td>2</td><td>Failed</td><td>Failed</td><td>Initiate treatment for CPE installation failure.</td><td> The functionality of closing the child work order when the parent work order closes is not yet implemented in WFM. Hence, Telflow can ignore the 'failed' notification sent by WFM for Mesh Extender work order. </td></tr></table> In case of Open Access, Installation failure scenario is applicable for the Mesh Extender except the below Mesh Extender specific reason codes, 2101 - Mesh Device is not available 2089 - SANAD is rejected by customer 2102 - Customer refused to sign PN	S. No	Scenario		Outcome	Comments		CPE Installation	ME Installation			1	Success	Failed	Order can proceed further.	Following reason codes are expected from WFM for Mesh Extender which can be ignored, 2101 - Mesh Device is not available 2089 - SANAD is rejected by customer 2102 - Customer refused to sign PN 07-Apr-2026 - Based on Abdalla's feedback, in 4A following is the only error code that shall be handled in Telflow for Mesh Extender since the expected error codes 2101 / 2089 / 2102 are not configured in WFM system. In future, when WFM configure these missing error codes, Telflow can handle it accordingly. 2041 - Customer Want to Cancel Telflow, upon receiving this error code from WFM, ignore this and completes the order successfully.	2	Failed	Failed	Initiate treatment for CPE installation failure.	The functionality of closing the child work order when the parent work order closes is not yet implemented in WFM. Hence, Telflow can ignore the 'failed' notification sent by WFM for Mesh Extender work order.
S. No	Scenario		Outcome	Comments																			
	CPE Installation	ME Installation																					
1	Success	Failed	Order can proceed further.	Following reason codes are expected from WFM for Mesh Extender which can be ignored, 2101 - Mesh Device is not available 2089 - SANAD is rejected by customer 2102 - Customer refused to sign PN 07-Apr-2026 - Based on Abdalla's feedback, in 4A following is the only error code that shall be handled in Telflow for Mesh Extender since the expected error codes 2101 / 2089 / 2102 are not configured in WFM system. In future, when WFM configure these missing error codes, Telflow can handle it accordingly. 2041 - Customer Want to Cancel Telflow, upon receiving this error code from WFM, ignore this and completes the order successfully.																			
2	Failed	Failed	Initiate treatment for CPE installation failure.	The functionality of closing the child work order when the parent work order closes is not yet implemented in WFM. Hence, Telflow can ignore the 'failed' notification sent by WFM for Mesh Extender work order.																			
2	Update Order Complete	Completed	Notify WFM that the provisioning is success and initiate UAT. The 'Completed' notification from WFM represents that the work order is completed with either success/failure.																				

3	Task Cancellation Request	Following are the applicable notification status, 1. OK ?? 2. Aborted Any failure notifications applicable? Check with Abdalla.	To cancel the work order in WFM. Note: 1. Cancelling the CPE work order implicitly cancels the child work order if any.
4	Reschedule Appointment		To reschedule the appointment in WFM. This is a synchronous API call and WFM sends the appointment Id in the response.

Nokia FlowOne

TMF 641 (Asynchronous)

The product order that's been created in Telflow shall be decomposed into TMF 641 service order and send it to NFO to provision the FTTH service in the network. NFO sends both intermediate and completed TMF 641 notifications to Telflow that shall be handled and notified to SingleView thus communicating the latest update on the order.

The authorisation information shall be handled by ESB internally and hence no need to send the information while initiating the TMF 641 service order request. The existing TMF 641 Integration Cartridge shall be reused for this integration with requisite enhancement based on the NFO EICD document. ESB sends the TMF 641 service order state change notification to the callback URL exposed by Telflow thus communicating the latest status of the service order.

Following are the applicable APIs,

S. No	API Name	Expected Nokia FlowOne Notifications	Description
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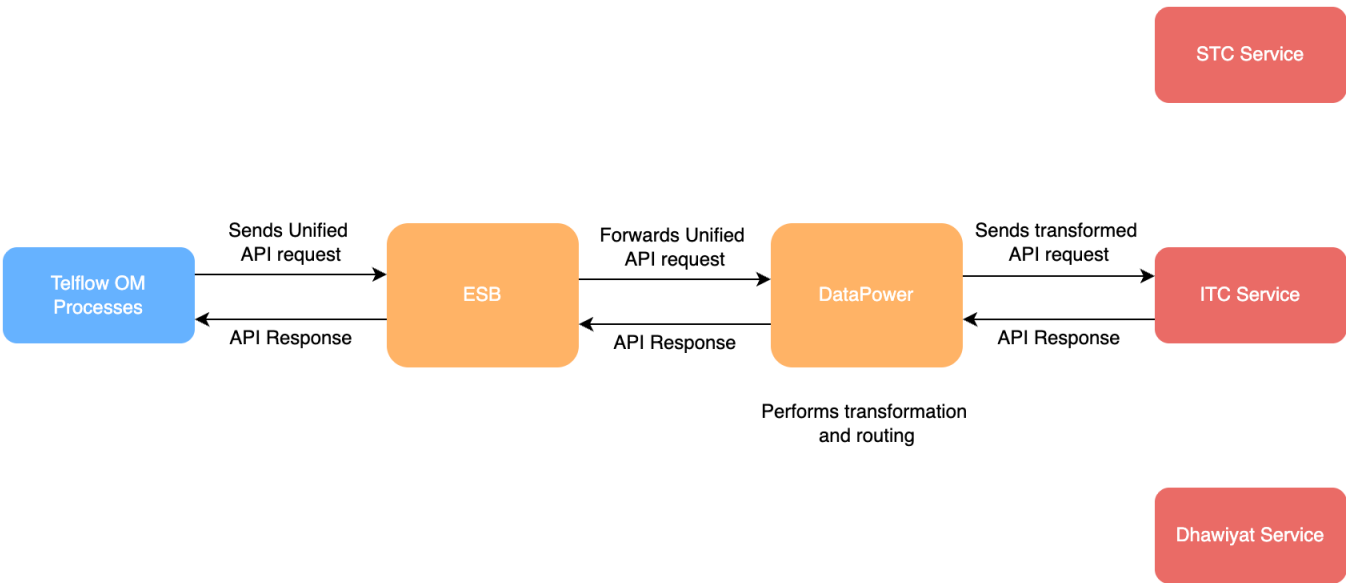
1	Create Service Order	Following are the applicable notification status, 1. Acknowledged 2. InProgress 3. Completed 4. Failed	Provisions FTTH service in the core network. Key Points: 1. The Telflow order Id shall be sent in the TMF 641 request as 'externalId' to NFO. 2. The customer, ODB, equipment and the service related information shall be sent in the TMF 641 request. 3. For Royal RCY customers, 'isRcy' flag must be sent in the request. 4. In the synchronous acknowledgement, NFO sends the service order Id and the same shall be persisted in Telflow against the product inventory. 5. Following are the CFS defined in NFO, a. Data CFS_FTTH_DATA b. Voice CFS_FTTH_VOICE 6. Provider MOBILY / STC / ITC / DAWIYAT / ACES must be sent in the TMF 641 request for FTTH service. 7. During termination, if the customer has both FTTH and Fixed Voice, Telflow sends both of the details in the request. 8. During suspend/resume scenario, only Fixed Voice shall be suspended/resumed in the network. 9. Following details must be specified in the TMF 641 request for FTTH service. <table><tr><th>Use Case</th><th>Action</th><th>Type</th></tr><tr><td>FTTH Activation</td><td>Activate</td><td>Provide</td></tr><tr><td>Device Swap</td><td>CPESwap / HAGSwap / ONTSwap</td><td>Modify</td></tr><tr><td>Bandwidth Upgrade / Downgrade</td><td>Upgrade / Downgrade (ModifyQoS ?)</td><td>Modify</td></tr><tr><td>FTTH Termination</td><td>Deactivate</td><td>Cease</td></tr><tr><td>FTTH Suspend</td><td>Suspend</td><td>Modify</td></tr><tr><td>FTTH Resume</td><td>Resume</td><td>Modify</td></tr><tr><td>FV Suspend</td><td>Bar</td><td>Modify</td></tr><tr><td>FV Resume</td><td>UnBar</td><td>Modify</td></tr><tr><td>FV Termination</td><td>Delete</td><td>Cease</td></tr><tr><td>FTTH Relocation</td><td>Relocation</td><td>Modify</td></tr><tr><td>Package Conversion (Postpaid to Prepaid)</td><td>PostToPre</td><td>Modify</td></tr><tr><td>Package Conversion (Prepaid to Postpaid)</td><td>PreToPost</td><td>Modify</td></tr><tr><td>FV Number Change</td><td>NumberSwap</td><td>Modify</td></tr><tr><td>Change Owner</td><td>ChangeOwner</td><td>Modify</td></tr></table>	Use Case	Action	Type	FTTH Activation	Activate	Provide	Device Swap	CPESwap / HAGSwap / ONTSwap	Modify	Bandwidth Upgrade / Downgrade	Upgrade / Downgrade (ModifyQoS ?)	Modify	FTTH Termination	Deactivate	Cease	FTTH Suspend	Suspend	Modify	FTTH Resume	Resume	Modify	FV Suspend	Bar	Modify	FV Resume	UnBar	Modify	FV Termination	Delete	Cease	FTTH Relocation	Relocation	Modify	Package Conversion (Postpaid to Prepaid)	PostToPre	Modify	Package Conversion (Prepaid to Postpaid)	PreToPost	Modify	FV Number Change	NumberSwap	Modify	Change Owner	ChangeOwner	Modify
Use Case	Action	Type																																														
FTTH Activation	Activate	Provide																																														
Device Swap	CPESwap / HAGSwap / ONTSwap	Modify																																														
Bandwidth Upgrade / Downgrade	Upgrade / Downgrade (ModifyQoS ?)	Modify																																														
FTTH Termination	Deactivate	Cease																																														
FTTH Suspend	Suspend	Modify																																														
FTTH Resume	Resume	Modify																																														
FV Suspend	Bar	Modify																																														
FV Resume	UnBar	Modify																																														
FV Termination	Delete	Cease																																														
FTTH Relocation	Relocation	Modify																																														
Package Conversion (Postpaid to Prepaid)	PostToPre	Modify																																														
Package Conversion (Prepaid to Postpaid)	PreToPost	Modify																																														
FV Number Change	NumberSwap	Modify																																														
Change Owner	ChangeOwner	Modify																																														

Data Power (Open Access Service Provider)

Bespoke Integration (Asynchronous)

During order provisioning, Telflow shall send the service installation request to Open Access for installing the service in the Open Access network. Data Power acts as an API aggregator between Telflow/ESB and Open Access provider. It performs the request transformation based on the infra provider details sent by Telflow in the API request and route the request to the requisite Open Access provider.

This integration is a REST based interface, asynchronous and of non-TMF standard (OAuth based custom integration).



Following are the applicable APIs,

S. No	API Name	Infrastructure Provider	Expected Open Access Notifications	Description																																													
1	Service Qualification	STC	Following are the applicable notification status, 1. Ordered 2. Completed 3. Closed	This API shall be invoked by Telflow to perform service qualification if the infra provider is STC. Key Points: - Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. - Telflow must send STC Order Id in the service qualification request as per the below format, a. F-MY<8-digit number> - Reservation/Feasibility Id shall be sent by Open Access provider in the 'Completed' notification. Note: - In case of bandwidth change, no need to perform service qualification for STC provider.																																													
2	Service Installation Request	STC	Following are the applicable notification status, 1. WO Created in WFMS 2. Technician Assignment 3. Technician Working 4. Activate ONT 5. Closed	This API shall be invoked by Telflow to perform service installation in the STC network. Key Points: - Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. - Telflow must send STC Order Id in the Installation request as per the below format, a. I-MY<8-digit number> - In the synchronous acknowledgement response, STC doesn't send their internal order Id. - ONT and/or HAG serial number is being sent by STC in the intermediate notification (post 'Activate ONT'). ONT serial number shall be persisted in Telflow as additional info. - Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView. Note: - STC sends notifications for the line items that starts with F001-*, F002-* and F005-*. Telflow shall process the notifications of the line item F001-* ignoring the others. - In the notification received from Data Power, Telflow shall parse the following fields to determine whether the notification must be processed or not. <table><tr><th rowspan="2">S. No</th><th colspan="4">Conditions</th><th>Match and Extract</th></tr><tr><th>Order Status =</th><th>Line Item Identifier =</th><th>Line Item Status =</th><th>Provider Status =</th><th>Provider Milestone</th></tr><tr><td>1</td><td>ORDERED</td><td>F001-*</td><td>IN-PROCESS</td><td>ORDERED</td><td>WO Created in WFMS</td></tr><tr><td>2</td><td>ORDERED</td><td>F001-*</td><td>READY</td><td>ORDERED</td><td>TECHNICIAN ASSIGNMENT</td></tr><tr><td>3</td><td>ORDERED</td><td>F001-*</td><td>READY</td><td>ORDERED</td><td>Technician Working</td></tr><tr><td>4</td><td>ORDERED</td><td>F001-*</td><td>READY</td><td>ORDERED</td><td>ACTIVATE ONT</td></tr><tr><td>5</td><td>Closed</td><td></td><td>Closed</td><td>Closed</td><td></td></tr></table> Note: Updated as per samples shared by Decoupling team. Should validate during testing. Telflow must extract the serial number by matching the below data in the received notification, <table><tr><td>FuncId =</td><td>Provider Milestone =</td></tr><tr><td>STC_CPE_UPDATE</td><td>TEST_FTTH_LINK</td></tr></table>	S. No	Conditions				Match and Extract	Order Status =	Line Item Identifier =	Line Item Status =	Provider Status =	Provider Milestone	1	ORDERED	F001-*	IN-PROCESS	ORDERED	WO Created in WFMS	2	ORDERED	F001-*	READY	ORDERED	TECHNICIAN ASSIGNMENT	3	ORDERED	F001-*	READY	ORDERED	Technician Working	4	ORDERED	F001-*	READY	ORDERED	ACTIVATE ONT	5	Closed		Closed	Closed		FuncId =	Provider Milestone =	STC_CPE_UPDATE	TEST_FTTH_LINK
S. No	Conditions					Match and Extract																																											
	Order Status =	Line Item Identifier =	Line Item Status =	Provider Status =	Provider Milestone																																												
1	ORDERED	F001-*	IN-PROCESS	ORDERED	WO Created in WFMS																																												
2	ORDERED	F001-*	READY	ORDERED	TECHNICIAN ASSIGNMENT																																												
3	ORDERED	F001-*	READY	ORDERED	Technician Working																																												
4	ORDERED	F001-*	READY	ORDERED	ACTIVATE ONT																																												
5	Closed		Closed	Closed																																													
FuncId =	Provider Milestone =																																																
STC_CPE_UPDATE	TEST_FTTH_LINK																																																

		ITC	Following are the applicable notification status, 1. Create Task 2. Assigned 3. Accepted 4. Set Off 5. Success 6. Leave	This API shall be invoked by Telflow to perform service installation in the ITC network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. 3. ONT/HAG serial number is being sent by ITC in the intermediate notification (post 'Set Off'). 4. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
		Dawiyat	Following are the applicable notification status, 1. InProgress 2. Acknowledged 3. InProgress-Dispatch 4. InProgress-Departure 5. InProgress-Arrival 6. InProgress-HAG Activation Successful 7. Completed	This API shall be invoked by Telflow to perform service installation in the Dawiyat network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. 3. ONT/HAG serial number is being sent by Dawiyat in the intermediate notification (post 'InProgress-Arrival'). 4. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
		ACES	Following are the applicable notification status, 1. Acknowledged 2. Received 3. Accepted 4. InProgress 5. PartialCompleted 6. Completed	This API shall be invoked by Telflow to perform service installation in the ACES network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ACES sends 'Provider Service Number'. 3. ONT/HAG serial number is being sent by ACES in the intermediate notification (post 'InProgress'). 4. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
3	Rewiring Request	Dawiyat	Note: Rewiring is not applicable for STC / ITC. Following are the applicable notification status, 1. InProgress 2. Acknowledged 3. InProgress-Dispatch 4. InProgress-Departure 5. InProgress-Arrival 6. InProgress-HAG Activation Successful 7. Completed <Abdalla to provide the details, send mail to Venkat>	This API shall be invoked by Telflow to perform rewiring in the Open Access provider network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. 3. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView. 4. Device serial number notification is not applicable for the Rewiring request.
		ACES	Following are the applicable notification status, 1. Acknowledged 2. Received 3. Accepted 4. InProgress 5. PartialCompleted 6. Completed	This API shall be invoked by Telflow to perform rewiring in the Open Access provider network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ACES doesn't send their internal order Id. 3. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView. 4. Device serial number notification is not applicable for the Rewiring request.
4	Device Swap	STC	Following are the applicable notification status, 1. WO Created in WFMS 2. Technician Assignment 3. Technician Working 4. Activate ONT 5. Closed	This API shall be invoked by Telflow to perform device swap in the STC network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. Telflow must send STC Order Id in the Installation request as per the below format, a. 'C'-MY<8-digit number>_1 (Format based on Mobily confirmation) 3. In the synchronous acknowledgement response, STC doesn't send their internal order Id. 4. ONT serial number is being sent by STC in the intermediate notification (post 'Activate ONT'). ONT serial number shall be persisted in Telflow as additional info. 5. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
		ITC	Following are the applicable notification status, 1. Create Task 2. Assigned 3. Accepted 4. Set Off 5. Success 6. Leave	This API shall be invoked by Telflow to perform device swap in the ITC network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. 3. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView. 4. ITC doesn't send the ONT serial number in the notification.
		Dawiyat	Following are the applicable notification status, 1. InProgress 2. Acknowledged 3. InProgress-Dispatch 4. InProgress-Departure 5. InProgress-Arrival 6. InProgress-HAG Activation Successful 7. Completed	This API shall be invoked by Telflow to perform device swap in the Dawiyat network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. 3. ONT serial number is being sent by Dawiyat in the intermediate notification (post 'InProgress-Arrival'). 4. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.

		ACES	Following are the applicable notification status, - Acknowledged - Received - Accepted - InProgress - PartialCompleted - Completed	This API shall be invoked by Telflow to perform service installation in the ACES network. Key Points: - Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. - In the synchronous acknowledgement response, ACES sends 'Provider Service Number'. - ONT/HAG serial number is being sent by ACES in the intermediate notification (post 'InProgress'). - Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
5	Service Modification Request (bandwidth change, FTTH AddOn)	STC	Following are the applicable notification status, Closed	This API shall be invoked by Telflow to modify service in the STC network. Key Points: - Telflow order Id, ODB, old/new bandwidth, etc., shall be sent in the request. - Telflow must send STC Order Id in the modification request as per the below format, - C-MY<8-digit number>_1 (Based on Mobily confirmation) - In the synchronous acknowledgement response, STC doesn't send their internal order Id.
		ITC	Following are the applicable notification status, Success	This API shall be invoked by Telflow to modify service in the ITC network. Key Points: - Telflow order Id, ODB, old/new bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, ITC sends 'Provider Service Number'.
		Dawiyat	Following are the applicable notification status, Completed	This API shall be invoked by Telflow to modify service in the Dawiyat network. Key Points: - Telflow order Id, ODB, old/new bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id.
		ACES	Following are the applicable notification status, Completed	This API shall be invoked by Telflow to modify service in the ACES network. Key Points: - Telflow order Id, ODB, old/new bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, ITC sends 'Provider Service Number'.
6	Service Termination Request	STC	Following are the applicable notification status, Closed	This API shall be invoked by Telflow to terminate service in the STC network. Key Points: - Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. - Telflow must send STC Order Id in the modification request as per the below format, - O-MY<8-digit number> - In the synchronous acknowledgement response, STC doesn't send their internal order Id. - During termination, if the customer has both FTTH and Fixed Voice, - Telflow sends both FTTH and FV details as a separate request. (How to handle FV request shall be shared later by Abdalla? Will be covered in Phase 4B.)
		ITC	Following are the applicable notification status, Success	This API shall be invoked by Telflow to terminate service in the ITC network. Key Points: - Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. - During termination, if the customer has both FTTH and Fixed Voice, - Telflow sends only FTTH details in the request.
		Dawiyat	Following are the applicable notification status, Completed	This API shall be invoked by Telflow to terminate service in the Dawiyat network. Key Points: - Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. - During termination, if the customer has both FTTH and Fixed Voice, - Telflow sends only FTTH details in the request.
		ACES	Following are the applicable notification status, Completed	This API shall be invoked by Telflow to terminate service in the ACES network. Key Points: - Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, ACES sends 'Provider Service Number'. - During termination, if the customer has both FTTH and Fixed Voice, - Telflow sends only FTTH details in the request.
7	Suspend Service Request	STC	Following are the applicable notification status, Closed	This API shall be invoked by Telflow to suspend service in the STC network. Key Points: - Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. - Telflow must send STC Order Id in the modification request as per the below format, - D-MY<8-digit number> - In the synchronous acknowledgement response, STC doesn't send their internal order Id. - Telflow suspends the Fixed Voice service in the Open Access network. (to be validated)
		ITC	Following are the applicable notification status, Success	This API shall be invoked by Telflow to suspend service in the ITC network. Key Points: - Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. - In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. - Telflow suspends the Fixed Voice service in the Open Access network. (to be validated)

		Dawiyat	Not Applicable Following are the applicable notification status, 1. Completed	This API shall be invoked by Telflow to terminate service in the Dawiyat network. Key Points: 1. Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. 3. Telflow suspends the Fixed Voice service in the Open Access network.
		ACES	Following are the applicable notification status, 1. Completed	This API shall be invoked by Telflow to suspend service in the ITC network. Key Points: 1. Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ACES sends 'Provider Service Number'. 3. Telflow suspends the Fixed Voice service in the Open Access network. (to be validated)
8	Resume Service Request	STC	Following are the applicable notification status, 1. Closed	This API shall be invoked by Telflow to resume service in the STC network. Key Points: 1. Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. 2. Telflow must send STC Order Id in the modification request as per the below format, a. E-MY<8-digit number> 3. In the synchronous acknowledgement response, STC doesn't send their internal order Id. 4. Telflow resumes the Fixed Voice service in the Open Access network. (to be validated)
		ITC	Following are the applicable notification status, 1. Success	This API shall be invoked by Telflow to resume service in the ITC network. Key Points: 1. Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. 3. Telflow resumes the Fixed Voice service in the Open Access network. (to be validated)
		Dawiyat	Not Applicable Following are the applicable notification status, 1. Completed	This API shall be invoked by Telflow to terminate service in the Dawiyat network. Key Points: 1. Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. 3. Telflow resumes the Fixed Voice service in the Open Access network.
		ACES	Following are the applicable notification status, 1. Completed	This API shall be invoked by Telflow to resume service in the ITC network. Key Points: 1. Telflow order Id, ODB, bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. 3. Telflow resumes the Fixed Voice service in the Open Access network. (to be validated)
9	Order Details Update (applicable for installation failures, open order update) Note: Order details update is not applicable for STC operator.	ITC	Not Applicable Following are the applicable notification status, 1. Success	This API shall be invoked by Telflow to perform order update for ITC orders. Key Points: 1. Telflow order Id, ODB, old/new bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'.
		Dawiyat	Not Applicable Following are the applicable notification status, 1. Completed	This API shall be invoked by Telflow to perform order update for Dawiyat orders. Key Points: 1. Telflow order Id, ODB, old/new bandwidth, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id.
		ACES	<Itd> <Venkat/Abdalla to provide the details>	<Venkat/Abdalla to provide the details>
10	Trouble Ticket (Create / ReOpen / Close SR) Note: Close SR is a synchronous API and notifications are not applicable.	STC	Following are the applicable notification status, 1. Resolved 2. Rejected 3. Closed	This API shall be invoked by Telflow to create/reopen SR in the Open Access provider network. Key Points: 1. Telflow order Id, ODB, etc., shall be sent in the request. 2. Telflow must send SR Id in the trouble ticket request as per the below format, a. 1-<12-digit number> As per Raqib's confirmation. 3. In the synchronous acknowledgement response, STC doesn't send their external SR Id. This will be shared in the later async notification. 4. In the synchronous acknowledgement response of Create SR, STC sends their external SR/TT Id. The SR /TT Id that was received shall be used in the Update/Close SR API request.
		ITC	Following are the applicable notification status, 1. Resolved 2. Rejected 3. Closed	This API shall be invoked by Telflow to create/reopen SR in the Open Access provider network. Key Points: 1. Telflow order Id, ODB, etc., shall be sent in the request. 2. Telflow must send SR Id in the trouble ticket request as per the below format, a. 1-<12-digit number> As per Raqib's confirmation. 3. In the synchronous acknowledgement response of Create SR, ITC sends SR/TT id. The SR/TT Id that was received shall be used in the Update/Close SR API request.

		Dawiyat	Following are the applicable notification status, 1. Resolved 2. Rejected 3. Closed	This API shall be invoked by Telflow to create/reopen SR in the Open Access provider network. Key Points: 1. Telflow order Id, ODB, etc., shall be sent in the request. 2. Telflow must send SR Id in the trouble ticket request as per the below format, a. 1-<12-digit number> As per Raqib's confirmation. 3. In the synchronous acknowledgement response of Create SR, Dawiyat sends their external SR/TT Id and guid. The SR/TT Id and the guid that were received shall be used in the Update/Close SR API request.
		ACES	<td> <Venkat/Abdalla to provide the details>	<Venkat/Abdalla to provide the details>
11	Cancel Service Installation Request	STC	Following are the applicable notification status, 1. ORDERED 2. InProgress 3. CANCELLED 4. Closed	This API shall be invoked by Telflow to cancel service installation request in the STC network. Key Points: 1. Telflow must send STC Order Id in the modification request as per the below format, a. I-MY<8-digit number> (Based on Mobily confirmation)
		ITC	Following are the applicable notification status, 1. Create Task 2. Assigned 3. Accepted 4. Set Off 5. Failure	This API shall be invoked by Telflow to cancel service installation request in the ITC network.
		Dawiyat	NA (Synchronous API)	This API shall be invoked by Telflow to cancel service installation request in the Dawiyat network.
		ACES	<td> <Venkat/Abdalla to provide the details>	<Venkat/Abdalla to provide the details>
12	AddOn Provisioning Request	STC	Following are the applicable notification status, 1. TBD	This API shall be invoked by Telflow to provision AddOn in the STC network. Key Points:
		ITC	Following are the applicable notification status, 1. TBD -	This API shall be invoked by Telflow to provision AddOn in the ITC network. Key Points:
		Dawiyat	Following are the applicable notification status, 1. TBD -	This API shall be invoked by Telflow to provision AddOn in the Dawiyat network. Key Points:
13	Relocation	STC	Following are the applicable notification status, 1. WO Created in WFMS 2. Technician Assignment 3. Technician Working 4. Activate ONT 5. Closed	This API shall be invoked by Telflow to perform relocation in the STC network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. Telflow must send STC Order Id in the Installation request as per the below format, a. 'I'-MY<8-digit number> (Format based on Mobily confirmation) 3. In the synchronous acknowledgement response, STC doesn't send their internal order Id. 4. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
		ITC	Following are the applicable notification status, 1. Create Task 2. Assigned 3. Accepted 4. Set Off 5. Success 6. Leave	This API shall be invoked by Telflow to perform relocation in the ITC network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. 3. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
		Dawiyat	Following are the applicable notification status, 1. InProgress 2. Acknowledged 3. InProgress-Dispatch 4. InProgress-Departure 5. InProgress-Arrival 6. InProgress-HAG Activation 7. Successful 7. Completed	This API shall be invoked by Telflow to perform service installation in the Dawiyat network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, Dawiyat doesn't send their internal order Id. 3. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.

		ACES	Following are the applicable notification status, 1. Acknowledged 2. Received 3. Accepted 4. InProgress 5. PartialCompleted 6. Completed	This API shall be invoked by Telflow to perform relocation in the ACES network. Key Points: 1. Telflow order Id, ODB, bandwidth, appointment details, etc., shall be sent in the request. 2. In the synchronous acknowledgement response, ITC sends 'Provider Service Number'. 3. Provider SAN sent by Open Access to Telflow shall be cascaded to SingleView.
14	Fixed Voice Termination Request	STC	TBD	TBD

Note:

The API shall be implemented in Telflow based on ESB EICD documentation.

Telflow - SingleView (Direct)

TMF 639 (Synchronous)

During order provisioning, Telflow shall query SingleView to fetch the CPE Mac Address of the installed CPE by passing in the serial number. This integration is a REST based interface, synchronous and of TMF 639 standard. The basic authentication information shall be passed while initiating the request. Based on the response from SingleView, the order shall proceed further.

Following details are expected in the response,

1. CPE/ONT/HAG Mac Address (xxxx.xxxx.xxxx format in lower case)
2. ~~CPE Model~~
3. ~~CPE Vendor~~
4. CPE Serial Number (~~HEX format~~)
5. Domain

Bespoke Integration (Asynchronous)

Following APIs are applicable,

S. No	Action API Description	Request	Response
1	ODB Patching Action	Following details are sent in the request, 1. Action Type 'ODB Patching' 2. SingleView Order Id 3. ODB ID 4. Service Address	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Notification: Following key details are sent by SingleView, 1. Action Id 2. Resolution Code 3. Resolution Text
2	Create action to Mobily / Open Access teams	Following details are sent in the request, 1. Action Type (refer the DMN table in the 'Handling Installation Failure' section) 2. SingleView Order Id 3. Reason Code 4. Reason Text 5. Comments (if any) 6. Engineer Details a. Id b. Name c. Phone Note: 1. Treatment category detail shall also be sent to SingleView.	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Notification: Following key details are sent by SingleView, 1. Action Id 2. Resolution Code 3. Resolution Text 4. Updated Order data a. ODB Id b. Contact Number 5. Comments (if any)

3	Book Appointment	Following details are sent in the request, 1. Action Type 'FTTH Appointment Capture' 2. SingleView Order Id 3. Reason Code 4. Reason Text 5. Comments (if any) 6. Engineer Details a. Id b. Name c. Phone	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Notification: Following key details are sent by SingleView, 1. Action Id 2. Resolution Code 3. Resolution Text 4. New Appointment details a. Appointment Id b. Slot Id c. Appointment Start Date d. Appointment End Date 5. Comments (if any)
4	Create Action in SV (for UAT Timer Expiry)	Following details are sent in the request, 1. Action Type 'Contact Customer for UAT' 2. SingleView Order Id	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Notification: Following key details are sent by SingleView, 1. Action Id 2. Resolution Code 3. Resolution Text
5	Initiate order verification (for cancellation)	Following details are sent in the request, 1. Action Type 'FTTH Order Verification' 2. SingleView Order Id 3. Reason Code 4. Reason Text 5. Comments (if any) 6. Engineer Details a. Id b. Name c. Phone	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Notification: Following key details are sent by SingleView, 1. Action Id 2. Resolution Code 3. Resolution Text 4. Comments (if any)

Bespoke Integration (Synchronous)

Following API is applicable,

S. No	Action API Description	Request	Response
1	Create Action for CPE / ME / ONT Installation	Following details are sent in the request, 1. Action Type 2. SingleView Order Id 3. Appointment Id 4. Task Status (Telflow order sub state)	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Note: Following are the supported action types, 1. CPE Installation 2. ME Installation 3. ONT Installation 4. CPE/HAG Swap 5. ONT Swap 6. Rewiring 7. Relocation

2	Update Action for CPE / ME / ONT Installation	Following details are sent in the request, 1. Action Type 2. SingleView Order Id 3. Appointment Id 4. Task Status (Telflow order sub state) 5. Partition 6. Region 7. Vendor Name 8. Engineer Details a. Id b. Name c. Phone 9. ODB Port 10. Reason Code 11. Reason Text 12. Comments (if any) 13. Serial Number	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Note: Following are the supported action types, 1. CPE Installation 2. ME Installation 3. ONT Installation 4. CPE/HAG Swap 5. ONT Swap 6. Rewiring 7. Relocation 8. Onsite Visit 9. Open SR Ticket 10. Re-Open SR Ticket 11. Close SR Ticket
3	Close Action for CPE / ME / ONT Installation	Following details are sent in the request, 1. Action Type 2. SingleView Order Id 3. Status 4. Appointment Id 5. Task Status (Telflow order sub state) 6. Partition 7. Region 8. Vendor Name 9. Engineer Details a. Id b. Name c. Phone 10. ODB Port 11. Reason Code 12. Reason Text 13. Comments (if any)	Acknowledgement: 1. SingleView sends the 'action Id' created in the acknowledgement response. Note: 1. Following are the supported action types, a. CPE Installation b. ME Installation c. ONT Installation d. CPE/HAG Swap e. ONT Swap f. Rewiring g. Relocation 2. 'Status' set to 'Complete' when UAT is completed.

Telflow - MapleWave Supply (Direct)

Bespoke Integration (Synchronous)

During order provisioning, Telflow shall send the request to update the CPE/Mesh Extender inventory status in MapleWave Supply. The request shall be sent to mark the CPE/ME as used for the customer's FTTH service. This integration is a REST based interface, synchronous and of non-TMF standard (OAuth based custom integration).